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INDONESIA ECONOMIC PROSPECTS

IEP

Digital Foundations for Growth



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1818 H Street NW
Washington DC 20433
Telephone: 202-473-1000
Internet: www.worldbank.org

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Preface

The Indonesia Economic Prospects (IEP) is a bi-annual World Bank report that assesses recent macroeconomic developments, the outlook, and risks, as well as specific development challenges for the Indonesian economy. The IEP aims to inform the public policy debate and serves a broad audience including the general public, government, private sector, civil society organizations, and other domestic and international stakeholders.

The IEP is a product of the World Bank Jakarta office and receives strategic guidance from an editorial board chaired by Carolyn Turk, Country Director for Indonesia and Timor-Leste. The report is prepared by the Economic Policy (EP) Global Practice team, under the guidance of Habib Nasser Rab (Practice Manager), Marco Hernandez (Practice Manager), James Neuman (Practice Manager) and David Knight (Lead Economist). The report is co-led by Wael Mansour (Senior Economist), Ratih Dwi Rahmadanti (Economist), and Jonathan Daniel Marskell (Senior Digital Specialist).

Deviana Djalil provided administrative support and coordinated the organization of the report launch event. The dissemination was organized by Gb Surya Ningnagara and Maulyati N. Slamet under the guidance of Lestari Boediono Qureshi. The report was designed and typeset by Arsianti.

Part A was prepared by Wael Mansour, Dwi Endah Abriningrum, Indira Maulani Hapsari, Mehwish Ashraf, Ratih Dwi Rahmadanti, and Rully Prassetya. Inputs were provided by Ramya Sundaram, Anastasia Denisova, Bridget Crumpton, Shreya Chatterjee, and Abror Tegar Pradana (Labor Markets), Ou Nie, Marlon Rawlins, Neni Lestari, and Putra Arsana (Financial Sector), Sailesh Tiwari, Ririn Salwa Purnamasari, and Samuel Nursamsu (Poverty Analysis). Part A benefitted from the comments of David Knight and was peer reviewed by Gitanjali Kumar.

Part B was prepared by Jonathan Marskell, Nadia Kartikasari, and Maria Tambunan, under the oversight of James Neumann. Contributions were made by Serene Ho, Mohammad Pasha, Jan van Rees, Christophorus Ivan Samuels, Scott Minehane, Marlon Rolston Rawlins, I Gede Putra Arsana, Gandhi Arssegi, Tony Seno Hartono, Maruska Vizek, Ivan Vigiono Pradana, Gracella Maureen, Yan Liu, Saloni Khurana, and Juan Manuel Porras Lopez. Part B benefited from comments by Mohamed Wahish, Camila Saad, Rong Chen, David Knight, Wael Mansour and Ratih Dwi Rahmadanti, and was peer reviewed by Thomas Chalumeau. Support from the Australia-World Bank-Indonesia Partnership (ABIP) Trust Fund is gratefully acknowledged.

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- December 2024: Funding Indonesia's Vision 2045
- June 2024: Unleashing Indonesia's Business Potential

To receive the IEP and related publications by email, please contact ddjalil@worldbank.org. For questions and comments, please contact wmansour@worldbank.org and ltriveno@worldbank.org.

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Abbreviations and Acronyms

3T	*Tertinggal, Terdepan, dan Terluar* Underdeveloped, Frontier, and Outermost)
AI	Artificial Intelligence
ALMPs	Active Labor Market Policies
APBI	Indonesian Coal Mining Association
APJII	*Asosiasi Penyelenggara Jasa* Internet Indonesia (Indonesian Internet Service Providers Association)
APS	Average Propensity to Save
ARPU	Average Revenue Per User
BAKTI	*Badan Aksesibilitas Telekomunikasi dan Informasi* (Telecommunication and Information Accessibility Agency)
BI	Bank Indonesia
BKPM	Investment Coordinating Board
BOP	Balance of Payment
BPD	Barrels Per Day
BPLAI	*Bantuan Pemerintah Layanan Akses Internet*
BPS	*Badan Pusat Statistik* (Central Bureau of Statistics)
BRTI	*Badan Regulasi Telekomunikasi Indonesia* (Telecommunication Regulatory Body)
BSSN	*Badan Siber dan Sandi Negara* (National Cyber and Crypto Agency)
BULOG	*Badan Urusan Logistik* (State Logistics Agency)
CAR	Adequacy Ratio
CBDC	Bank Digital Currency
CDNs	Content Delivery Networks
CEIC	CEIC Data
CIT	Corporate Income Tax
CLS	Cable Landing Station
CMEA	Coordinating Ministry for Economic Affairs
CPO	Crude Palm Oil
CPE	Customer Premises Equipment
CPI	Consumer Price Index
CTAS	Core Tax Administration System
CV	Coefficient of Variation
D2D	Direct-to-Device
DAPODIK	*Data Pokok Pendidikan*
DBH	*Dana Bagi Hasil* (Revenue Sharing Fund)
DEFA	Digital Economy Framework Agreement
DEN	*Dewan Ekonomi Nasional* (National Economic Council)
DPI	Digital Public Infrastructure
DTRR	Digital Trade Readiness
EAP	East Asia and the Pacific
EP	Economic Policy
ERP	Enterprise Resource Planning
EVs	Electric Vehicles
FDI	Foreign Direct Investment
FTTH	Fiber-to-the-Home
FTTP	Fiber-to-the-Premise
FWA	Fixed Wireless Access
GDP	Gross Domestic Product
GEO	Geostationary Earth Orbit
GDIP	Global Digital Inclusion Partnership
GNI	Gross National Income
GOI	Government of Indonesia
HIC	High-Income Countries
HH	Household
HET	*Harga Eceran Tertinggi* (Retail Ceiling Prices)
HPP	*Harga Pembelian Pemerintah* (Government Purchase Price)
IBRD	International Bank for Reconstruction and Development
ICT	Information and Communication Technology
ICLS	International Conference of Labour Statistics
IDA	International Development Association
IDC	International Data Corporation
IDR	Indonesian Rupiah
IEP	Indonesia Economic Prospects
IKD	*Identitas Kependudukan Digital* (Digital Population Identity)
ILO	International Labour Organization
INA	Indonesia Investment Authority
INPRES	*Instruksi Presiden* (Presidential Instruction)
IOH	Indosat Ooredoo Hutchison
ISCO	International Standard Classification of Occupations
ISP	Internet Service Provider

ITU	International Telecommunication Union
IXPs	Internet Exchange Points
JKK	*Jaminan Kecelakaan Kerja* (Work-Accident Insurance)
JKP	*Jaminan Kehilangan Pekerjaan* (Job Loss Insurance)
JSS	Jogja Smart Service
KIK	*Kontrak Investasi Kolektif* (Collective Investment Contracts)
Komdigi	*Kementerian Komunikasi dan Digital* (Ministry of Communication and Digital Affairs)
Kominfo	*Kementerian Komunikasi dan Informatika* (Ministry of Communication and Informatics)
KTP	*Kartu Tanda Penduduk* (National ID)
KUR	*Kredit Usaha Rakyat* (People's Business Credit)
L2INDO	Listening to Indonesia
LAR	Loan at Risk
LCR	Liquidity Coverage Ratio
LDR	Loan-to-Deposit Ratio
LEO	Low Earth Orbit
LMIS	Labor Market Information System
LNG	Liquefied Natural Gas
M2	Broad Money
MBG	*Makan Bergizi Gratis* (Nutritious Meal Program)
MC	Middle-Class
MCU	Mother and Child
MDES	Ministry of Digital Economy and Society
MEO	Medium Earth Orbit
MFMOD	Macro-Fiscal Model
MNO	Mobile Network Operator
MOA	Ministry of Agriculture
MOF	Ministry of Finance
MoHA	Ministry of Home Affairs
MoPSE	Ministry of Primary and Secondary Education
MPC	Marginal Propensity to Consume
MPS	Marginal Propensity to Save
MSMEs	Micro, Small and Medium Enterprises
NBTC	National Broadcasting and Telecommunications Commission
NEET	Not in Employment, Education, or Training
NFA	Net Foreign Assets
NPL	Non-Performing Loans
NSFR	Net Stable Funding Ratio
ODP	Optical Distribution Point
OECD	Organisation for Economic Co-operation and Development
OJK	*Otoritas Jasa Keuangan* (Financial Services Authority)
OSS	Online Single Submission
PDN	*Pusat Data Nasional* (National Data Centers)
PES	Public Employment Services
PMI	Purchasing Manager Index
PNBP	*Penerimaan Negara Bukan Pajak* (Non-Tax Revenue)
PP	Percentage Points
PPA	Power-Purchase Agreements
PPP	Public-Private Partnerships
PPP	Purchasing Power Parity
PSrE	*Penyelenggara Sertifikasi Elektronik* (Certification Authorities)
QRIS	Quick Response Interoperability Standard
ROA	Return-on-Assets
ROE	Return-on-Equity
RPJMN	*Rencana Pembangunan Jangka Menengah Nasional* (National Medium-Term Development Plan)
RRR	Reserve Requirement Ratio
SAL	*Sisa Anggaran Lebih* (Excess Financing Balance)
SATRIA-1	*Satelit Republik Indonesia*-1
SILPA	*Sisa Lebih Pembiayaan Anggaran* (Excess Cash Surplus)
SNAP	*Standar Nasional Open API Pembayaran* (National Standard for Payment Open API)
SOBs	State-Owned Banks
SOEs	State-Owned Enterprises
SPBE	*Sistem Pemerintahan Berbasis* Elektronik (Electronic-Based Government System)
SPC	Special Purpose Companies
SPHP	*Stabilisasi Pasokan dan Harga Pangan*
SPLP	*Sistem Penghubung Layanan* Pemerintah (Government Service Exchange System)
SRBI	*Sertifikat Rupiah Bank Indonesia* (Monetary Securities)
SupTech	Supervisory Technology
SUSENAS	*Survei Sosial Ekonomi Nasional* (National Survey of Socioeconomics)
TOT	TOT Public Company Limited
TVET	Technical and Vocational Education and Training
UMIC	Upper Middle-Income Countries
USO	Universal Service Obligation
VAT	Value-Added Tax
WB	World Bank
YOY	Year-on-Year

Executive Summary



Executive Summary

I. Economic Update

Indonesia's economy continues to demonstrate resilience amid heightened global uncertainty. Real GDP growth held at 5.0 percent year-on-year over the first three quarters of 2025, supported by robust investment and a notable, albeit temporary, contribution from net exports, which benefited from front-loaded goods trade and rising global demand for commodities, particularly palm oil, iron, steel, and gold. Growth is relatively broad based, with services contributing the most to economic activity and agriculture substantially rebounding following favorable climate conditions and government support programs. Inflation is picking up on the back of rallying food prices but remains within the central bank's target band. Meanwhile, financial buffers are adequate and the fiscal framework compliant with the national fiscal rules. Yet, with private consumption—traditionally the main driver for growth—softening, the durability of growth increasingly hinges on strengthening household purchasing power, including through improved job quality and policies that boosts investment returns and productivity.

While macroeconomic stability remains, challenges arise from the labor market with impact on household welfare. Private consumption's contribution to growth moderated slightly to 2.7 percentage points (pp) (53.3 percent of GDP growth), down from 2.8 pp in 2024 (54.8 percent of GDP growth). This is consistent with labor market indicators pointing to persistent challenges in job quality. Employment rose by 1.3 percent between August 2024 and August 2025, but gains were concentrated in lower-paying sectors such as low-value services and agriculture. Real wages have trended down since 2018, and the structure of employment shows signs of polarization, with middle-skilled jobs shrinking relative to both low- and high-skilled segments. Youth entrants are disproportionately finding low-tier informal jobs. This is consistent with survey evidence of a growing disconnect between objective welfare gains and subjective perceptions: despite continuing declines in monetary poverty, a rising share of households' report feeling poor. This sense of economic insecurity is more pronounced among the middle class, amid higher income volatility and subdued real wage dynamics. These patterns weigh on household consumption and reinforce precautionary saving, even as headline macroeconomic indicators remain solid.

Government policy has responded with a mix of targeted support and prudence, though effectiveness is constrained by weak revenue. The fiscal deficit widened to 2.0 percent of GDP by October 2025, up from 1.4 percent of GDP same period last year, amid weaker revenues linked to easing commodity prices, accelerated tax refunds, and SOE dividends retained by Danantara. Despite this constrained environment, the government delivered quarterly stimulus packages totaling 0.5 percent of GDP, aimed at bolstering consumption and job creation. Additional spending was possible due to a budget efficiency drive and underspending in some of the priority programs. Accommodative financing conditions helped contain borrowing costs, but interest payment to revenue remains high at 20.5 percent (year-to-October). Looking ahead, the 2026 Budget Law targets a 2.7 percent deficit with ambitious revenue goals; achieving these will require strong tax administration and policy efforts in a less favorable commodity prices setting to safeguard space for growth-enhancing spending.

Monetary policy is becoming more accommodative to support private sector credit, but transmission channels have been uneven. With inflation contained, Bank Indonesia (BI) lowered the policy rate by a cumulative 150 basis points since September 2024 to 4.75 percent, and complemented rate cuts with new macroprudential incentives aiming to encourage faster pass-through to lending rates. Nonetheless, market lending and deposit rates have been slow to adjust, pointing to friction in monetary transmission channels. Despite adequate system liquidity, nominal credit growth slowed to 7.4 percent year-on-year in October 2025, suggesting constraints arising from demand for bankable projects and cautious risk appetite. Easing has also coincided with capital outflows and Rupiah pressures, requiring market interventions from BI to

maintain stability. Strengthening credit uptake will therefore depend on improved investment sentiment and credit infrastructure.

The outlook remains resilient, with risks broadly balanced. Growth is projected to hold at 5.0 percent in 2026, rising to 5.2 percent in 2027 on a gradual pickup in investment anchored by state-led investment via Danantara, monetary easing to stimulate private sector credit, and Foreign Direct Investment (FDI). Low inflation and fiscal stimulus will support private consumption, but it is expected to contribute less than previous years to overall growth amid weaker labor income especially for middle-class consumers. Inflation is expected to remain within target, though food and energy price volatility present recurring challenges. Downside risks include further softening of real wages that could dampen consumption further, heightened global trade tensions, and reversal of external financial conditions. Domestically, weaker-than-planned revenues could test adherence to fiscal rules and constrain spending. While upside potential risk arises from stronger demand in major trading partners and accelerated implementation of ongoing business, trade and investment deregulation reforms.

A calibrated policy package can address near-term pressures while lifting medium-term prospects. Business enabling policies, which reverse declining productivity growth and mobilize private capital into high-value investment, remain core to creating good jobs. In tandem, supply-side measures can start upskilling the future labor force. Targeted interventions to ease skills mismatches and improve job quality are essential: strengthening labor market information systems, expanding active labor market policies with a focus on lifelong learning and digital upskilling, and improving school-to-work transitions—especially for youth and women—would raise employability and support middle-class job creation. On fiscal policy, a bolder revenue strategy—centered on quick wins in tax administration and digitalization, broadening the tax base through adjusted thresholds, and enhanced compliance—can create fiscal space to expand growth programs in infrastructure, human capital, and climate resilience. On finance, well-designed, time-bound credit guarantees and risk-sharing facilities, underpinned by robust digital credit infrastructure, can catalyze lending to viable MSMEs and tradable sectors, help translate monetary easing into investment, and crowd in private capital with modest fiscal outlays.

Taken together, these measures can sustain Indonesia's resilience and gradually lift potential growth, ensuring that more households experience tangible welfare gains as the economy navigates a more uncertain global environment.

II. Digital Foundations for Growth

Indonesia's digital infrastructure—the foundation of its emerging digital economy—has expanded rapidly, but its quality has not kept pace with rising demand and its usage remains uneven. Internet access has expanded quickly: 72.8 percent of Indonesians used the internet in 2024, up from 54 percent in 2020. Yet connectivity remains "broad but shallow." The domestic broadband backbone now reaches 98 percent of districts, but only half of villages have fiber optic distribution points. Eastern Indonesia and some rural areas elsewhere mostly rely on costly and capacity-constrained microwave or satellite links. Performance lags far behind regional peers: Indonesia ranks last among ASEAN upper middle-income economies in fixed broadband penetration and in both mobile and fixed broadband speeds. Just 22 percent of schools have a 100 Mbps or faster internet connection and 24 percent of health clinics report having good internet, hampering government efforts to strengthen education and healthcare service delivery. The result is a growing divide between a digital Indonesia—connected but constrained—and an unconnected Indonesia that risks being left further behind.

Achieving near-universal mobile network coverage in Indonesia has been a success story but mobile internet speeds are among the slowest in ASEAN. Tower sharing reforms and strong competition have enabled 4G coverage above 99 percent of the population and some of the region's most affordable prices. Yet internet speeds and reliability are poor because Indonesia assigns barely half as much spectrum for mobile internet as its neighbors—ranking last in the G20—and has been delayed in release new bands. As

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a result, Indonesia's 5G transition has been among the slowest in the region's major economies, which limits the quality, energy and cost-efficiency gains that 5G can deliver. Indonesia also maintains some of the region's highest spectrum fees—about 11 percent of mobile network operator (MNO) revenues—limiting resources available for network upgrades. Reduced fees could be tied to clearer coverage and performance obligations. In addition to fair allocation of more spectrum to MNOs, improving quality of mobile broadband will eventually also require accelerating tower fiberization, given rising data traffic and the limits of satellite and microwave links that connect many towers in rural areas.

Fixed broadband remains Indonesia's most significant connectivity weakness, limiting productivity gains. Only one-fifth of households have a fixed broadband connection. Coverage is concentrated in urban areas, as high deployment costs, complex permitting, and limited infrastructure sharing slow expansion to rural areas. Telkom's two-thirds market share and the difficulty smaller internet service providers face in accessing affordable backbone capacity reduce competition, keeping prices high and quality of service low. As a result, fixed broadband remains unaffordable for many households, costing about twice as much relative to income as in Malaysia, Thailand, and Viet Nam. Telkom's planned spinoff of its fiber assets into a wholesale subsidiary offers an opportunity to move toward a more open, shared national backbone and strengthen competition across the sector.

Reducing the high cost of rural fiber deployment will require more infrastructure sharing—including access to PLN's poles—and complementary use of fixed wireless access (FWA) and satellite. In eastern Indonesia, high backbone costs and weak interconnections across networks, especially on the Palapa Ring, keep internet prices elevated and reduce resilience. The recent release of FWA spectrum and announced plans to offer packages at less than half of current averages could support wider uptake, though inherent capacity limits mean FWA will likely play a transitional role and complement—rather than replace—the need for a national fiber network. To maximize impact, the government's SATRIA-1 satellite should be focused on the most remote areas of Indonesia—especially in eastern Indonesia—where neither fiber nor FWA is feasible. Satellites also provide critical redundancy during disasters or network outages, making targeted use even more important. Ensuring the financial sustainability of BAKTI—the government's universal service agency—by exploring a shift from operating networks to acting as a market enabler will also be important for extending connectivity to remote areas.

Indonesia's data center capacity is expanding, but growth is constrained by a complex investment environment. Local data centers matter because data-intensive services and newer AI applications need computing capacity close to users. They also attract investment, support technology transfer and local innovation, and create skilled digital jobs. Despite being ASEAN's largest digital economy, Indonesia trails Malaysia and Singapore by a wide margin in data center capacity—Malaysia has around twenty times more capacity per capita—and growth in 2025 was outpaced by Thailand. Relatively slow approval processes, complex and uncoordinated permitting, uncertainty over data governance, and challenges in securing reliable and renewable electricity and water connections deter investors. Greater adoption of cloud by the public sector and Danantara's interest in sustainable data centers offer timely opportunities for the government to mobilize private capital and accelerate expansion of domestic capacity.

A more predictable and supportive regulatory environment is essential to closing infrastructure gaps and accelerating digital economy growth. Key regulatory areas remain at an emerging stage: implementation of the Personal Data Protection Law, enactment of a cybersecurity law, rules governing digital platforms, and clarity on cross-border data flows—all of which affect investor confidence and integration into international digital supply chains. Restrictions on where new international submarine cables may land deter global cloud providers and carriers from investing, and excluding Jakarta reinforces a single-corridor dependency on the Singapore-Batam route for most international bandwidth.

Indonesia can help unlock faster productivity growth and expand economic opportunities by strengthening the quality and reach of its digital infrastructure. Importantly, stronger digital infrastructure will create new job opportunities—both

in the digital sector and across the wider economy—particularly for youth and women in less advantaged areas who are currently most constrained by limited connectivity. Achieving this will require multiplying mobile broadband speeds, making fixed broadband universal and affordable, expanding sustainable data center and cloud capacity, and strengthening and diversifying the national backbone. Continued progress in digital skills, digital trust, GovTech/digital government, and digital public infrastructure will help Indonesia translate better infrastructure into productivity gains.

To seize this opportunity, the report recommends the following practical, high-impact actions for the government to take in the near term:

1. Assign available 700 MHz and 2.6 GHz band spectrum to mobile broadband, and release and assign 3.5 GHz band capacity in urban areas to accelerate 5G deployment.
2. Mobilize private investment in rural fixed broadband by aggregating predictable, multi-year demand from schools, health clinics, and village offices to increase commercial viability.
3. Promote fair open access to wholesale fiber networks and passive infrastructure—including PLN's poles—and standardized rights-of-way procedures to improve competition and reduce deployment costs.
4. Streamline data center permitting and improve coordination across special economic zone (SEZ) policy, electricity provision, and incentive frameworks to reduce delays and improve the investment environment.
5. Set clear, risk-based rules for data storage and cross-border data flows—including through issuance of the Government Regulation on Personal Data Protection and alignment of other regulations—while promoting secure government adoption of cloud services to anchor demand.
6. Amend Minister of Maritime Affairs and Fisheries Decree No. 14 of 2021 to allow additional international submarine cable landing sites, including Jakarta and other major commercial hubs.
7. Review Palapa Ring wholesale tariffs to ensure open and sustainable access, particularly to improve affordability and connectivity in eastern Indonesia.



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A. Economic Update



A. Economic Update

1. Recent Economic Developments

Indonesia's economy has been resilient despite global uncertainty, with steady growth driven by investment and commodities exports, but consumption is softening amid souring perceptions of economic performance.

The global economy in 2025 faced heightened policy uncertainty, and elevated geopolitical and trade tensions. Stockpiling, supply-chain adjustments, and easing global financing conditions have partially cushioned economic activity from these shocks. In East Asia and the Pacific (EAP), economic activity remains strong, supported by robust trade, tourism, and consumption. But, as some of these temporary benefits of accelerated trade flow and favorable financing subside, and as tariffs are passed through to end consumers, global growth is expected to moderate. Greater trade barriers externally, as well as limited structural reform domestically present challenges to create better jobs in EAP, while automation and Artificial Intelligence (AI) are rapidly changing the shape of work (EAP Economic Update, October 2025).

Amid this uncertainty, Indonesia's economy has remained resilient, buoyed by investment and net exports. GDP growth reached 5.0 percent year-on-year (yoy) in the first three quarters of 2025 (3Q-25), at par with recent years (Figure A.1). Domestic demand remains the main growth driver, although the contribution of net exports rose from -0.1 in 2024 to 21 percent, marking its greatest contribution for five years. This effect is temporary though as it was driven by front-loaded goods exports and rising global demand for commodities, particularly palm oil, iron steel, and gold. Private consumption growth inched down, contributing 2.7 percentage points (pp) to GDP (53.3 percent of GDP growth), down from 2.8 pp in 2024 (54.8 percent of GDP growth). Low core inflation, sluggish car and retail sales, and subdued credit growth all point to moderating private consumption. Public consumption's contribution to growth dropped sharply from 0.5 pp in 2024 to 0.1 pp in 2025 (9.6 to 1.9 percent of GDP growth), due to budget reallocations and slow disbursement in priority programs. In line with robust capital goods imports, investment remains strong, contributing 1.4 pp to growth (29 percent of GDP growth), unchanged from 2024, driven by public defense and industrial sectors.

Growth is relatively broad based, with services supported by fiscal stimulus and agriculture by favorable climate conditions. Services contributed 57.4 percent to growth in 3Q-25, up from 56.2 percent in 2024, led by education, health, business services, and ICT (Figure A.2). Services growth was boosted by public spending on flagship education and health programs. Fiscal stimulus also benefited wholesale, retail, and hospitality sectors including leisure activities in tourist-led regions such as Bali and Nusa Tenggara. Manufacturing also performed relatively well, led by the basic metal, pharmaceutical, and food and beverage industries. Manufacturing benefited from external demand for palm oil products, chemical materials, and iron and steel products. Those three sub-sectors' contribution rose by 2.3 pp from last year to 15.7 percent. Agriculture output hit a decade high thanks to favorable climate conditions. In addition, private investment in palm oil (CPO) production and downstreaming accelerated. Growth in the mining sector was subdued, affected by lower coal exports due to rising production in China and India and the suspension of PT Freeport Indonesia's operations following a landslide at the Grasberg Mine in Papua.

Figure A.1: Growth remains resilient, as weak consumption was offset by strong net exports
(percentage points contribution to yoy growth)

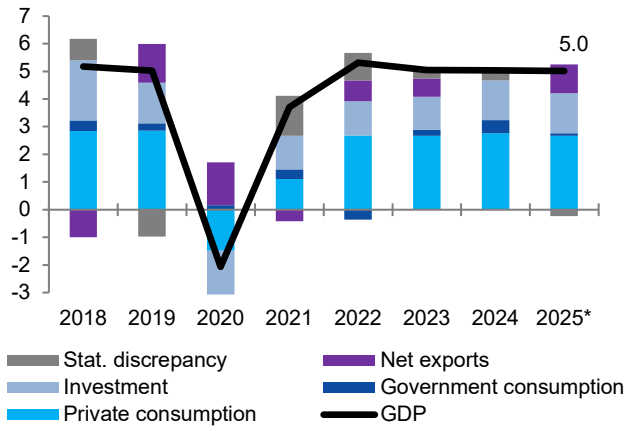
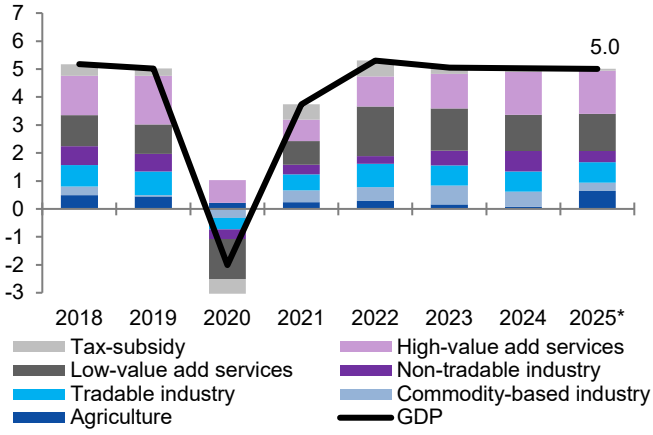


Figure A.2: Growth was led by services and manufacturing, while agriculture rebounded
(percentage points contribution to yoy growth)

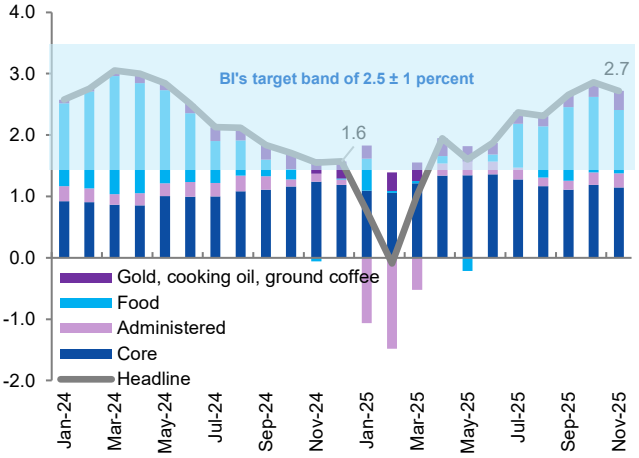


Source: BPS, CEIC, WB Staff calculations.
Note: 2025* is three quarters GDP in 2025. High-value-add services are ICT, financial & insurance, real estate, business, public admin, defense & social security, education, health, and other services. Low-value-add services are retail & trade, transport & storage, and hospitality. Non-tradable industry is utilities and construction. Tradable industry is manufacturing excl. commodity-based industries. Commodity based industry is mining & quarrying, coal and O&G refinery, rubber products, non-metallic quarrying, and basic metals manufacturing.

Slower consumption growth and subsidies have partially offset inflationary pressure, which continues to rise on the back of rising food and commodities prices.

Inflation is picking up on the back of rising food and commodity prices. Consumer price inflation increased from 1.6 percent to 2.7 percent yoy over the course of the year to November, still within Bank Indonesia's (BI) target inflation band (Figure A.3). The pick-up was mostly led by food price inflation, which reached 5 percent yoy in September. Slow distribution from the State Logistics Agency's (BULOG) warehouses (see Box A.1) and elevated demand for poultry products—stemming from the nutritious meal program—amid escalating chicken feed prices, have contributed to food price inflation. Rising global demand for gold, palm oil, and ground coffee impacted inflation as exports of these commodities accelerated, causing prices to go up.

Figure A.3: Inflation rose due to rallying food and commodity prices
(percentage points contribution to yoy growth)



Source: BPS, CEIC, WB Staff calculations.

Meanwhile, core and administered price inflation remained stable amid moderate consumption growth and ongoing subsidies[1]. Core inflation edged up to 1.9 percent in November from 1.8 percent in December 2024, a modest change associated with moderating household consumption and anchored inflation expectations. Administered prices climbed from 1.1 to 1.6 percent between September and November, due to recent upward adjustments in airline tickets and tobacco prices. However, the government took measures to stabilize transport and electricity prices through various subsidy programs. This came at a fiscal cost of 0.16 percent of GDP (year-to-September).

1. The share of the total consumption basket of the CPI is divided as follows: Core (62 percent), Administered Prices (16 percent), Volatile elements (22 percent). Core inflation is computed by the World Bank excluding volatile items like gold, ground coffee, and cooking oil; all items added to the official definition used by the Central Statistics Agency.

BOX A.1: Inflation and Rice Food Policy

In 2025, food inflation was partly affected by policies that increased rice prices. BULOG is mandated to secure national food availability and price stability by conducting procurement, storage, and distribution of key staples, especially rice. It also carries out government-assigned tasks such as managing public food reserves, implementing market operations to stabilize prices, and supporting food assistance programs for vulnerable households. In 2025, there was a major policy shift in Indonesia's food trade system. To boost farmers' income and ensure high absorption during harvest season, the government mandated BULOG to purchase all unhusked rice (*gabah*) at a higher price of IDR6,500 per kilogram, regardless of quality. Previously, BULOG would offer reduced prices (i.e., price deductions (*rafaksi*)) for lower quality rice and could also opt not to purchase any additional quantities. High crop production coupled with the new policy resulted in BULOG's rice stocks surging to a record 4.2 million tons by July, up from 1.7 million tons in January, and remaining through October.

This policy imposes a considerable fiscal cost. In 2025, BULOG's allocation for procuring rice reached 0.07 percent of GDP (IDR16.6 trillion). This figure accounts for 11.5 percent of the government's total food security budget for 2025. For comparison, BULOG's allocation is equivalent to around56 percent of the Ministry of Agriculture's (MOA) overall budget. By early September, all the allocation for BULOG's rice procurement had been fully disbursed.

A rice surplus does not automatically ensure price stability. Rice, which accounts for 12.2 percent of the food and beverages consumption basket, is a key inflation driver for Indonesia. Despite a five-year high production of 35 million tons, resulting in rice self-sufficiency, food prices have steadily increased since February. By September, rice prices in 214 districts surpassed the government's Retail Price Ceiling. As a result, the Ministry of Home Affairs (MOHA) instructed local leaders to step up inflation monitoring, provide rice aid, and accelerate rice supply distribution operations (*Stabilisasi Pasokan dan Harga Pangan*, SPHP) to curb further price pressures.

However, policies aimed at stabilizing prices placed strain on distribution channels with unintended consequences of exacerbating food price inflation. Narrowing margins between the Government Purchase Price (*Harga Pembelian Pemerintah*, HPP or farmgate prices) and the regulated Retail Prices Ceiling (*Harga Eceran Tertinggi*, HET), coupled with high logistics costs, are discouraging rice movement and affecting small and medium mills, resulting in slower private supply. In addition, uncertainty regarding revised quality classifications and potential new ceiling prices is leading producers and distributors to delay procurement or retain stocks. This further restricts market supply.

These policies may also lead to greater risk of stock spoilage. Rising rice stockpiles has created new challenges like warehouse capacity constraints and quality issues. Greater variation in rice quality, such as increased moisture and impurity levels, increases the risk of spoilage during storage. The MOA has reported that a large amount of rice in the warehouses has been damaged due to warehouses' overcapacity and distribution bottlenecks. Addressing issues related to price controls at the various levels of the supply chain coupled with tackling bottlenecks are important to ensure a smooth supply of rice and align incentives of farmers, distributors, and consumers.

Employment growth masks persistent challenges in job quality and labor market vulnerabilities.

Overall employment, formal and informal, rose in 2025, but job quality has not improved. Between August 2024 and August 2025, there was a net addition of 1.9 million jobs (equating to annual employment growth of 1.3 percent), a slower pace than the previous year's 4.8 million net additional jobs (3.4 percent). Labor force participation and unemployment were both unchanged, at 70.6 and 4.8 percent, respectively, over the latest year. Female labor force participation edged up by 0.21 pp (now at 56.6 percent), while male participation dipped by 0.26 pp (now at 84.4 percent). The quality of employment has not improved though. Most new jobs created were in agriculture (up 0.49 million) and accommodation and food services (up 0.42 million) which, at about IDR2.55 million monthly, both pay well below the overall national average of IDR3.33 million.

Real wages have been declining. Between 2018 and 2024, real wages declined economy-wide by 1.1 percent annually. High-skilled workers experienced the greatest declines, of 2.3 percent annually, followed by middle-skilled workers, whose wages decreased by 1.1 percent annually (Figure A.4). On the other hand, real wages grew slightly for low-skilled workers, at 0.3 percent annually. The only sectors which did not record a drop in real wages for high-skilled workers are high-value added manufacturing, utilities (specifically electricity), information and communication technology (ICT).

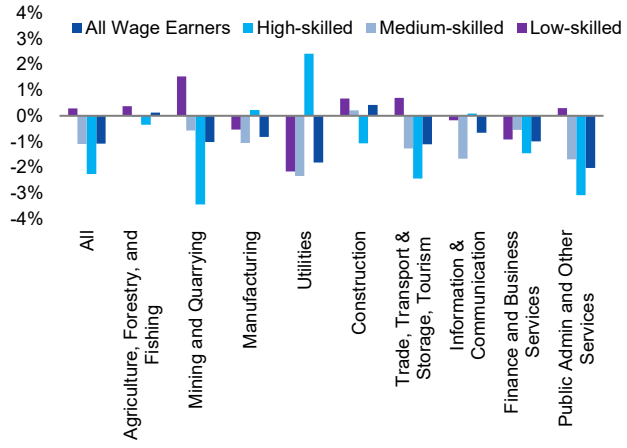
Not only are real wages declining but the skill mix is also changing. The share of middle-skilled employment dropped from 71.1 percent in 2018 to 68.3 percent in 2024 Figure A.5) offset by an increase in low-skilled employment (by 2.3 percentage point). High-skilled employment share also increased by 0.5 percent points over this period (Figure A.6). The agriculture sector remains the largest employer of low-skilled workers and the second largest employment of middle-skilled workers, after the trade, transport and storage sector. In the manufacturing and utilities sectors, only the share of low-skilled employment grew, whereas the mining, construction and ICT sectors saw growth in the share of middle-skilled employment, bucking the overall trend.

Youth in Indonesia face a difficult school-to-work transition, and while most men are employed by age 30, women experience persistently high inactivity. The first transition occurs at around age 19, when individuals leave high school. At this stage, 16.6 percent of young women are not in employment, education, or training (NEET), more than double the rate for young men (7.9 percent) (Figure A.7 and Figure A.8). A second transition takes place around age 24, after college. By then, about 66 percent of the population is employed, but women are more than ten times as likely as men to be NEET (39.2 percent vs. 3.8 percent). From age 29 to 54, men's employment rates remain above 90 percent. In contrast, over 43 percent of women remain inactive after age 29, and even as inactivity declines from age 36 onward, it never falls below 28 percent. Women's inactivity rates at prime working ages (25–54) stand at 36.2 percent—significantly higher than the averages in upper-middle-income (29.0 percent) and high-income economies (19.7 percent). Meanwhile, inactivity among men in the same age group (2.9 percent) is well below both UMIC and HIC averages (9.4 percent and 7.9 percent, respectively).

Young people secure employment but they are less likely to have a stable job that offers protection. Individuals aged 15-24 who enter the workforce are primarily employed in low-tier informal employment (Figure A.9).[2] Less than 15 percent of this group were engaged in formal employment. A large majority of people aged 15-35 remain in low-tier informal employment for most of their work careers. Additionally, being formal workers does not mean being fully protected. Only 11 percent of formal wage employment in the age group 15-35 had complete social insurance (health, work accident, death, old-age, pension) and a written contract.

Figure A.4: Real wages declined for high- and middle-skilled workers

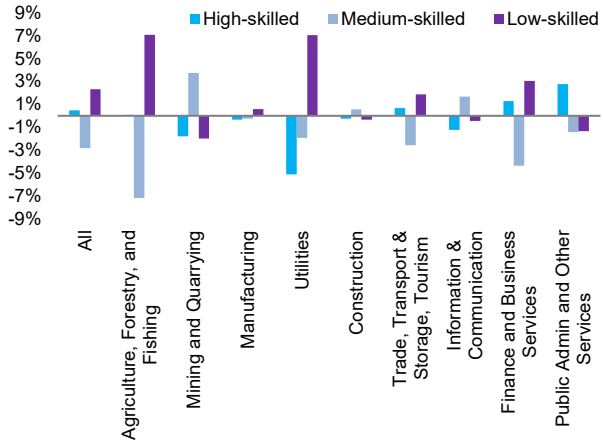
(Cumulative average growth rate of real wages, 2018-2024)



Source: Sakernas August 2018 and 2024

Figure A.5: Many sectors are seeing early signs of hollowing-out of middle-skilled employment

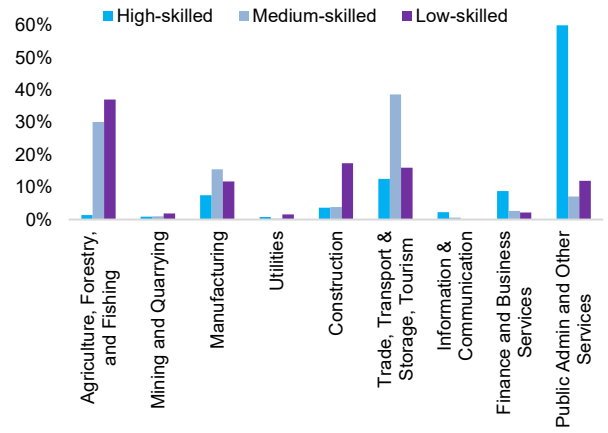
(change in share of skill level by sector, 2018-2024)



Source: Sakernas August 2018 and 2024

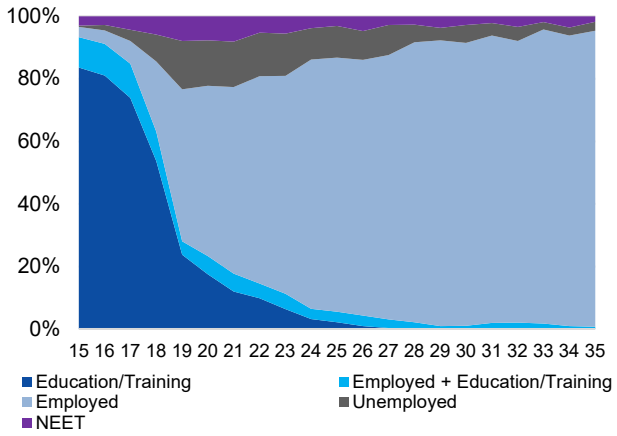
2. The definition of informal jobs follows the new formality definition by the International Conference of Labour Statistics (ICLS)-17. An individual in a high-tier job works in the higher-skilled codes of the International Standard Classification of Occupations (ISCO)— managers, professionals, technicians and associate professionals, and clerical support workers, while low-tier jobs are services and sales workers, skilled agriculture workers, craft and related trades workers, plant and machine operators and assemblers, and elementary occupations.

Figure A.6: Public administration & other services employ the largest share of high-skilled workers
(employment of skill level by sector, 2024)



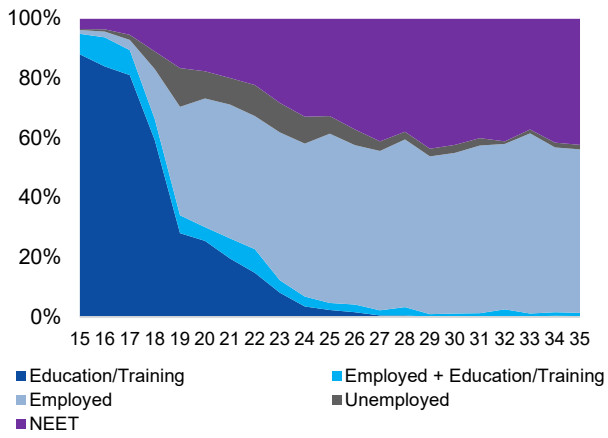
Source: Sakernas August 2024

Figure A.7: Employment rates are relatively high among prime-age men
(school-to-work transition, male, percent)



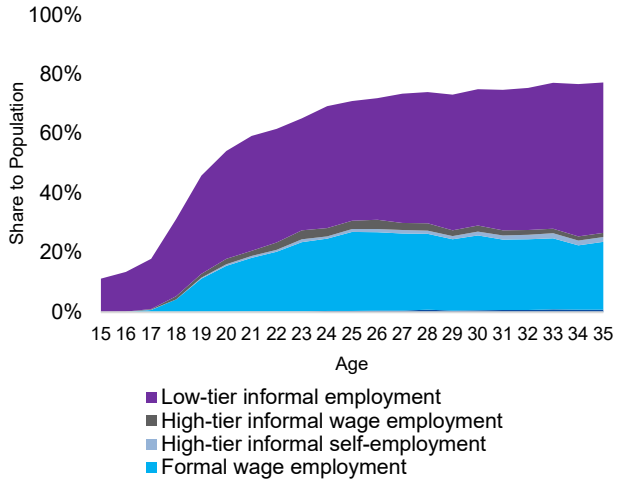
Source: Sakernas February 2025

Figure A.8: While one-third of prime-age women are not in employment, education or training
(school-to-work transition, female, percent)



Source: Sakernas February 2025

Figure A.9: Youth entering the workforce are mainly employed in low-tier employment
(type of jobs youth enter, percent)



Source: Sakernas August 2024

Growth and jobs have reduced poverty but the lack of good jobs is fueling economic insecurity and loss of confidence, especially among the aspiring middle class.

Economic growth has helped reduce poverty. The national poverty rate declined by 0.5 pp over the previous year reaching 8.5 percent in March 2025. At USD4.2/day (2021 PPP)[3], poverty stood at 19.9 percent, 27.8 pp lower than a decade ago. Recent growth has been relatively pro-poor, with the consumption of those in the bottom 40 percent of the distribution rising by 3.5 percent on average, compared to an overall growth of 2.5 percent between 2023 and 2024[4].

However, growth in incomes and consumption has been more modest for the middle class. Between 2023 and 2024, households in the 40th–90th consumption percentiles saw a relatively modest consumption growth of 2 percent, with income gains benefiting poorer

[3] The World Bank recently updated the international poverty line standard following the new 2021 PPP. The World Bank has also adopted a higher poverty line of $8.30/day (2021 PPP), which it will start using for countries classified as upper middle-income countries (UMIC). For more details refer to: https://www.worldbank.org/en/news/factsheet/2025/06/13/updated-global-poverty-lines-indonesia.
[4] This metric refers to a survey-based household consumption and is different from per capital GDP growth captured in national accounts.

households the most. According to the Listening to Indonesia survey data from September 2024 to August 2025 (see Box A.2), median real income declined by 0.2 percent monthly, driven by a drop among the middle 40 household group. Poorer households (the bottom 40) experienced much stronger gains with monthly real income up 1.1 percent,[5] compared to just 0.2 percent for the top 20. This distributional pattern of income and welfare growth is consistent with continued trend that has been observed based on the National Survey of Socioeconomics (SUSENAS) since the pandemic and likely reflects weak labor market outcomes and persistent barriers in creating better quality jobs.

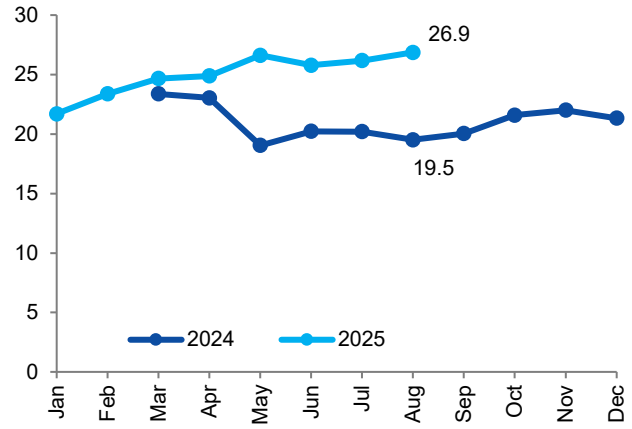
Although monetary welfare improved, a notable disconnect persists: more households now perceive themselves as poor, and self-classified middle class increasingly express concerns about their economic situation. In August 2025, 27 percent of households identified themselves as poor, a rise of 7 pp from the previous year (Figure A.10), with this upward trend in subjective poverty accelerating since January. Perceptions of worsening economic conditions have also increased, as the share of households reporting deteriorating circumstances rose by 5 pp to 24 percent, largely driven by concerns among the self-classified middle class households (Figure A.11). While lower-income households have benefited from welfare gains and now view their economic situation more positively, the middle class's growing sense of insecurity highlights a widening gap between objective progress and subjective well-being.

These perceptions of poverty and insecurity coincide with rising income volatility. Economic vulnerability is evident in growing expectations of financial deterioration especially among the poorest. Poorer households experience substantially greater income fluctuations, with the lowest quintile facing around 36 percent higher volatility[6] in both 2024 and 2025. Overall, income volatility increased by 4.3 percent in 2025 compared to 2024. This instability, combined with limited savings, constrains households' ability to cope with financial shocks. On average, the poorest households can only save for 1 to 2 months a year, while the richest (top 20 percentile) can save for 3 to 4 months. Hence, the poorest households are often ill-equipped to manage unexpected financial needs, reinforcing their vulnerability and insecurity.

In response to evolving expectations, households have exhibited markedly different financial behaviors. Wealthier households, anticipating continued uncertainty, have increased precautionary savings, thereby enhancing their financial buffers. The marginal propensity to save (MPS)[7] among higher-income households (top 20) rose in 2025, with these households saving around 9 cents for every additional Rupiah earned, compared to 7 cents in 2024. In contrast, for middle-income households (middle 40), MPS declined to 2 cents or below in 2025, while poorest households (bottom 40) remained around 1 cent, indicating a reduced capacity to save and greater vulnerability to financial shocks. This divergence is reflected in emergency preparedness. By August 2025, 80 percent of the richest quintile were able to cover unexpected expenses, while the share among the poorest quintile was only around 38 percent.

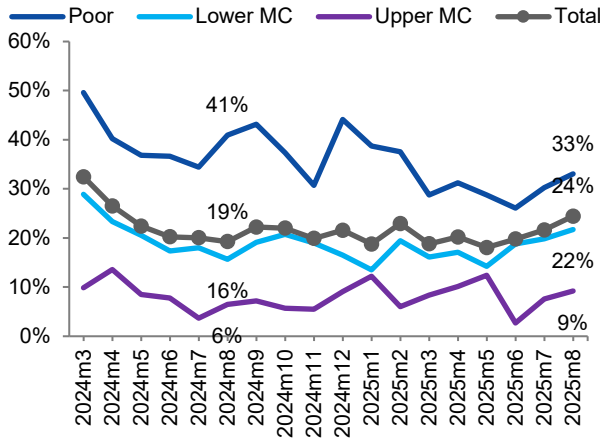
5. Household groupings, including decile, are based on spatially and temporally deflated per capita income. Temporal deflator is based on the official monthly CPI with the base period of February 2024, while spatial deflation is calculated from the consumption module of March 2024 SUSENAS in district level.
6. The analysis covers March to August in each year due to data availability. The coefficient of variation (CV) of income is defined as the fraction of standard deviation and mean of real income of households between March and August. A CV of 58.7 percent means the standard deviation is around more than half of the size of the mean, higher number means higher volatility.
7. Marginal propensity to save is calculated as the product of the income-savings elasticity and the average propensity to save (APS) for each income decile, based on data from the March – August periods of 2024 and 2025. The APS is the average share of household savings relative to income. The income-savings elasticity is derived from estimations of log savings on log income, interacted by deflated per capital income quintile and year, controlling for household and month fixed effects as well as changes in household size. Both income and savings data are sourced from the L2INDO survey

Figure A.10: More HHs identified themselves as poor
(subjective poverty, percent)



Q: How would you classify your household (HH) today?

Figure A.11: Perception of worsening economic conditions also increased
(percent of negative answers)



*Q: Do you believe that the current economic conditions in the neighborhood where you live are good, or not? *MC = middle-class*

BOX A.2: Listening to Indonesia Survey

Listening to Indonesia (L2INDO) is a nationally representative, high-frequency phone survey conducted monthly since March 2024. L2INDO is a joint Bappenas-World Bank program that tracks household welfare, economic conditions, and public sentiment through a nationally representative monthly panel survey since March 2024. This survey tracks the social and economic wellbeing of households across Indonesia, capturing public views on economic conditions and policy with only a short lag. By blending welfare monitoring with polling-style questions, L2INDO enables a closer examination of the drivers behind changing conditions and perceptions. The survey gathers perspectives from people from different socio-economic backgrounds and regions providing a comprehensive view of how households experience and respond to economic and policy shifts.

Each month, the survey aims to interview up to 2,000 households, randomly selected from the first baseline survey of 5,000-household conducted in October–November 2023 across 500 communities. The same households are contacted in every round, with replacements drawn from the same community when respondents drop out. As of today, over 80 percent of current participants have completed more than 15 rounds. A second baseline, conducted in July–September 2025 with the same sample size, will further expand the sample pool.

Each round includes core modules on economic and social vulnerability, economic activities, income, financial behavior, and perceptions of economic conditions and governance. Rotating additional modules cover digital activities, subsidy views, coping strategies, trust in taxation and governance, and other emerging topics such as housing, environmental conditions, and inequality. The second baseline also includes Indonesia's first national measure of lead (Pb) exposure, indoor air quality monitoring and time use.

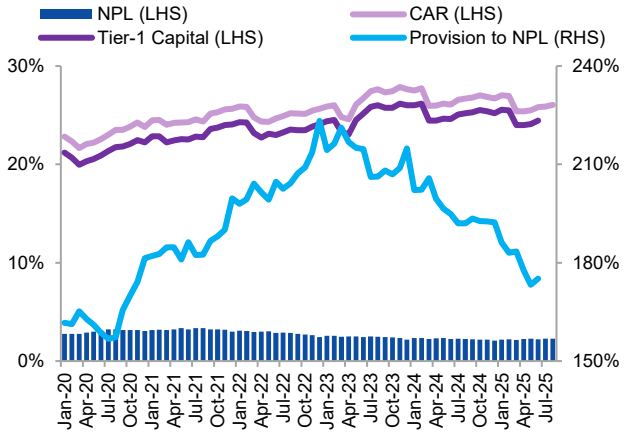
The survey is expected to catalyze change by aligning development strategies with public views and concerns, thereby strengthening the social contract through the integration of public sentiment into policy formulation. It also aims to address knowledge gaps related to volatility and vulnerability.

Strong asset quality and robust buffers keep banks resilient even as credit slows due to a limited pipeline of bankable projects.

Bank asset quality is healthy, with results from a recent stress test showing enough buffers to withstand potential adverse shocks. As of August 2025, the non-performing loans (NPL) ratio remains low at 2.3 percent of total loans. The capital adequacy ratio (CAR) stood at 26 percent, well above the regulatory minimum of 10.5 percent. The level of provisioning was 175 percent of the value of NPLs, providing ample loss-absorption capacity. The loan at risk (LAR) ratio[8] stands at 9.7 percent, less than half its level during pandemic years (Figure A.12). A reverse stress test[9] on 136 Indonesian banks using balance sheet data suggests that the system can withstand a significant credit shock: NPLs would need to rise to 18 percent of total loans before the weakest 20 percent of banks fall below minimum capital adequacy. This "distance to distress" remained consistently high during and after the pandemic.

Profitability in the banking sector remains stable, although lending growth rates have slowed down amid potentially weak demand for credit and a risks mismatch between demand and supply. As of August 2025, return-on-assets (ROA) and return-on-equity (ROE) stood at 2.5 percent and 14.3 percent, respectively, a slight uptick over the last couple of months. The credit-to-GDP ratio stood at 39.8 percent as of Q1-25 and has been below its long-run trend since Q3-22[10], suggesting low level of financial intermediation and credit extension relative to peers and to Indonesia's own historical averages. Lending to the private sector expanded by 7.4 percent yoy in October, a slower pace than most other months in the last three years. The relatively lower credit growth may be partially caused by weak demand for credit due to factors such as a lack of bankable projects or the use of own retained earnings to finance working capital, in addition to credit supply side factors such as low-risk appetite, lack of liquidity, and high lending rates. The slowdown is driven primarily by very low growth in the loans used for working capital (2.8 percent yoy), while investment loan growth remains robust (14.4 percent yoy), consistent with a pick-up in investment. Interest rates on both investment loans and working capital loans have been edging down in 2025[11], suggesting that credit demand factors may have a more important role, although both supply and demand factors, including a risks mismatch, are at play.

Figure A.12: With low NPL and high CAR, bank asset quality remains healthy
(percent)



Source: The Financial Services Authority (OJK).

System-wide funding and market liquidity remain adequate, and the government has injected liquidity into large state-owned banks (SOBs) to boost credit growth. The liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR), designed to gauge bank liquidity conditions in times of stress, stood at 200 and 130 percent respectively, as of June 2025. Both are above the 100 percent regulatory minimum. The Loan-to-deposit (LDR), an indicator of funding liquidity, has gradually increased (meaning declining liquidity) to 86 percent as of mid-2025 compared to below 80 percent during the economic recovery. Since September, the Ministry of Finance (MOF) has placed around Rp276 trillion in local currency deposits (equivalent to 1.2 percent of GDP or 3.5 percent of total credit to private sector) in several major SOBs to boost lending and support the economy. This intervention has so far had no discernible impact on credit growth. Its potential effectiveness remains uncertain, as it assumes that lending is constrained by funding liquidity and that there is sufficient unmet demand for credit.

8. LAR is a forward-looking indicator of bank asset quality defined as the sum of NPLs, restructured loans, and special mention loans.
9. The team conducted a reverse stress test to identify the rise in NPLs necessary to trigger a bank's undercapitalization and system-wide stress. Data used is for June 2025. See Feyen and Mare (2022) methodology details. Results should be interpreted with caution as: 1) the available data do not allow for the consideration of collateral in determining the appropriate level of provisions; 2) NPL data are backward-looking and come with a lag; 3) policy support such as forbearance measures and government guarantees are not accounted for.
10. Bank for International Settlements, credit-to-GDP gaps.
11. While trending downwards, interest rates on investment loans became cheaper than working capital loans. Interest rate on investment loans declined from 8.71 to 8.25 percent between December 2024 and September 2025, while interest on working capital dropped from 8.62 to 8.42 percent over the same period.

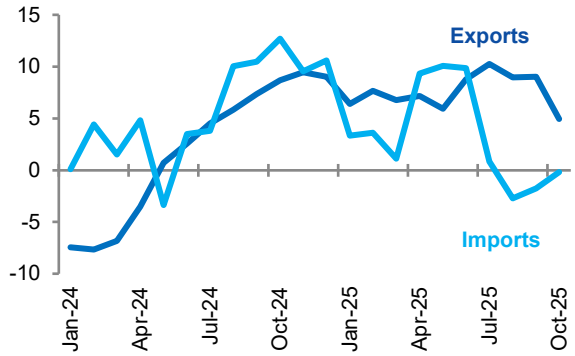
Banks should pay adequate attention to loan quality, apply market-based criterion for selection of beneficiaries and provide strong and timely reporting and auditing when they extend the proceeds from the government liquidity injection. These measures help mitigate potential credit risks and market distortion related to directed lending. However, if the challenge is the lack of bankable projects, then the government's support could be better targeted to supporting bankability through de-risking mechanisms such as credit guarantees.

Indonesia's current account posted a surplus thanks to commodity windfalls and decelerating imports, but weak FDI and financial outflows continue to weigh on the overall balance of payment.

Broad-based deceleration in imports coupled with commodity exports expanded the goods trade surplus. Goods imports increased at a slower pace than exports (Figure A.13). Intermediate goods imports fell, consistent with slower credit growth and a contraction in the Purchasing Manager Index (PMI) from April through July. The import of consumer goods experienced slower growth, corresponding to moderating retail sales since July. Meanwhile, commodity prices drove exports growth (Figure A.14), largely from CPO exports to India and stronger global demand for biodiesel, as well as steel exports to China. The front-loaded export orders, ahead of the new tariffs in August, also boosted exports, namely for items like electrical machinery. Consequently, the goods trade surplus widened to 2.6 percent of GDP for the year-to-October, up 0.8 pp from the same period in 2024. However, export drivers are temporary, with growth dipping in October as the exports front-loading starts to unwind in the last quarter of the year.

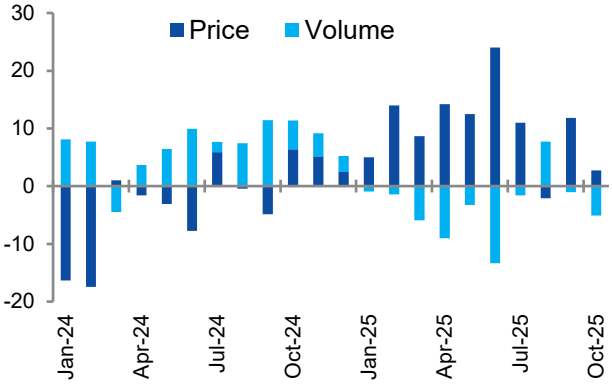
The current account posted a surplus of 0.1 percent of GDP in the first three quarters of the year (3Q-25) (Figure A.15). A robust goods trade surplus offset widening deficits in both services and income accounts (by 0.2 pp of GDP). Despite a slightly wider services trade deficit, the travel services balance improved between July and September, as inbound international travelers to Indonesia rose by 11.5 percent. Meanwhile, a larger income account deficit driven by a growing primary income deficit, reflecting elevated profit repatriation by foreign investors.

Figure A.13: Goods imports grew at a slower pace than exports primarily due to a decline in intermediate goods imports
(yoy growth, 3MMA)



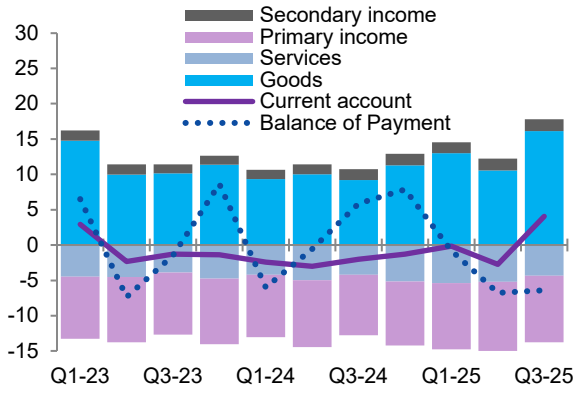
Source: BPS, World Bank staff calculations.

Figure A.14: Goods exports remain resilient, reflecting strong terms of trade rather than volume growth
(contribution to percent yoy growth)



Source: BPS, World Bank staff calculations.

Figure A.15: The current account balance switched to a surplus due to strong goods trade surplus
(USD billion)



Source: BI, World Bank staff calculations.

However, the capital account deteriorated, owing to weak Foreign Direct Investment (FDI) and large portfolio outflows. FDI in 3Q-25 declined to 0.9 percent of GDP, from 1.4 percent of GDP in 3Q-24. This decline is attributed to domestic and global uncertainty. To attract more investment, the Coordinating Ministry for Economic Affairs (CMEA) and Ministry of Finance are preparing a new regulatory debottlenecking reform plan as part of the President's drive for deregulation. In addition, portfolio investment in 3Q-25 saw a deficit of 1.0 percent of GDP, driven by the maturing of foreign-owned monetary securities (SRBI) and outflows from government bonds. As a result, the BOP deficit widened to 1 percent of GDP in 3Q-25, up from 0.05 percent of GDP in 3Q-24.

2. The Policy Stance

Fiscal stimulus to support domestic demand is constrained by declining revenue and underspending.

The fiscal deficit widened as revenue weakened amid structural constraints. The fiscal deficit reached 2.0 percent of GDP as of October 2025, 0.6 pp higher than the same period last year (Figure A.16). Revenue dropped to 8.9 percent of GDP, down 1.3 pp from October 2024.[12] Three factors drove this decline. First, easing commodity prices reduced related revenues.[13] Second, the revised tax refund mechanism accelerated refunds and led to a contraction in net tax collections (MOF 2025a) (Figure A.17).[14,15] Third, State-Owned Enterprises (SOEs) dividends, equivalent to 0.3 percent of GDP, were transferred to Danantara[16] in March 2025.

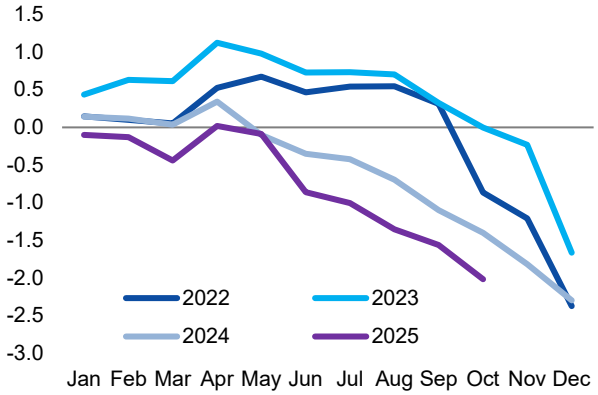
Public spending as a share of GDP declined to accommodate the falling revenue envelope. Central government spending (including subnational transfers) fell by 0.7 pp of GDP compared to last year. This contraction was broad-based (Figure A.18), reflecting budget efficiency measures implemented in early 2025.[17] There has also been underspending in some priority programs (Figure A.19).[18] Interest payments, at 1.8 percent of GDP as of October (21 percent of total revenue), have remained high and at par with the previous year. At the subnational level, although transfers declined only slightly, actual spending contracted by 7 percent yoy, likely due to changes in local leadership and budget efficiency measures introduced in early 2025[19] (MOF 2025b). Meanwhile, the government delivered quarterly fiscal stimulus targeting social assistance and worker incentives to support domestic demand (see Box A.3).

Budget financing conditions have remained favorable. With a larger deficit, net financing requirements reached 86 percent of the budget target by October 2025, up from 73 percent in the same period last year. Government bond yields fell across all tenors amid monetary policy easing. The 10-year benchmark yield dropped to 6.07 percent in late October 2025, its lowest since mid-2013, before rebounding slightly to 6.25 in late November. Investor demand remained strong, with the bid-to-cover ratio rising to 4.6 in October from 1.8 a year earlier. The demand is mostly from domestic investors, most notably commercial banks, insurance companies, pension funds, and mutual funds. Nonresident investors, however, recorded net outflows in September to November, reducing year-to-date inflows and their share in tradable bonds.[20]

12. As of October, revenue declined by 6.0 percent year-on-year and reached only 74 percent of the target set in the 2025 outlook. This is lower than the 85 percent realized on average by the end of October over the past three years.
13. As of October 2025, the average Indonesian Crude Oil Price (ICP) for January–October had declined by 13 percent yoy, the coal reference price (HBA) fell by 8 percent, and nickel by 10 percent. These declines were slightly offset by a 14 percent increase in the crude palm oil price. Oil lifting increased by 1 percent yoy, gas output remained broadly unchanged, while coal production declined by 7 percent yoy.
14. The Directorate General of Taxes Regulation PER-6/PJ/2025, issued in May 2025, expands Indonesia's accelerated tax refund system to Special Purpose Companies (SPC) and Collective Investment Contracts (KIK), unifying refund procedures, and clarifying eligible input taxes—creating a more transparent and automated national framework.
15. As of end-October 2025, gross corporate income tax collection still grew by 5.3 percent yoy; however, net collections contracted by 9.6 percent. Meanwhile, gross and net sales tax collections (VAT plus luxury goods sales tax) declined by 2.1 and 10.3 percent, respectively.
16. Danantara is the new sovereign wealth fund and superholding company for Indonesia's SOEs.
17. For details, please see World Bank, Indonesia Economic Prospects – June 2025.
18. Of the 1.1 percent of GDP budget for 11 priority programs in 2025, only 0.4 percent of GDP has been spent as of October 2025.
19. As of October 2025, the disbursement of transfers from the central to subnational governments reached 83 percent of the MOF budget outlook for 2025, slightly lower than 85 percent last year (1.2 percent contraction yoy). However, subnational government spending stood at 62 percent of the budget (data as of December 9, 2025), down from 65 percent in 2024, reflecting a 7 percent contraction compared to last year.
20. The share of nonresidents in government bonds declined from 14.5 percent at the start of 2025 to 13.3 percent (as of December 5, 2025), while the share held by commercial banks increased from 17.5 percent to 22.4 percent.

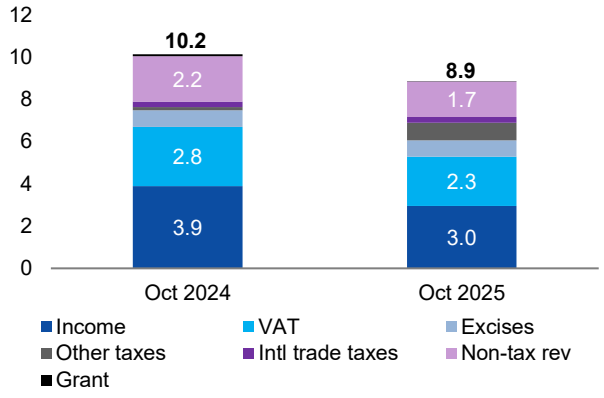
Strengthening cash management can reduce costs from excess reserves. Since the pandemic, the GOI has maintained a high stock of excess financing balance (SAL) equivalent to nearly 2.5 percent of GDP.[21] These funds earn a low interest rate at BI, below the cost of funds. Maintaining large cash balances in excess of prudent levels bears higher interest costs that could be avoided (Figure A.20). Monthly cash accumulation predates the COVID-19 period but has accelerated in the years following it as revenue realization outperformed budget projections given commodities' windfalls and tax reforms[22]. With revenues weakening in 2025, the cash surplus (SILPA) has gradually declined in the past five months (Figure A.21). Improved cash flow forecasting and coordination between borrowing and spending could hence lower financing costs.

Figure A.16: The fiscal position has weakened compared to recent years
(Fiscal balance (percent of GDP))



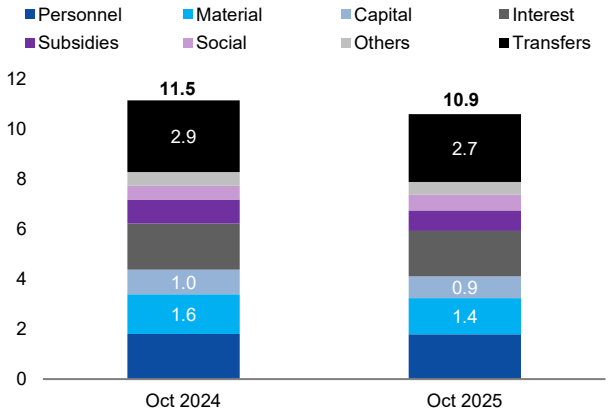
Source: Ministry of Finance, CEIC, World Bank staff calculations.

Figure A.17: This was driven by lower tax and non-tax revenue collections
(Revenue collection (percent of GDP))



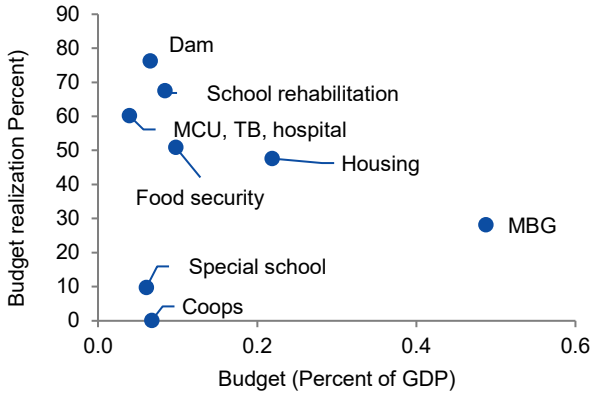
Source: Ministry of Finance, CEIC, World Bank staff calculations.

Figure A.18: Expenditure was reduced accordingly as the revenue envelope tightened
(Government spending (percent of GDP))



Source: Ministry of Finance, CEIC, World Bank staff calculations.

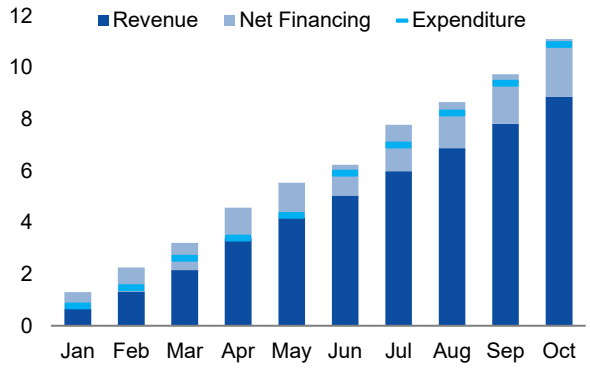
Figure A.19: Unspent budget in key priority programs also helped contain the overall deficit
(Priority programs allocated and realized budget as of October 2025)



Source: Ministry of Finance, World Bank staff calculations.
Note: Budget allocation is based on the budget outlook. Realization of cooperatives is based on September data. MBG = free nutritious meal program; Coops = red and white cooperatives program; Special school = people's school (sekolah rakyat) program and Garuda excellence school.

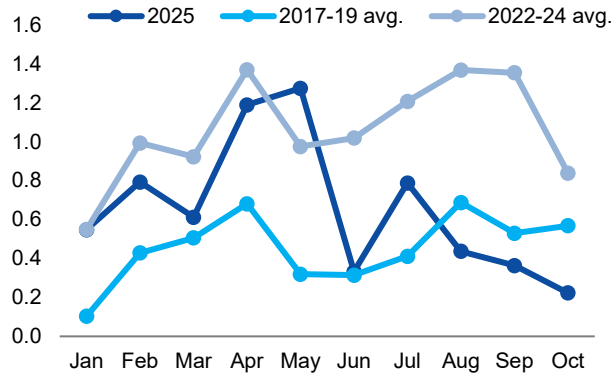
21. In September and November 2025, MOF has placed part of the SAL (IDR 276 trillion or 1.2 percent of GDP) in SOBs to stimulate credit.
22. Revenue realization averaged 119 percent of the initial budget in 2022-2024. This performance is linked to commodities windfalls and the implementation of the tax harmonization law (2022) which included VAT rate hike and other reforms.

Figure A.20: Excess borrowing kept cash balances high amid structurally weak revenue mobilization
(Revenue, financing, and expenditure in 2025 (percent of GDP))



Source: Ministry of Finance, World Bank staff calculations.

Figure A.21: Cash accumulation predates COVID-19 but has recently accelerated
(Excess cash surplus (SILPA) (percent of GDP))



Source: Ministry of Finance, CEIC, World Bank staff calculations.
Note: Revenue + Net financing > Expenditure = Cash surplus (SILPA).

BOX A.3: 2025 Fiscal Stimulus

In 2025, the government introduced a fiscal stimulus package focusing on boosting household consumption and providing labor-market incentives to stimulate job creation. The stimulus amounts to 0.5 percent of GDP (Figure A). The measures are grouped into six major categories (Figure B): social transfers, worker incentives, auto-sector incentives, housing-sector incentives, MSME tax incentives, and other programs.

Support to stimulate consumption is delivered through social transfers and targeted incentives in priority sectors. Social transfers (0.28 percent of GDP) include cash transfers, rice aid, and electricity-tariff discount for low-income households. Measures also included discounts on public transports tickets as well as tickets for trains, flights, and ferries. Sector incentives included VAT subsidy on automobile purchases (0.06 percent of GDP) and housing (0.02 percent of GDP); as well as luxury goods tax reduction and zero percent import duty for EVs.

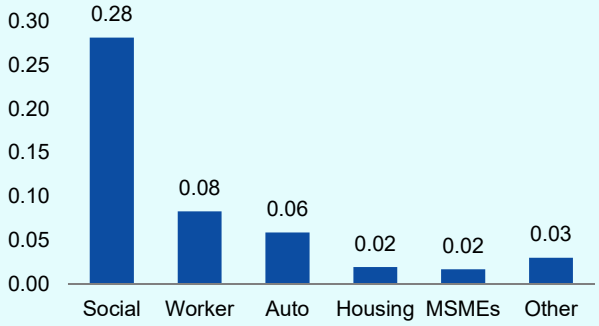
Labor-market incentives focus on workers, MSMEs, and employers. The worker-incentive programs (0.08 percent of GDP) include easier access to Job Loss Insurance (JKP), wage subsidies for low-income workers and contract teachers, cash for-work daily-wage jobs, and six-months paid internship programs for 100,000 recent graduates to help provide work experience and improve employability. Tax incentives include an income tax waiver to reduce labor costs and temporary income tax relief for the tourism sector. Meanwhile, MSMEs continue to benefit from the 0.5 percent final income tax, with full exemption for very small firms, while employers receive interest-subsidy support for machinery revitalization and temporary reductions in workplace-accident premiums (JKK).

Figure A. The government delivered 0.5 percent of GDP in fiscal stimulus during 2025
Fiscal stimulus in 2025 (percent of GDP)



Source: Coordinating Ministry of Economic Affairs, World Bank staff calculations.
Note: The quarter shown reflects when the stimulus measure was primarily implemented, although full implementation may extend into subsequent quarter(s).

Figure B. The stimulus, largely social and labor aid, supported aggregate demand
Fiscal stimulus in 2025 (percent of GDP)



Source: Coordinating Ministry of Economic Affairs, World Bank staff calculations.

Monetary easing has progressed, but muted demand and tradeoff between growth and external pressures constrain its impact.

Low inflation has enabled BI to maintain an accommodative policy stance, but monetary easing has had limited pass-through so far. Since September 2024, BI has cut its policy rate by 150 bps, from 6.25 percent to 4.75 percent as of October 2025, the lowest level since 2022. Despite those cuts, average bank deposit and lending rates have only declined by 83 and 15 bps, respectively (Figure A.22). This is in part due to the special (higher) rates provided to large depositors, which account for 26 percent of banks' total third-party funds. Moreover, the limited pass-through suggests friction in monetary transmission—stemming from shallow financial markets, prevailing informality, and price controls—which limit the domestic response of policy rate adjustments (see IEP December 2023). Concurrently, BI has reduced the issuance of short-term monetary securities (SRBI), lowering their outstanding stock by 0.9 percent of GDP year-to-September. This policy also reflects a more accommodative stance aimed at boosting liquidity. As a result, broad money (M2) growth has picked up recently (Figure A.23), but it was mostly led by higher net foreign assets (NFA) from trade surpluses.

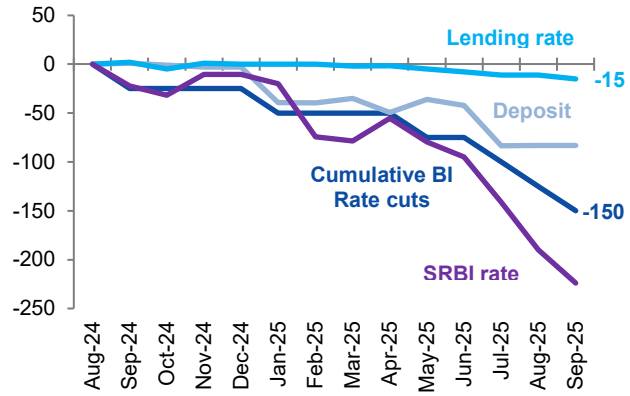
BI is utilizing macroprudential incentives to further improve monetary policy transmission. Starting in December, BI will introduce two policy changes for its Reserve Requirement Ratio (RRR) incentive scheme. First, it will change the criteria for those benefiting from the 5 pp RRR reduction. Eligibility will now require banks' commitment to future lending in targeted sectors rather than realized loan growth. Second, it will provide an additional cut of up to 0.5 pp, on top of the existing 5 pp reduction, depending on how quickly banks lower lending rates relative to BI rate cuts. Commercial banks with a lending rate cut elasticity of 0.3–0.6 will get an additional 0.4 pp RRR reduction, while those above 0.6 elasticity receive an extra 0.5 pp RRR cut (Table A.1). This new incentive is expected to unlock IDR123 trillion of liquidity (1.5 percent of credit to the private sector). In implementing these incentives, BI and OJK will periodically assess banks' credit growth commitments and their realization.

Table A.1: The new macroprudential incentives through interest rate channel

Elasticity of new lending rates to BI Rate cuts	Incentives (pp)
< 0.3	0.0
≥ 0.3 and < 0.6	0.4
≥ 0.6	0.5

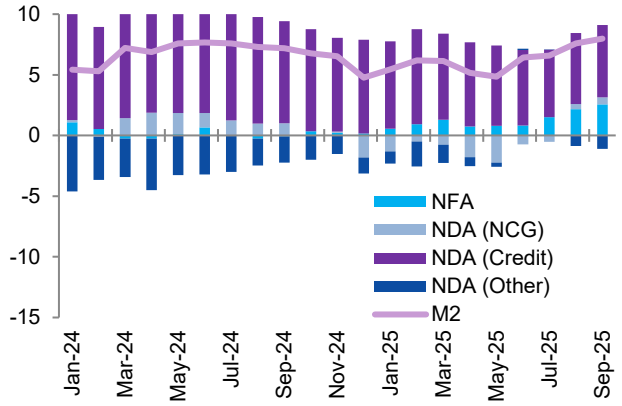
Lower policy rates have translated into lower yields on government securities and narrower spreads against US Treasury yields, contributing to capital outflows and pressures on the Rupiah. The accommodative monetary policy has resulted in a downward shift of the yield curve (Figure A.24). Hence, the yield spread between Indonesia 10-year government bonds and US Treasury securities has significantly narrowed by 42 bps year-to-November. Falling yields led foreign investors to divest from government bonds. Between July and November, portfolio investment saw net outflows of US$7.8 billion (0.5 percent of GDP), while peers such as Thailand and the Philippines posted inflows (Figure A.25). As a result, the Rupiah fell 2.7 percent, compared to a 0.4 percent drop in the JP Morgan Emerging Market Currency Index. Nominal effective and real effective exchange rates also point to a depreciation of the Rupiah. To mitigate currency pressures, BI intervened by purchasing IDR123.5 trillion (US$7.4 billion) in government securities on the secondary market between July and November. As a result, gross international reserves dipped to a one-year low in September but rebounded in November after the issuance of sovereign global bonds. Reserves remain adequate and cover 6.2 months of imports of goods and services.

Figure A.22: Lending rate rigidity amid aggressive monetary easing
(basis points)



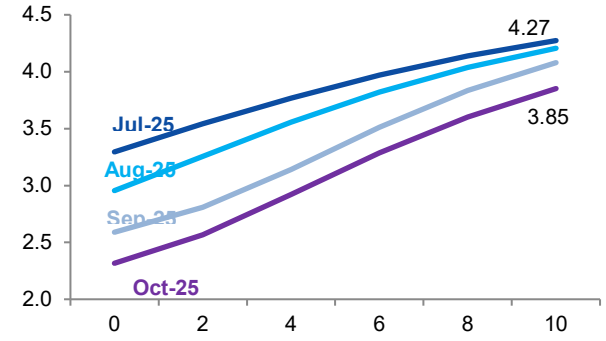
Source: Bank Indonesia, World Bank staff calculations.

Figure A.23: Broad money growth picked up as monetary policy eased
(contribution to growth, yoy)



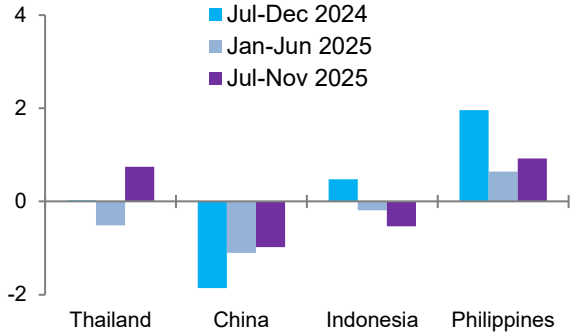
Source: Bank Indonesia, World Bank staff calculations.

Figure A.24: Monetary easing has pushed the yield curve downward…
(X-axis: govt bond tenor; Y-axis: inflation-adjusted yield)



Source: IBPA, CEIC, World Bank staff calculations.
Note: inflation adjustment is based on current inflation rate over the maturity period.

Figure A.25: …contributing to portfolio outflows from Indonesia in the second half of 2025
(percent of GDP)



Source: IIF, Bank Indonesia, World Bank staff calculations.

3. Outlook and Risks

Indonesia's economy is projected to remain resilient. Growth is projected at 5.0 percent per year in 2025-26, rising to 5.2 percent in 2027 (Table A.2). Private consumption will contribute on average 53 percent of overall GDP growth, less than previous years, amid weaknesses in labor income. This is expected to be partially offset by low inflation and targeted fiscal stimulus that could boost consumption. Investment growth is projected to gradually increase, averaging 6.2 percent between 2025 and 2027. This will be supported by three drivers: a scale up of state-led investment via Danantara[23] (Table A.3), further monetary easing to stimulate private sector credit, and FDI through downstreaming, deregulation, and special economic zones reform targeting energy, natural resources, manufacturing, and services.[24] Rising investment should offset weaker net-exports contributions, which are estimated to average 5.5 percent as terms-of-trade decline. Commodity based manufacturing, agribusiness, construction, and services are projected to drive the supply-side, reflecting government priorities. Rising investment levels are expected to boost imports and contribute to a widening current account deficit (up to 1.3 percent of GDP in 2027), which is expected to be financed through FDI and portfolio flows. Gross international reserve

23. Danantara will raise financing from SOE dividends, domestic markets, and international investors to implement a series of planned projects in infrastructure, natural resources downstreaming, renewable energy, and food security.
24. As a result, FDI is projected to return to its pre-covid level of around 1.5 percent of GDP.

levels should remain adequate at around 6 months of imports of goods and services. At 2.6 percent average in 2026-27, inflation is projected to remain anchored within BI's target range of [2.5 percent ±1].

The fiscal deficit is projected to remain stable over the medium term, though both revenue and spending levels are lower than planned. The 2025 budget deficit is projected at 2.8 percent of GDP, consistent with the government's outlook but with weaker revenue and expenditure performance.[25] The deficit is expected to stay at 2.8 percent in 2026 and edge up slightly to 2.9 percent of GDP in 2027. Revenue may fall short of the ambitious 2026 target (see Box A.4), which would require spending adjustment to keep the deficit within the ceiling. A slightly higher primary deficit in 2025–27 relative to 2024 is expected to increase debt levels, and gross financing needs are expected to increase from 5.5 percent of GDP in 2024 to around 7 percent by 2027, reflecting higher deficits and debt repayments.[26] The public debt-to-GDP ratio is projected to rise gradually from 40 percent in 2024 to 42 percent in 2027.

The risks to the outlook are balanced. On the downside, failure to implement structural reforms will exacerbate labor market challenges, including the creation of middle-class jobs and growth in real wages, adding pressure on household consumption. Escalated trade policy uncertainty and heightened geopolitical tensions could weigh further on terms-of-trade, spur inflation, hurt business confidence and investment, and limit fiscal space. Tightened financial conditions from global inflationary pressures could impact portfolio flows, put pressure on the Rupiah, and increase borrowing costs. In addition, unfavorable climate conditions and weather-related disasters could also dampen economic activity. On the upside, higher prices for commodity exports, stronger recovery in major trading partners like China as well as accelerating the implementation of Indonesia's new trade deals (with EU, Canada, US and others) can support terms-of-trade and therefore boost growth.

BOX A.4: The 2026 Budget Law: Balancing Fiscal Prudence and Government Priorities

The budget deficit is set at 2.7 percent of GDP in the 2026 Budget Law. The law also projects revenue to increase by 0.2 pp above its 2025 projection, while expenditure is expected to rise by 0.1 pp.

Ambitious revenue growth, given easing commodity prices and limited tax policy reforms, will require strong tax administration effort. The government projects a 10 percent nominal increase in revenue, outpacing nominal GDP growth of 8 percent and contrasting with a 6 percent contraction year-to-date in 2025 (Figure B). This challenge is greater given the expected easing in commodity prices (World Bank, 2025a). Much of the anticipated gains stem from income tax and VAT, anchored around improved implementation of the Core Tax Administration System (CTAS) and joint inter-agency initiatives to enhance tax and non-tax revenues compliance.

Total expenditure is set to increase based on higher allocations for central government spending on priority programs. Total expenditure is projected to grow by about 9 percent (compared to 5 percent in the 2025 outlook). The increase is driven primarily by higher central government material spending, reflecting a larger allocation for the free meal program (MBG), which more than offsets a 20 percent decline in transfers to regions (Figure C). While disbursement and implementation have progressed at a gradual pace, less than one-third of the allocated budget of this program was executed as of October 2025. The budget envelope for priority programs is set to double to 2.2 percent of GDP (or 15 percent of total expenditure). On the other hand, capital spending is set to decline, for a second consecutive year, this time by 0.3 pp (to 1.1 percent of GDP).

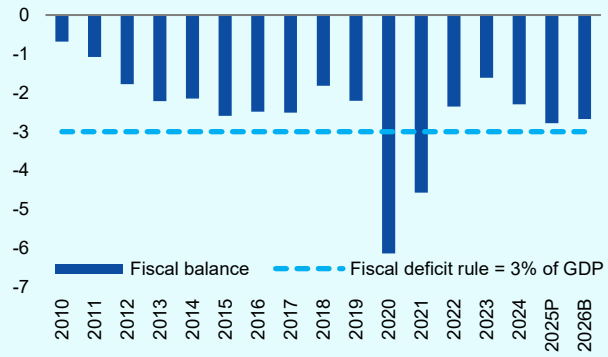
The budget law also reflects a shift toward greater centralization and discretionary spending. The sharp cut in transfers to regions, mainly due to a large reduction in revenue-sharing funds,[27] compounds fiscal strain on subnational governments already affected by budget cuts in 2025, with recent attempts to raise local taxes sparking local protests.[28] A notable share of the proposed budget remains unallocated, increasing the level of discretionary spending. This is

25. Revenue is projected to fall to 11.6 percent of GDP—0.4 pp below the 2025 outlook and 1.2 pp below 2024 level. Meanwhile, expenditure is expected to decline to 14.4 percent of GDP—0.4 pp lower than the 2025 outlook and 0.8 pp lower than the 2024 level.
26. The primary deficit in 2025–27 is projected to hover between 0.4 and 0.5 percent of GDP, compared to 0.1 percent in 2024.
27. Revenue sharing fund (DBH) is a fiscal transfer mechanism from the central government (predominantly) to the region where those revenues (certain taxes and natural resource revenues) originate from.
28. For instance, in early to mid-August, there were protests against local and building tax increase in Pati (Central Java), Bone (South Sulawesi), and Cirebon (West Java). The tax increase was subsequently canceled in response to public opposition.

classified as "other" spending, which has risen from 3 to 8 percent of the budget (Figure D). The sharp increase in 2025 and 2026 may indicate potential future budget reallocations or a larger share of discretionary spending, including possible financing of local infrastructure projects under the President's instruction programs.[29]

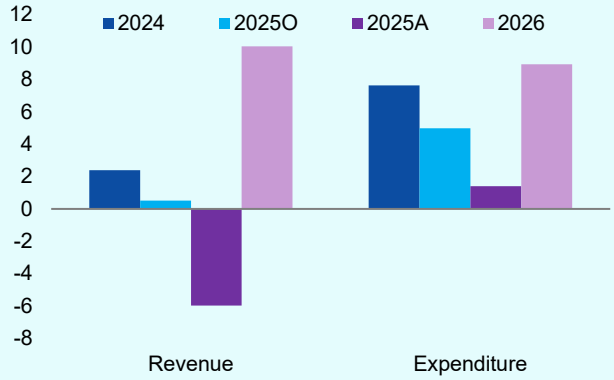
The budget faces two key risks: (i) revenue underperformance against ambitious targets, which reduces space and (ii) implementation bottlenecks leading to underspending and hence impact on growth. Moreover, the heavy tilt toward social transfers and discretionary spending raises questions about allocative efficiency and limited growth multipliers. Addressing implementation bottlenecks for priority programs and public investment could reduce the need for fiscal stimulus to boost economic activity.

Figure A. The 2026 budget targets a relatively high deficit—among the largest outside COVID years
Fiscal balance (percent of GDP)



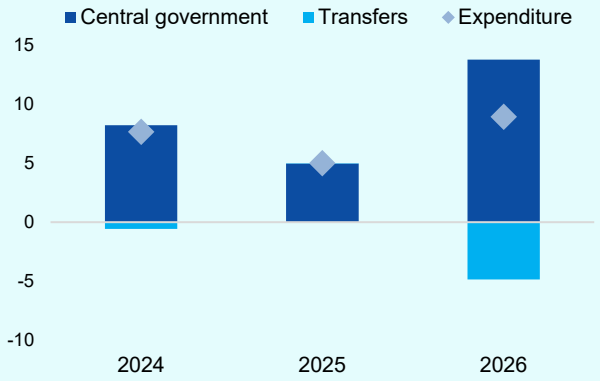
Source: Ministry of Finance, CEIC, World Bank staff calculations.
Note: 2025P = government budget outlook; 2026B= 2026 budget.

Figure B. Despite weaker revenue collection in 2025, the budget targets a sharp rebound
Revenue and expenditure growth rate target (percent)



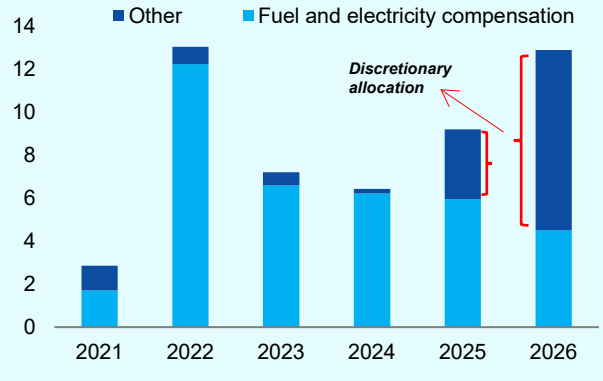
Source: Ministry of Finance, CEIC, World Bank staff calculations.
Note: The 2024, 2025O, 2025A and 2026 numbers refer to 2024 budget realization, 2025 outlook, 2025 actual realization as of October 2025, and 2026 budget, respectively. The 2026 growth rate is relative to the 2025 outlook.

Figure C. Rising central spending on priority programs more than offsets falling regional transfers
Contribution to expenditure growth (percent)



Source: Ministry of Finance, World Bank staff calculations.

Figure D. An increasingly large share of the budget is categorized as "other spending"
Components of "Other spending" category (percent of total expenditure)



Source: Ministry of Finance, World Bank staff calculations.
Note: 2025 data refer to the budget outlook. 2026 data based on proposed budget but the difference between proposed and actual budget is small.

Source: World Bank staff elaboration based on the approved budget and financial notes of Budget Bill.

29. On June 23, 2025, the President issued Presidential Instruction (Inpres) 11/2025 on the acceleration of regional road connectivity improvement to support food and energy self-sufficiency.

4. Policy Priorities

Strengthening active labor market programs and information systems as well as investing in foundational skills and teaching quality is key to addressing job polarization and skills mismatch.

Indonesia has experienced 5 decades of strong growth reaching Upper Middle-Income in 2023. Growth has delivered on jobs and higher living standards. The country now faces challenges common to Middle-Income Countries. Growth is stagnant (at around 5 percent) when it needs to accelerate to achieve high income by 2045, and the economy is struggling to create good and middle-class paying jobs. The demand for these jobs will come through policies that reverse the decline in productivity growth. As such, a stronger push is necessary on business enabling policies and predictability of the regulatory environment. This includes reforms to deregulate, promote competition, support firm entry-exit, and encourage quality investment to expand good jobs. Moreover, mobilizing private capital will be key. Business enabling policies should boost higher-value investment, requiring more private capital to support businesses with loans, equity, guarantees, and risk insurance (World Bank Indonesia Country Growth and Jobs Report, forthcoming).[30] While these reforms focus on the binding constraints for boosting the demand for good jobs, it is also important to start preparing supply-side interventions to boost the skills of the future labor force.

Supply-side interventions are also needed to improve the quality of jobs. Globalization, greening, and technology have changed the location, content, and structure of work resulting in rapid shifts in demand for occupations and skills. Across the world, routine manual, and cognitive skills—abundant in Indonesia—are facing declining demand. Instead, advanced cognitive skills such as problem-solving and verbal ability acquired through strong foundational education are facing increasing demand, as technical skills learned through the TVET (Technical and Vocational Education and Training) system, universities, and on-the-job training. Further, Indonesia will be aging in the next decade, and the cohorts entering working age is forecast to start shrinking in the early 2030s. Policies that encourage women to enter the labor force and remain active during prime age together with improved labor productivity, will be needed to continue to accelerate economic growth.

Indonesia can adopt a combination of short-and-medium term supply-side interventions to address growing job polarization and skill mismatches.

Short-term:

a. **Address information failures to support optimal decision making by individuals and firms through the development of a world-class labor market information system.** Evidence shows that better allocation of talent across occupations leads to economic growth and higher individual earnings.[31] Providing information on the labor market returns of different fields can also reduce the gender imbalance in career and college major choices.[32] As such, a modernized Labor Market Information System (LMIS) and robust Public Employment Services (PES) can significantly reduce frictional unemployment by lowering search costs and improving matching quality. Evidence suggests that effective PES can increase the exit rate from unemployment by streamlining vacancy referrals and reducing mismatch (refer to *World Bank, 2020; Card et al., 2010*) especially at the lower end of middle-skilled job matches where private sector employment services might be less utilized. Strengthening Indonesia's digital employment platforms (e.g., KarirHub) to provide real-time signaling on skills demand—rather than just job listings—will help align the education system's output with market realities and minimize the "scarring effects" of early-career unemployment (see *ILO, 2024; Hensvik et al, 2019*).
b. **Develop effective active labor market policies (ALMPs) with a renewed focus on lifelong learning.** Evidence highlights that well-targeted ALMPs—such as re-skilling and up-skilling programs aligned with employer needs, career counseling services based on real-time labor market intelligence and wage subsidies for displaced middle-skill workers—can facilitate smoother transitions into emerging sectors (*ILO, 2021; OECD, 2023*). Denmark's flexicurity model blends flexible labor laws with generous unemployment benefits and publicly funded upskilling, enabling rapid reallocation of workers toward emerging job opportunities. Singapore emphasizes lifelong learning and adaptability through its SkillsFuture initiative, which offers

30. Demand side interventions are discussed in detail in the forthcoming World Bank Indonesia Country Growth and Jobs Report.
31. Hsieh, C; Hurst, E; Charles, J; Klenow, P. 2019; Matthew W & Basit, Z, 2021.
32. See: Matthew W & Basit, Z, 2021.; Hastings et al., 2015; Kudashvili, N and Todua, G.2022; Gassier, M; Rouanet, LM; Traore, L. 2022.

individualized learning credits, modular training pathways, and strong industry partnerships to align skills development with labor market needs (*Truman et al, 2019*). South Korea, a pioneer in skills and industry partnerships, has established an integrated labor market information system (Work-net), robust mechanisms for assessing training outcomes and strong links between industry and training providers.

c. **Digital upskilling programs in coordination with private sector** could help workers, especially in middle-skill roles, to cope with changing technology and increasing automation. The provision of such training (e.g., digital inventory management, tech-enabled logistics) complement existing middle-skill roles. Successful public private partnerships have been utilized in Korea for the same purpose.

In Indonesia, strengthening ALMPs in tandem with dynamic labor market information systems and modular, demand-responsive training pathways could play a key role in reallocating labor toward high-productivity, future-oriented sectors while enhancing the employability of young and displaced workers.

Short to Medium-term: It is vital to enhance foundational skills to promote a future ready workforce able to adapt to a rapidly evolving labor market. Indonesia scores lower than regional peers on foundational skills, with more than half of Indonesian youth not meeting minimum proficiency standards in literacy, numeracy, digital and socioemotional competencies, that are pivotal to future learning. Key amongst effective approaches is investing in early childhood education and improving teaching quality through structured pedagogy and scaffolding as well as targeting teaching at the right level (World Bank 2023). Vietnam and the municipality of Sobral in Brazil, for example, show that significant improvements in learning outcomes can be achieved with proven education reforms that are underpinned by strong political commitment and robust accountability. Vietnam's success derives from prioritizing foundational learning at the national level, setting clear targets for literacy and learning, investing in teacher quality and leveraging assessment data to both improve instruction and monitor results. In contrast, reform at the municipal level in Sobral propelled it from one of the lowest to the highest performing systems in the country. This was spearheaded through a sequenced set of reforms, structured pedagogy and a focus on accountability where performance-based incentives and regular monitoring of schools proved effective in holding teachers and school leaders accountable for student learning outcomes.

Those reforms and programs are largely brought by and financed, but not exclusively, by the public sector; and hence will require doubling down on current government efforts to create fiscal space.

A bold revenues strategy coupled with ringfencing growth-enhancing public spending can help mitigate fiscal risks, boost fiscal space, and support government growth ambitions.

Fiscal risks need to be considered going forward. If revenue performance remains weak, the 2025 fiscal deficit could approach the statutory limit under full budget execution. Similar risks apply to the 2026 budget.[33] Lower transfers to subnational governments—down about 20 percent—could constrain service delivery and affect local development outcomes. Persistent revenue shortfalls and underperforming transfers would limit fiscal space and test adherence to deficit rules in the years ahead.

A bolder and more proactive revenue strategy is needed to safeguard fiscal sustainability. With rising risks of revenue shortfalls and wider deficits in 2026-27, stronger revenue mobilization is urgent. The government could focus on quick wins in tax administration and digitalization. Expanding e-services, enhancing data integration and simplifying filing procedures can mobilize around 1 percent of GDP in additional revenue, at least for VAT and CIT[34]. Revenue could also be strengthened by lowering VAT and corporate income tax (CIT) thresholds, which are set significantly higher than in comparable countries. Broader CIT bases—through phasing out reduced rates, special regimes and other incentives—would also enhance fairness and efficiency. Reducing VAT exemptions in selected sectors could further improve revenue performance while preserving neutrality. These policy measures focusing on VAT and CIT could raise up to an additional 1.5 percent of GDP. Finally, tax reforms to impose standardized tax treatment on capital gains could mobilize additional revenue in an economically efficient way, while supporting greater equality.

33. Revenue realization in 2025 is projected to fall to 11.6 percent of GDP, while achieving the 2026 target of 12.3 percent of GDP would require a 0.7 pp increase—a level of improvement typically seen only during commodity price booms.
34. The estimated revenue gain is based on the assumption that half of the identified VAT compliance potential is realized (World Bank 2025b).

Reorienting spending toward growth-enhancing investments is key to stronger medium-term outcomes. The composition of public spending needs to reflect Indonesia's development priorities. Greater emphasis on capital spending would raise the economy's productive capacity and support regional infrastructure development. However, infrastructure projects need to be guided by sound public investment management practices. Improved project selection, faster implementation, addressing spending bottlenecks, and stronger monitoring would raise returns and reduce inefficiencies. Capital spending is projected to decline to 1.3 percent of GDP over the medium term, despite Indonesia's fiscal multiplier for capital expenditure being relatively higher than that of peer emerging markets.[35]

Fiscal space and labor market outcomes improve as firms, including MSMEs, formalize and increase productivity. One area to support those firms is through boosting access to credit and encouraging bankable projects through credit guarantees.

Credit guarantee programs paired with robust credit infrastructure are effective instruments to de-risk projects, improve bankability, and mobilize both domestic and international private capital.

Credit guarantee programs paired with robust credit infrastructure are effective instruments to de-risk projects, improve bankability, and mobilize both domestic and international private capital. Properly designed credit guarantees such as partial credit guarantees[36], portfolio guarantees, and first-loss facilities reduce both expected losses and uncertainty for lenders. By sharing risk, they enable banks to extend credit to MSMEs that lack collateral or credit history, offer larger loan sizes and longer tenors, and ultimately make small-business lending commercially viable.

Public-sector intervention through guarantees is often justified where information asymmetries and market failures constrain credit allocation. Guarantees diversify risks more efficiently than individual lenders and mobilize private capital at a fraction of the fiscal cost of direct lending. Channeling public resources into partial credit guarantees is typically less costly than providing direct liquidity while maintaining strong incentives for prudent lending and borrower discipline. Guarantees mobilize private capital without requiring full public funding up front, and they mobilize private capital rather than replacing it. As a result, well-designed guarantee schemes can catalyze substantial private investment with relatively modest public budgets.

The government-funded credit guarantee program—KUR—plays a central role in expanding MSME finance. The People's Business Credit (*Kredit Usaha Rakyat*, KUR) program has been instrumental in unlocking credit to the unbanked and underbanked as evidenced by more than 85 percent of KUR borrowers are first-time recipients of formal credit. The KUR program funds the guarantee fees and subsidizes interest rates while leaving loan capital provision to financial institutions. The credit guarantee is provided by OJK-licensed public and private guarantors companies for a fee. However, enhancing graduation incentives, targeting rules, and enforcement of inclusivity criteria is essential to reduce long-term subsidy dependence and ensure the program consistently reaches priority segments. Guarantee institutions also need clearer mandates and product innovation to expand their risk appetite beyond government programs such as KUR and support more complex, productivity-enhancing investments.

Guarantee programs are most effective when anchored in strong digital credit infrastructure. Indonesia has made progress with expanding number of credit reporting service providers, but further harmonization, data-sharing reciprocity, and integration of alternative data are needed. A robust open finance framework—linking payment data, utilities, business registry information, land certification, and the movable collateral (fidusia) registry—can sharply improve credit scoring accuracy, reduce borrowing costs, and unlock access to a wider range of financial services. Complementary reforms to insolvency, debt resolution, creditor rights, and the fidusia framework will further strengthen lender confidence and expand the pool of bankable MSMEs.

Together, well-designed guarantees and modern credit infrastructure form a mutually reinforcing foundation for Indonesia's digital financial ecosystem—mobilizing private capital, boosting productivity, and supporting inclusive, sustainable growth.

35. According to World Bank's MFMOD model estimates, Indonesia's capital expenditure multiplier is close to 1.4 in the short-term, higher than the 75th percentile of emerging markets and developing economies.
36. he World Bank's operational guidance on designing public credit guarantee schemes and its toolkit for evaluating them set out best-practices (eligibility rules, caps, pricing, monitoring) that preserve incentives for good lending while protecting public funds.

Table A.2: Selected Macroeconomic Indicators

	2022	2023	2024	2025	2026	2027
	Actual			WB projections		
Real GDP growth and inflation, percent change						
Real GDP	5.3	5.0	5.0	5.0	5.0	5.2
Private Consumption	5.0	4.9	5.1	4.9	4.9	4.9
Government Consumption	-4.4	3.0	6.6	0.1	5.8	2.3
Gross Fixed Investment	3.9	3.8	4.6	6.1	6.2	6.3
Exports	16.2	1.3	6.5	7.0	5.6	6.9
Imports	15.0	-1.6	7.9	5.9	6.7	6.5
Consumer Price Inflation (average, %)	4.1	3.7	2.3	2.0	2.6	2.6
Fiscal accounts, central government, percent of GDP						
Revenues	13.5	13.3	12.8	11.6	11.8	12.2
of which Tax Revenue	10.4	10.3	10.1	9.4	9.7	10.1
Expenditures	15.8	14.9	15.1	14.4	14.6	15.1
Primary Balance	-0.4	0.5	-0.1	-0.5	-0.4	-0.5
Fiscal Balance	-2.4	-1.6	-2.3	-2.8	-2.8	-2.9
Central Government Debt	39.7	39.2	39.8	40.5	41.1	41.5
Balance of Payments, percent of GDP unless indicated otherwise						
Current Account Balance	1.0	-0.1	-0.6	-0.3	-1.1	-1.3
Exports, Goods and Services	23.9	21.2	21.5	22.0	22.1	22.1
Imports, Goods and Services	20.7	19.2	20.0	20.1	21.0	21.3
Net Foreign Direct Investment	1.4	1.1	1.0	1.1	1.2	1.5
Gross Reserves (months of imports of goods & services)	6.0	6.7	6.6	6.2	6.1	6.0
Memorandum items						
Nominal GDP (IDR trillion)	19,588	20,892	22,139	23,750	25,156	26,727
Nominal GDP growth	15.4	6.7	6.0	7.3	5.9	6.2

Table A.3: Danantara Fund and Investment Plans

Funding Target	Value	Investment Plan	Value
IDR trillion unless indicated otherwise			
Credit facilities from SOE banks	210.0	Merah Putih Village Cooperatives	210.0
Foreign loan facility	165.0	ACWA Power (Saudi Arabia partnership)	163.7
SOE dividend	116.0	Waste to Energy Project	91.0
Foreign investment	71.0	Hajj Village (Saudi Arabia)	33.0
Patriot bonds	50.0	Garuda Indonesia	30.5
		Vale Indonesia	23.3
		Chandra Asri Petrochemical (co-investment with INA)	13.0
		Pertamina Geothermal Energy & PLN Indonesia Power	1.9
		Aneka Tambang Dragon Project	0.8
		Agrinas Pangan Nusantara	0.7
Total	**683.0**	**Total**	**567.9**
Total as percent of GDP	***2.9***	***Total as percent of GDP***	***2.4***

Source: Media outlets, Mandiri Sekuritas Economic Research.

B. Digital Foundations for Growth



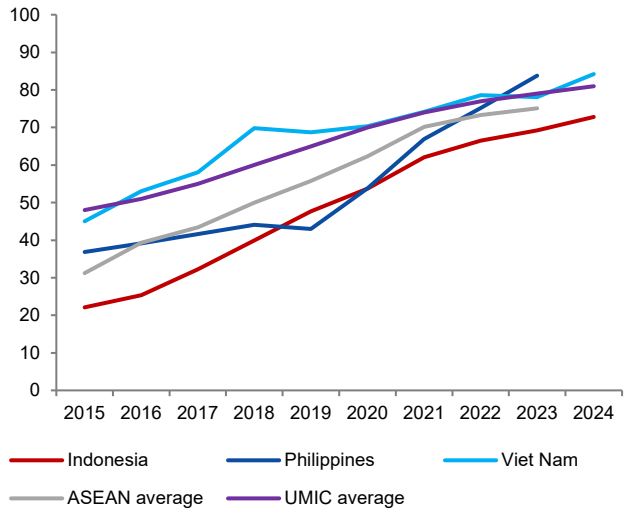
B. Digital Foundations for Growth

1. The Imperative to Expand and Upgrade Indonesia's Digital Infrastructure

Indonesia can transform widespread digital access into increased productivity and better-quality jobs by improving the quality, reliability, and reach of its digital infrastructure. Digital technologies can have important impacts on how economies grow, firms compete, governments function, countries respond to shocks, and how people work, learn, communicate, and access information. But realizing their full potential depends on the foundations of fast, reliable, and affordable internet and modern data centers with cloud capabilities, for which international evidence shows substantial gains in productivity, employment, and economic growth (see Box B.1). Improved connectivity enables broader industrial upgrading to support automation, advanced manufacturing, and more digital services. Artificial intelligence (AI), for example, can only generate productivity gains where reliable broadband and data center capacity allow data to move and be processed efficiently. For Indonesia, stronger digital infrastructure can play a key role in driving the next generation of economic growth—anchored in higher productivity, greater resilience, and more inclusive participation in the digital economy.

Internet use in Indonesia has expanded rapidly—from 54 percent of the population in 2020 to 72.8 percent in 2024[37]—yet connectivity remains "broad but shallow", with speeds and modes of access limiting how productively it can be used. Despite the progress, internet use in Indonesia is behind the ASEAN and upper-middle-income countries averages (75 and 81 percent, respectively) (Figure B.1).[38] Strong competition and infrastructure sharing have helped expand mobile coverage to almost all populated areas and kept data affordable—a remarkable achievement given Indonesia's geography. By contrast, weaker competition and fragmented infrastructure in fixed broadband have constrained coverage, quality, and affordability: only 22 percent of households[39] have fixed broadband connections and 22 percent of schools have 100+ Mbps internet connections.[40] Indonesia continues to significantly lag regional and income-level peers on average broadband speeds. Furthermore, only 10 and 3 percent of Indonesians use a laptop or desktop computer,[41] compared with 26 percent in Thailand and 80 percent in Malaysia,[42] limiting engagement in more complex or data-intensive activities.

Figure B.1: Share of people using the internet in ASEAN



Source: International Telecommunication Union (ITU), 2025. *ITU DataHub.*

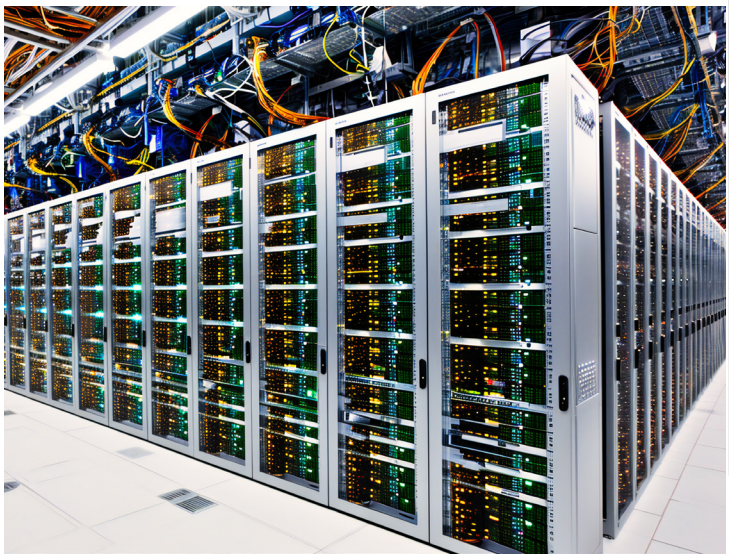
37. Central Bureau of Statistics (BPS), 2025. *National Social and Economic Survey (Susenas) 2024).*
38. A 2025 survey by the Indonesian Internet Service Providers Association (APJII) found that 80.7 percent of Indonesians using the internet. This analysis relies on *Susenas* because it provides nationally representative, comparable data across provinces and population groups.
39. Telegeography, 2025. *GlobalComms Database – Indonesia.*
40. World Bank staff calculations based on H1 2025 data provided by the Ministry of Primary and Secondary Education.
41. BPS, 2025.
42. International Telecommunication Union (ITU), 2025. *ITU DataHub – Individuals using a computer.*

While improving internet quality is critical, significant digital divides remain, creating two Indonesias: a digital Indonesia—connected but constrained—and an unconnected Indonesia still struggling to get online. As described in the 'Who is connected and who is not' section and throughout this chapter, these disparities reflect differences in digital skills, device access, and the uneven reach and performance of networks. In the leading regions—Java, Bali, and parts of Sumatra—digital tools are already being used but face congestion and low speeds. On the supply-side, fiber optic distribution points (ODP) exist in only about half of all villages (*desa*) (see 'Backbone' section for more analysis). These divides reinforce a dual challenge: expanding access for inclusion while improving quality to boost productivity and growth.

Improving the quality of digital infrastructure has also become increasingly urgent for Indonesia's regional competitiveness. A World Bank survey of multinational enterprises operating in Indonesia found 'inadequate digital/ICT infrastructure' as a leading reason for not expanding investment in the past 3-5 years.[43] ASEAN's digital economy is growing rapidly, and the recently negotiated Digital Economy Framework Agreement (DEFA) will accelerate this further through more cross-border trade. An earlier World Bank study found that unreliable internet was a key barrier for Micro, Small and Medium Enterprises (MSMEs) to adopt digital tools.[44] Yet much of Indonesia's growing digital demand for cloud and data processing is still served by data centers providers abroad, signaling untapped domestic potential. Regional peers like Malaysia, Thailand, Viet Nam, and recently the Philippines have made boosting internet speeds and attracting investment in AI-ready data centers as high-level national economic priorities.

The government has recognized the opportunities of digital transformation and the need for better digital infrastructure. The Asta Cita framework and National Medium-Term Development Plan (RPJMN) 2025-2029 recognize digital transformation as key to achieving the country's development goals, including the *Indonesia Emas 2045* vision of high-income status. Connectivity is seen as an enabler of the government's flagship programs, especially improved education, healthcare, and public services. The RPJMN sets ambitious targets: extending fiber networks to 90 percent of sub-districts (*kecamatan*), doubling fixed broadband subscriptions, and increasing data center capacity fivefold. The Government has also announced plans to double average internet speeds to 100 Mbps and halve the price of fixed broadband to IDR 100,000 per month. Achieving these goals will require mobilizing far greater levels of private investment, as well as increased competition in the case of fixed broadband. This will depend on bold reforms to reduce regulatory and deployment barriers, provide long-term policy certainty, and create commercially viable markets, supported by a more strategic public role in aggregating broadband and data center demand across government and public facilities.

Although this special focus chapter focuses on digital infrastructure, progress will also rely on strengthening digital skills, trust, governance, and digital public infrastructure (DPI). Firms can only benefit from new technologies if managers and workers have the competencies to use them effectively, and citizens will only engage in meaningful online transactions if they feel safe and secure. Expanding digital skills programs and improving digital trust by strengthening governance, data protection, and cybersecurity will therefore be essential to maximize the impact of enhanced infrastructure. A clear regulatory and governance framework will provide more certainty to citizens and investors. And scaling DPI will help translate enhanced broadband access into more meaningful productive digital transactions. Together, these efforts can ensure that Indonesia's digital transformation is not only faster but also safer, more inclusive, and more productive.



43. World Bank Group analysis for the Indonesia Country Private Sector Diagnostic: Survey of Multinational Enterprises. "Improve digital/ICT infrastructure" was cited by 29.6 percent of dynamic investors (4th overall) and 47.6 percent of stagnant investors – the leading factor among the latter, though this result should be treated as anecdotal given the smaller sample.
44. World Bank, 2021. *Beyond Unicorns: Harnessing digital technologies for inclusion in Indonesia.*

BOX B.1: Summary of empirical evidence on the economic benefits of better internet

There is a substantial and growing body of evidence linking improvements in broadband coverage and speed to positive economic and social outcomes. A 2024 global meta-review concluded that higher broadband availability and quality typically raise GDP and productivity.[45] Relevant recent empirical studies include:

- A 2019 firm-level study of 20 European Union (EU) countries (2010-2015) found that a 10-percentage-point increase in broadband adoption raises total factor productivity by 3.5-5.8 percent over five years, with particularly strong effects in manufacturing and digital-intensive sectors.[46]
- A 2020 study of Turkish manufacturing firms (2012-2015) found that improving broadband speeds above 100 Mbps led to a 5-6 percentage point increase in firm-level productivity.[47]
- A 2021 ITU study of 139 countries (2010-2020) found that 10 percent rise in fixed broadband penetration increased GDP per capita by 0.85 percent in middle-income and 1.25 percent in high-income countries.[48] A 2025 update covering 107 countries (2010-2023) shows stronger effects at 1.29 and 1.43 percent respectively, with even higher gains in Asia Pacific rising from 1.53 to 1.84 percent.[49]
- A 2022 cross-country study of 116 economies (2014-2019) found that a 10 percentage-point increase in mobile-broadband speed is associated with about 0.2 percentage-point higher labor productivity in the following year, based on fixed-effects panel estimations.[50]
- A 2022 firm-level study (2006-2018) analyzing the staggered 2014-2016 "Broadband China" rollout as a quasi-natural experiment found that broadband expansion raised firm productivity by 4 percent.[51]
- A 2023 study in Peru found that connecting schools to broadband led to gradually rising student test scores, from near zero impact in the first year to statistically significant learning gains after five years.[52]

There is comparatively less evidence related to increased data center capacity. However, a 2022 study using cross-country data found that a 10 percent increase in data center capacity raises exports of data-related services by about 1.6 percent, while a similar increase in secure internet servers raises such exports by 2.1 percent—highlighting the importance of digital infrastructure for competitiveness in global services trade.[53]

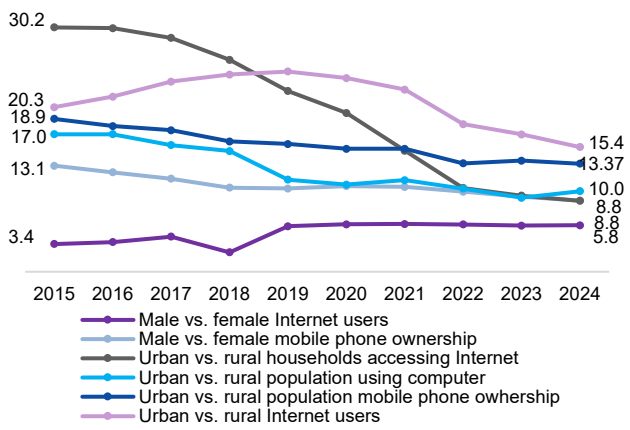
45. Briglauer, W., Krämer, J., and Palan, N., 2024. *Socioeconomic Benefits of High-Speed Broadband Availability and Service Adoption: A Survey. Telecommunications* Policy. 48(7).
46. Gal, P., Nicoletti, G., Renault, T., Sorbe, S., & Timiliotis, C., 2019. *Digitalisation and productivity: In search of the holy grail – firm-level empirical evidence from EU countries. In OECD economics Department working papers (Vol. 1533).*
47. Dalgıç, B., and Fazlıoğlu, B., 2020. *The Impact of Broadband Speed on Productivity: Findings from Turkish Firms.* Applied Economics Letters. 27(21).
48. International Telecommunication Union (ITU), 2021. *The economic impact of broadband and digitization through the COVID-19 pandemic: Econometric modelling.*
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52. Lakdawala, L., Nakasone, E., and Kho, S., 2023. *The Effects of School Internet Access on Educational Outcomes: Evidence from Peru.* American Economic Journal: Economic Policy. 15(3).
53. van der Marel, E., 2022. *Offshore Data and Cross-Border Trade.* European University Institute Working Paper RSC 2022/43.

2. State of Indonesia's Digitalization

Who is connected and who is not

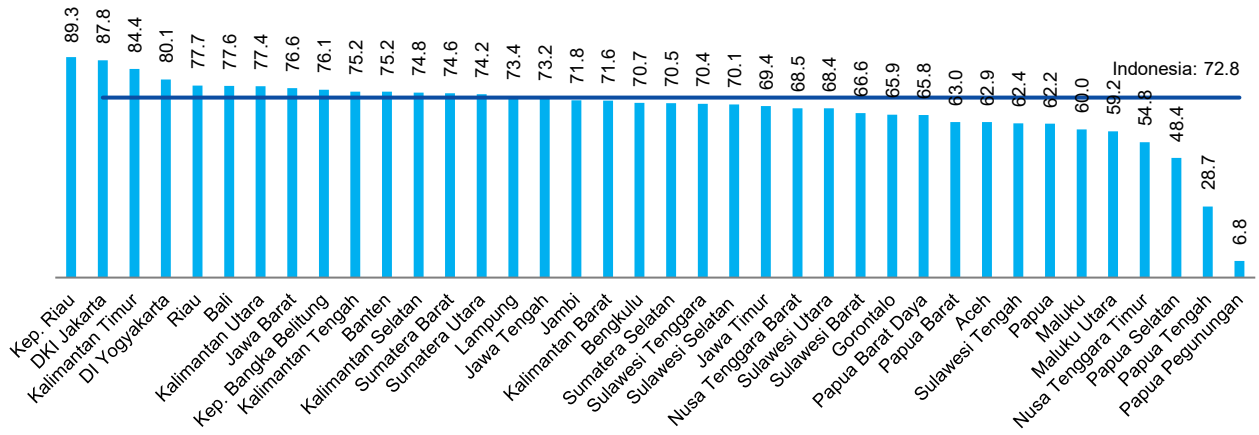
Internet access in Indonesia has become more inclusive, even as important disparities persist. Internet use has expanded across all groups, with connectivity gains strongest in rural areas. Figure B.2 shows that urban-rural gap in internet use has narrowed from 20.3 percentage points in 2015 to 15.4 in 2024. Men are 5.8 percentage points more likely than women to use the internet—a gap slightly widening in the past ten years—but the difference in mobile phone ownership has fallen from 13.1 to 8.8 points. Access is also converging across income and education groups. Nonetheless, provincial differences remain significant, with usage ranging from 89.3 and 87.8 percent in Riau Islands and Jakarta to 28.7 and 6.8 percent in Central Papua and Papua Highlands, respectively (see Figure B.3).

Figure B.2: Changes in internet access disparities



Source: World Bank staff calculations based on ITU, 2025. *ITU DataHub*. & BPS, 2025. *Susenas 2024*.

Figure B.3: Internet access by province

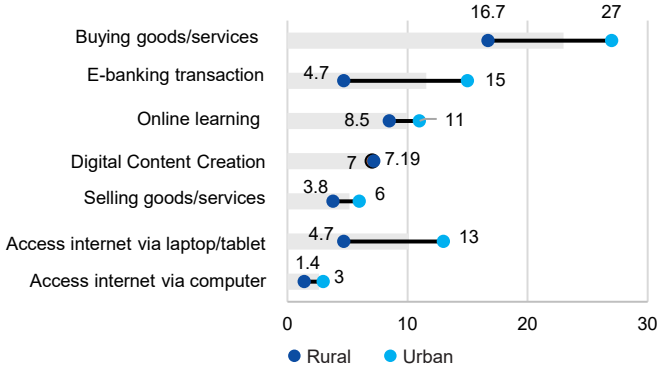


Source: BPS, 2025. Susenas 2024.

What Indonesians use the internet for

Most Indonesians still use the internet primarily for leisure instead of productive or income-enhancing activities. About 78 percent of Indonesia's internet users go online for social media and 85 percent for entertainment, while only a small share use it for productive purposes,[54] such as online learning (10 percent), buying or selling goods and services (23 and 5 percent, respectively), email exchange (13 percent), digital content creation (7 percent), or remote work (1.6 percent).[55] Another survey found that only 23 percent of Indonesians use the internet for economic and business activities. These levels of productive uses remain below regional peers. For example, 27 percent of Malaysians take online courses, 71 percent use

Figure B.4: Purposes of Internet Use in Indonesia by Area (in percentage of population)



Source: BPS. 2025. *Susenas 2024*.

54. Indonesian Internet Service Providers Association (APJII), 2025. *Internet Penetration and User Behavior Survey.*
55. BPS, 2025. *Susenas 2024.*

the internet for online purchases, and 37 percent for work-related document editing.[56] Rural internet users are even less engaged in these activities (see Figure B.4) and only few access the internet via laptops or computers that enable higher productivity. While over 40 percent of the population in Java use the internet for buying or selling goods and services, only about 25 percent of the population in Sulawesi and far fewer in Maluku and Papua do so. Disparities also appear in online banking and learning.

Poor internet quality, limited skills, and low-value content reinforce each other to constrain digital productivity. Many platforms are built for lower internet speeds, reducing functionality and demand for investment in better infrastructure. Breaking this vicious cycle can start with stronger digital infrastructure, supported by advances in skills and government-led efforts to create richer, more data-intensive public content and services.

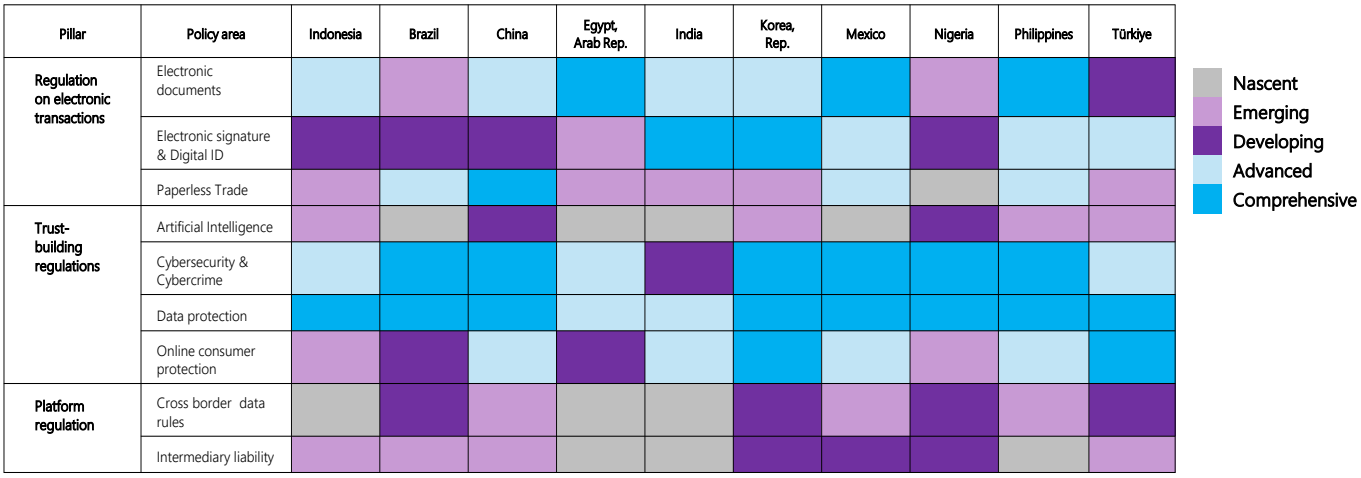
Digital economy

Indonesia has ASEAN's largest digital economy with gross merchandise value (GMV) reaching US$99 billion in 2025, well ahead of Vietnam (US$39 billion), Malaysia (US$39 billion), and the Philippines (US$36 billion).[57] On GMV per-capita terms, Indonesia (around US$350) is in line with its level of development, ahead of the Philippines (US$310) and behind Viet Nam (US$386) and Malaysia ($1,080). Much of Indonesia's digital economy growth comes from widespread consumption—primarily e-commerce (US$71 billion)[58]—which is further described in the 'What Indonesians use the internet for' section. There is an opportunity to accelerate growth by shifting focus from consumption to digital productive use, enabling firms, workers, and regions outside of Java to capture the next wave of digitally driven growth and job creation.

While Indonesia's digital economy continues to expand, its regulatory framework remains a work in progress. Most policy areas are at the developing stage, showing steady progress but not yet reaching the level seen in South Korea, China or Türkiye (Figure B.5). Greater consistency and coordination across policy areas is essential to build digital trust and support the country's transition from a consumption-driven to a productivity-enhancing digital economy. For example, the cross-border data flows framework remains restrictive and underdeveloped, creating uncertainty for firms and investors (also discussed in the 'Data Centers' section), with the emphasis on data localization limiting efficiency and regional integration. A more conducive regulatory framework would strengthen investor confidence, enable Indonesia to better integrate with regional and global digital markets, and capture larger gains from cross-border digital trade.

The ICT services sector in Indonesia remains small and undiversified, contributing six percent of GDP and dominated by telecommunications services.

Figure B.5: Digital Regulation Readiness of Indonesia and Peers



Source: World Bank Digital Trade Readiness (DTRR) database.[59]

56. ITU, 2025. *ITU DataHub: Digital Development Indicators – Individuals using the Internet, by type of activity.*
57. Google, Temasek, and Bain & Company, 2025. *e-Conomy SEA 2025.*
58. ibid
59. Data are from the World Bank Digital Trade Readiness (DTRR) database, which assesses countries' regulatory frameworks across three pillars: electronic transactions, trust-building regulations, and platform regulation. It provides qualitative assessments of nine policy areas and over 110 regulatory measures central to digital trade, capturing mainly de jure readiness. The DTRR rates each area into five readiness categories (quintiles) based on the share of positive regulatory responses.

Its linkages to the broader economy are limited, while its contribution to GDP and employment remain modest. Its impact on domestic investment is relatively stronger, reflecting ongoing network expansion and infrastructure upgrades. However, weak FDI and export performance highlights the early stage of development in higher-value segments such as software and cloud services. Most ICT-related jobs remain concentrated in lower-skill telecommunications and distribution activities.

Increasing value addition in services is closely linked to the pattern of both domestic and foreign investment in services. Domestic investment in services (Figure B.6) and foreign investment in services (Figure B.7) shows that most funds still flows to low-skill domestic and tradable services[60] which generate limited productivity gains to the economy. In contrast, investment in higher-value segments, such as global innovator services (e.g. telecommunication and digital services) remain modest, even though these activities support stronger knowledge transfer and closer links to manufacturing and modern tradeable sectors. Attracting more investment toward higher-value services would help raise the sophistication of services and deepen value addition.

Most firms, especially MSMEs, remain at an early stage of digital adoption. Fewer than half maintain a business website, around 40 percent use accounting software, and only one in five employ enterprise resource planning (ERP) or other specialized management systems. This limited use of digital tools means that for most firms, digitalization stops at basic connectivity and communication, such as using messaging apps or social media for sales, without extending to data-driven operations or integrated supply chains. Such shallow adoption is among many factors that constrain productivity growth, participation in formal financial systems, and participation in regional and global value chains. Firms that operate largely offline face higher transaction costs, weaker inventory and financial management, and limited access to digital finance or e-commerce platforms.

Indonesia's rapid digitalization is reshaping labor demand, but the gains remain concentrated in urban Java. Use of the internet for work almost tripled to 47.1 percent in 2023, up from 17.6 percent in 2018.[61] The rise of platform work—engaging an estimated 1.5 million workers in 2023, half of whom entered the platform workforce within the previous four years—has widened entry points for youth and women, but participation remains concentrated in Java's urban centers.[62] Only 53 percent of Indonesians report basic digital skills and just 4 percent advanced,[63] and most employers struggle to find candidates with the capabilities they need. Stronger digital infrastructure is therefore essential—not only to expand access to digital services but to enable higher-productivity work,

Figure B.6: Domestic investment in global innovator services (e.g. telecommunication, digital services) remain modest.

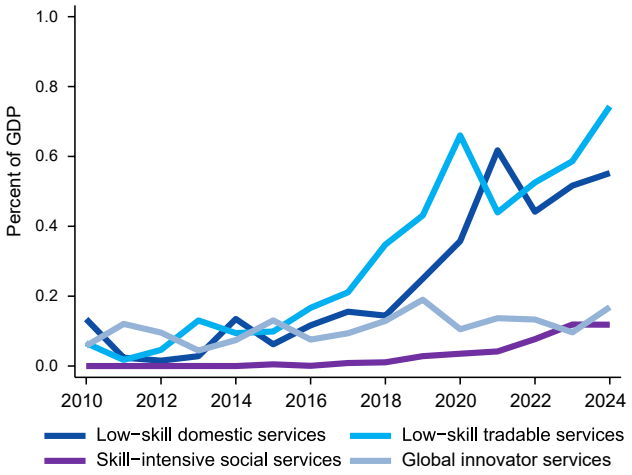
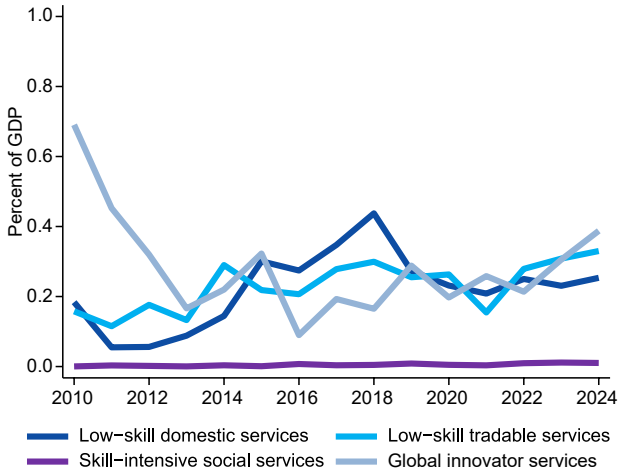


Figure B.7: While foreign investment in global innovator has shown only modest trend upward.



Source: Ministry of Investment/Investment Coordinating Board, WB Indonesia Country Growth and Jobs Report (forthcoming).

60. Low-skill domestic services includes, for example, retail and wholesale, while low-skill tradable services include accommodation and food and beverage services.
61. World Bank calculation based on National Laborforce Surveys covering 2018-2023.
62. World Bank, forthcoming. *The Labor Market Potential of Indonesia's Digital Transformation.*
63. ITU, 2024. *Measuring Digital Development: Facts and Figures 2023.*

online learning, and the use of cloud and AI tools. Without more reliable and affordable connectivity beyond major cities, better-paid digital work will remain geographically concentrated, reinforcing regional inequality.



Indonesia has made significant strides in expanding access to financial services, with 56 percent of adults holding an account at a formal financial institution in 2024.[64] Higher usage of some form of financial service is reported, reflecting the impact of DPI and the rapid expansion of fintech. However, active usage of digital financial services remains limited. As noted in the 'DPI' section, usage of digital payment adoption is uneven by income and rural-urban divides. MSMEs, which represent 61 percent of GDP and 99 percent of firms, receive just 19.6 percent of total bank lending, and many still rely predominantly on cash transactions, limiting their ability to build credit histories and access financing. These gaps highlight the need to move beyond access toward driving meaningful, sustained use of DFS, especially among women, rural populations, and MSMEs. As discussed further in Box B.2, with the right reforms, Indonesia's digital financial sector can serve as a catalyst for inclusive growth, job creation, and deeper integration into the global digital economy.

BOX B.2: Digital financial services: The next growth frontier in Indonesia

Indonesia is well positioned for the next phase of digital financial innovation, but realizing this potential will require a more integrated and interoperable ecosystem. Fragmentation across platforms and bilateral APIs continue to limit scale. The National Standard for Payment Open API (SNAP) has made progress but could further evolve into a full-fledge Open Finance framework covering more services such as savings, credit and insurance, to help households and firms manage risk and build resilience. A structured Open Finance framework, which enables consumer-consented data sharing under robust governance and protection could unlock innovation such as data-driven credit scoring and alternative data use. International experience shows that harmonization and interoperability are prerequisites for scaling digital payments, supporting cross-border transactions, and integrating Indonesia into regional and global value chains.

Improving credit infrastructure is also critical. Strengthening the secured transaction regulatory framework and development of the moveable collateral registry (fidusia) would unlock lending for MSMEs and households, while harmonization and interoperability of existing credit reporting systems—to expand data coverage for borrowers' creditworthiness assessment—could be a "quick win" with significant impact.

The next frontier for Indonesia's digital financial services lies in harnessing advanced technologies—particularly AI, open finance, and cross-border payment innovations—to drive inclusion, efficiency, and trade competitiveness. The Financial Services Authority (OJK) and Bank Indonesia (BI) are piloting AI and machine learning tools to enhance supervisory technology (SupTech), risk-based supervision, and consumer protection, with ongoing World Bank support for scaling these solutions. On the trade front, BI's Project Garuda is piloting a central bank digital currency (CBDC)—the Digital Rupiah—designed to support programmable, cross-border payments and integration with regional payment systems, lowering transaction costs and facilitating MSME trade. To realize these gains, Indonesia must harmonize regulatory frameworks, ensure interoperability across platforms, and strengthen financial consumer protection and anti-scam frameworks to foster public trust and confidence.

64. World Bank, 2025. *Global Findex Survey – 2024.*

Digital public infrastructure

Indonesia's digital public infrastructure (DPI)[65] is taking shape, but adoption needs to accelerate to deliver broader impact. As connectivity improves, Indonesia has an opportunity to build inclusive and dynamic digital identity, digital payments, data sharing, and other foundational platforms that can underpin seamless service delivery, expand economic participation, and accelerate the country's digital transformation. Momentum around DPI grew in 2025 under the leadership of the National Economic Council (DEN), which coordinated a pilot to apply DPI for social assistance delivery in Banyuwangi regency that will be scaled nationally in 2026. This "use case" approach—focused on solving real problems—offers a practical model for aligning institutions and accelerating adoption, with potential expansion to business licensing (OSS), financial inclusion, taxation, and other public and private services.

With 97 percent of adults having a national ID (KTP)[66], Indonesia is now extending this foundation into the digital space through the Digital Population Identity (IKD) app, launched by the Ministry of Home Affairs in 2022. The app contains a digital wallet that allows citizens to present their KTP and other official documents, and as a digital identity to sign into online services such as the *SatuSehat* digital health application and digital banking apps.[67] 17 million Indonesians have signed up for IKD so far,[68] and the Government aims to reach 100 million (half of all adults) by 2029. By enabling remote identity verification, IKD can reduce the need for physically visiting offices or submitting documents, lower transaction costs, and expand access to higher-value online services such as financial products, property registration, and business licensing. Continued effort is needed to catch up with ASEAN leaders, Singapore and Viet Nam, where 91 and 69 percent of adults have Singpass and VNeID, respectively.[69] Efforts to increase the adoption of IKD in both public and private sectors need to be accelerated as current usage is limited. Indonesia also still lacks a trusted digital identity for businesses, which could help streamline e-invoicing and participation in digital trade. As ASEAN moves toward interoperability of digital identities under the DEFA, conforming with future regional standards will enable cross-border transactions.

Indonesia's digital payments ecosystem has expanded rapidly, anchored by Bank Indonesia's (BI) BI-FAST and Quick Response Interoperability Standard (QRIS) systems. BI-FAST was launched in 2021 as a payment system that enables low-cost real-time transfers across banks and non-banks, while QRIS—introduced in 2019—now serves more than 58.3 million users and 41.2 million merchants.[70] Despite these advances in access, usage remains uneven: almost half of adults in the richest 60 percent use digital payments, while only 34 percent among the poorest 40 percent do so.[71] Regional disparities are even sharper: Jakarta alone accounts for about 74 percent of mobile money (server-based e-money) accounts, and the rest of Java representing another 15.6 percent.[72] Expanding MSME digitalization programs and integration between payment systems could help broaden adoption and productivity gains. At the same time, the inability of social assistance beneficiaries to choose their own accounts to receive transfers[73] reduces opportunities to deepen financial inclusion among low-income households. Stronger consumer protection enforcement and sustained financial literacy efforts are needed to deepen trust and safeguard users. Cross-border interoperability is also advancing, with QRIS now able to be used in Malaysia, Singapore, and Thailand and soon China, Japan and South Korea.

Data exchange and electronic signatures are the least developed components of Indonesia's DPI. The Government Service Exchange System (SPLP) provides a platform for data sharing within the government as is aligned with international good practices. Uptake, however, has been limited so far: only a small number

65. DPI refers to digital platforms that facilitate essential functions at society scale and can be reused across sectors, including digital identity, digital payments, data sharing, and electronic signatures. DPIs enable public and private service providers to build on these systems, innovate, and roll out new services more quickly and efficiently. For more information, see: World Bank, 2025. *Digital Public Infrastructure and Development: A World Bank Group approach.*
66. World Bank, 2025. *Global Findex-ID4D Survey 2024.*
67. The World Bank is supporting development of IKD, as well as broader improvements to population and civil registration, through the ID for Inclusive Service Delivery and Digital Transformation in Indonesia Project.
68. Data provided by the Directorate-General for Population and Civil Registration, Ministry of Home Affairs.
69. World Bank, 2025. *Global Findex-ID4D Survey 2024.*
70. Data for end of October 2025 provided by Bank Indonesia
71. World Bank, 2025. *Global Findex Survey – 2024.*
72. Bank Indonesia, 2025. *Payment System and Market Infrastructure Statistics (Statistik Sistem Pembayaran dan Infrastruktur Pasar), September 2025.*
73. Presidential Regulation No. 63 of 2017 on Non-Cash Distribution of Social Assistance limits disbursement to accounts at SOE banks.

of ministries and local governments actively use it, and transaction volumes remain modest. Progress has been slowed by inadequate mandates and incentives for agencies to share data and integrate with SPLP. Electronic signatures, meanwhile, are legally recognized and regulated under the Electronic Information and Transactions Law (Law No. 11 of 2020), with 10 licensed certification authorities (PSrE). Wider use has likely been constrained by cost considerations and the absence of a clear risk-based framework to guide agencies and businesses to choose technologies appropriate to transaction sensitivity.

Digital government

Indonesia has invested heavily in digital government over the past decade, but progress has lagged expectations. The Electronic-Based Government System (SPBE) and *Satu Data* (One Data) initiatives were launched through respective Presidential Regulations in 2018 and 2019 to improve the quality of public services and data governance. While well-intentioned, agencies continue to operate separate databases and platforms, leading to fragmented systems, poor user experiences, and the proliferation of thousands of government applications. Coordination mechanisms and budget clearances for ICT spending remain largely procedural, rather than strategic. Despite the Electronic Information and Transactions Law giving legal effect to digital documents, outdated sector regulations still require citizens to submit physical documents, slowing the transition to fully digital services. There are bright spots, however, where strong leadership and collaboration have produced results at scale.[74]

Digital government not only depends on strong infrastructure—it also drives demand for it. As more services and processes become digital, ministries and local governments will need reliable broadband to connect public facilities so that staff can use digital tools, access cloud applications, and exchange data securely. Strengthening government-wide standards for interoperability, security, and data governance—and equipping agencies with shared tools and reusable components—would help integrate services and reduce duplicated investments. Expanding high-speed connectivity to schools, health centers, and local government offices will strengthen frontline service delivery while stimulating broader demand for last mile networks. There is also an opportunity to shift the approach to government data hosting from a model centered on government-managed national data centers (Pusat Data Nasional, PDN) to a hybrid one that also leverages commercial cloud and public-private partnerships (PPP), balancing data sovereignty with creating the predictable government demand needed to stimulate private investment in domestic data center capacity.

Digital trust

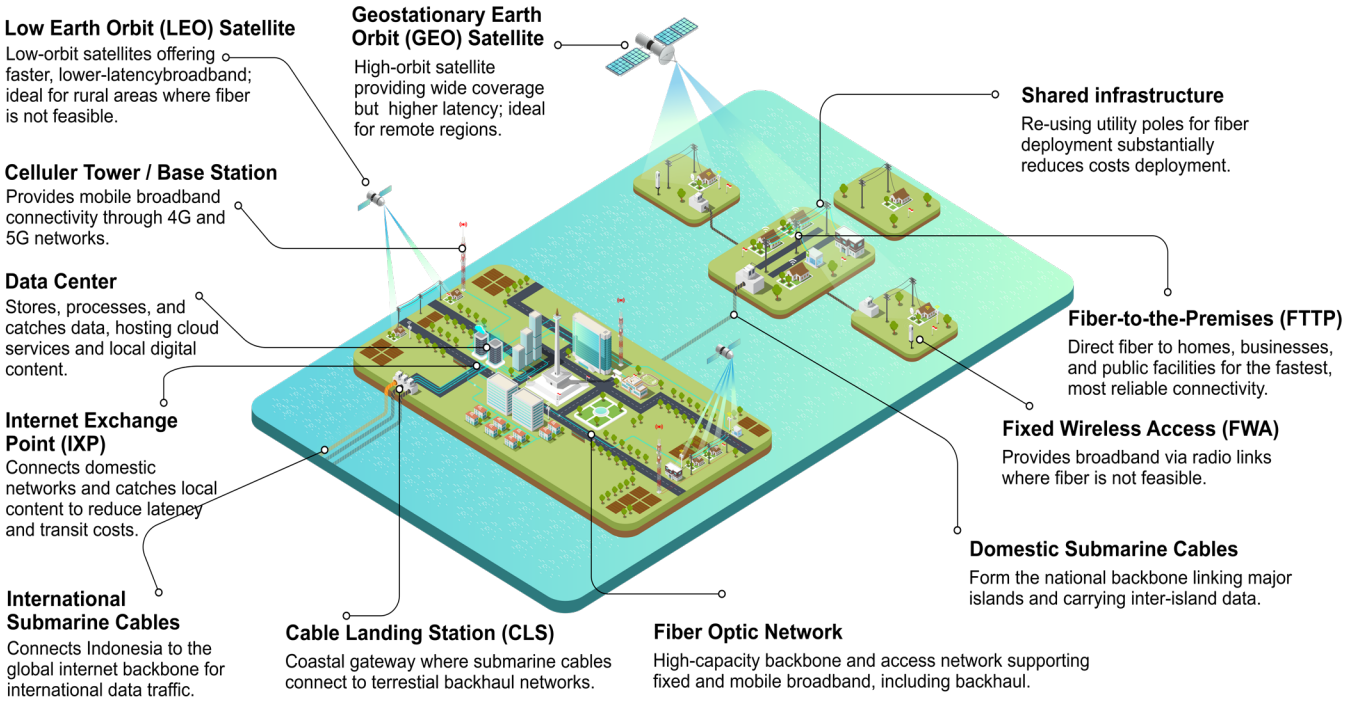
Giving citizens and firms confidence to transact and share data online is critical for turning digital access into productivity, yet Indonesia—despite making some progress—still faces important gaps in data protection and cybersecurity. The Personal Data Protection Law (Law No. 27 of 2022) was a major step toward a comprehensive privacy framework, but its enforceability and impact remain limited as the implementing Government Regulation and establishment of the independent supervisory authority are still pending. The absence of comprehensive cybersecurity law also means that government agencies and businesses face limited obligations for prevention, reporting, and coordination. Meanwhile, data breaches, hacking, and online scams have increased sharply. The National Cyber and Crypto Agency (BSSN) leads incident response and coordination, but preventive capabilities, such as systematic risk assessments, vulnerability management, and capacity-building, remain underdeveloped. Public awareness and cyber-hygiene training, particularly for government agencies and firms handling sensitive data, remain limited, reducing overall preparedness and resilience. As service digitalization and data volumes grow—and new threats such as AI-driven fraud, deepfakes, and ransomware emerge—these gaps heighten exposure to cyber incidents and erode confidence in digital platforms.

74. Examples include the Ministry of Health's *PeduliLindungi* app, which was originally developed for COVID-19 vaccination management and has evolved into the national *SatuSehat* health platform; the *Online Single Submission* (OSS) system that has streamlined business licensing across ministries and agencies; and integrated local service platforms such as Jakarta's *JAKI*, West Java's *Sapawarga*, and Yogyakarta's *Jogja Smart Service* (JSS).

3. Strengthening Indonesia's Digital Infrastructure

Digital infrastructure underpins how data moves, is processed, and ultimately reaches people and firms—connecting Indonesia to the world and carrying data across its vast archipelago. Such infrastructure can be viewed as a value chain that runs from the first mile—international connectivity—to the last mile, where households, schools, and firms go online. Indonesia's first mile connectivity is through submarine cables, before feeding into national backbones that carry data across islands through the middle mile. These domestic networks—combining terrestrial fiber, microwave, and satellite links—connect cities, towns, and villages across the archipelago to Internet Exchange Points (IXPs) and data centers, where data is cached and processed locally to improve performance and reliability, as well as to reduce costs of data transit from overseas. Each step in this chain involves infrastructure and interconnection costs that are ultimately reflected in what users pay for internet access. Finally, the last mile delivers access through fiber-to-the-home or -premise (FTTH/FTTP), and through towers that provide mobile broadband and fixed wireless access (FWA) over dedicated spectrum frequencies, complemented by satellite connections that enable reach to the most remote communities. Given Indonesia's geography of more than 17,000 islands, with sharp differences in terrain, density, and income that shape network economics, extending this infrastructure is costly and complex.

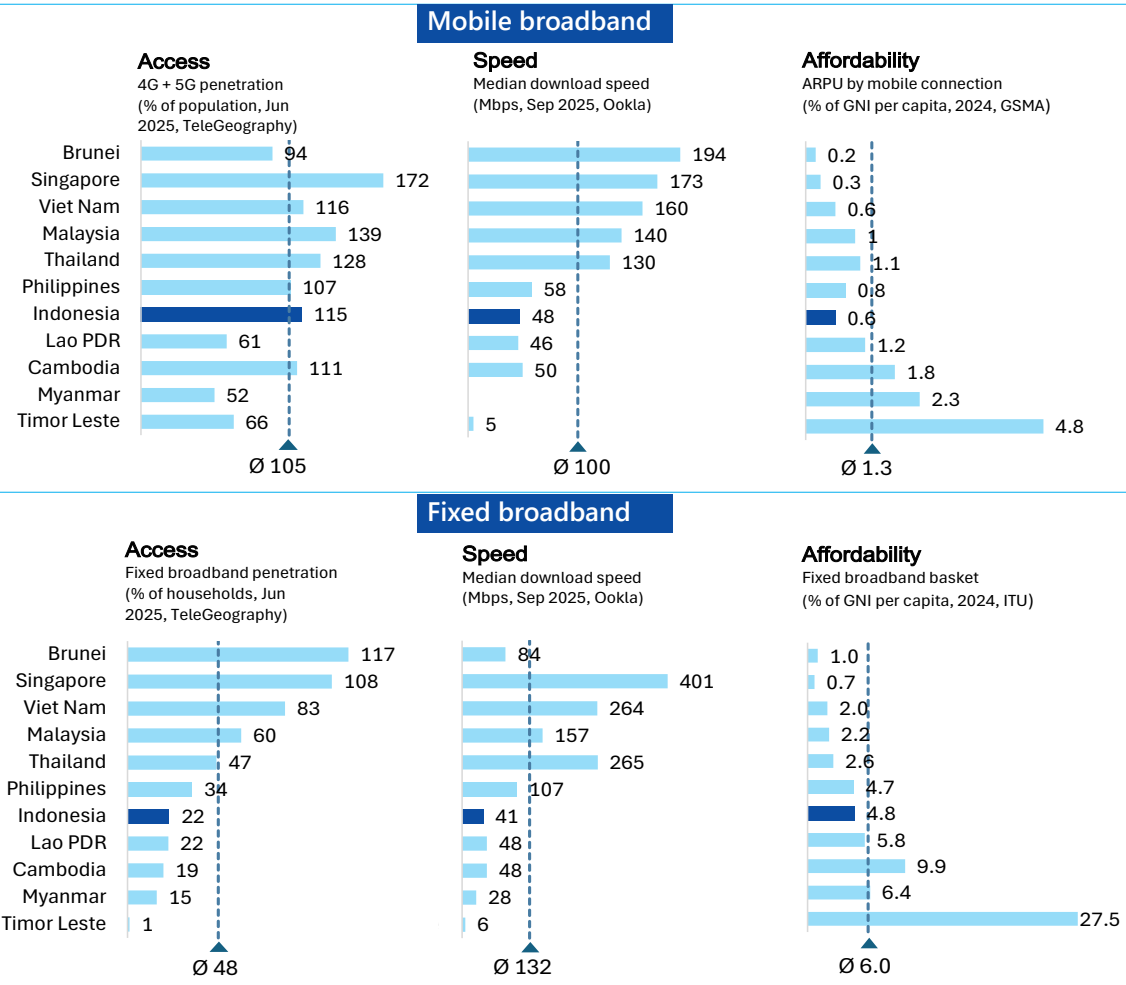
Figure B.8: Illustration of digital infrastructure in the Indonesian context



Source: World Bank original illustration (2025).

Indonesia leads ASEAN in affordable mobile access but lags sharply in broadband quality, especially fixed broadband, which will be critical for future growth (see Figure B.9). Mobile broadband in Indonesia is one of the most accessible and affordable in the region, with subscriptions reaching 115 percent of the population and average revenue per user (ARPU) at just 0.6 percent of GNI per capita. However, median mobile speeds are less than a third of Malaysia, Singapore, and Viet Nam. Fixed broadband has been challenged by low competition and limited infrastructure sharing, which have kept prices high and coverage low: only 22 percent of households are connected, and average speeds rank near the bottom in ASEAN. Without a concerted effort to open networks and expand fixed broadband access, Indonesia risks falling further behind regional peers in productivity, innovation, and digital trade.

Figure B.9: Coverage and quality of internet in ASEAN, ranked by fixed broadband penetration



Source: World Bank staff compilation of data cited in the graphic.[75]

Institutional arrangements

The institutional arrangements for digital infrastructure are relatively clear, with the newly renamed Ministry of Communication and Digital Affairs (Komdigi) playing the leading role. At the beginning of 2025, Komdigi created the Directorate General for Digital Infrastructure, merging policy and regulatory responsibilities for broadband networks and data centers, including licensing of operators, spectrum management, fiber and tower regulation, and oversight of submarine cables.[76] Other ministries also play enabling and regulatory roles: the Ministry of Home Affairs (MoHA) oversees how local governments issue permits and collects local revenues, the Ministry of Finance (MoF) manages non-tax revenue (PNBP) policies, including spectrum and license fees, and the Coordinating Ministry for Food Affairs and the Ministry of Marine Affairs and Fisheries coordinate submarine cable policies across utility sectors.

The Telecommunication and Information Accessibility Agency (*Badan Aksesibilitas Telekomunikasi dan Informasi,* BAKTI), which is attached to Komdigi, serves as the government's implementing arm for universal access. BAKTI uses revenue from the Universal Service Obligation (USO)—a 1.25 percent levy on the gross revenue of licensed telecommunications operators—to invest in mobile towers, satellite, and fiber networks in commercially unviable areas primarily in underdeveloped, frontier, and outermost regions (*tertinggal, terdepan, dan terluar,* 3T) regions. Unlike most USO agencies internationally, BAKTI acts as a network operator rather than a market enabler, directly building and managing infrastructure. It has also served as the public partner in PPPs such as the Palapa Ring (a partial fiber backbone) and SATRIA-1 (Indonesia's first sovereign satellite). However, BAKTI's long-term financial sustainability remains a challenge, given the high cost of its network

75. Broadband speeds in Timor-Leste are not reported by Ookla and thus retrieved from Digital 2024 Timor-Leste page by Data Reportal.
76. The Telecommunication Regulatory Body (BRTI), a semi-autonomous regulator, was disbanded in 2022, and its functions were subsequently absorbed into the then-Kominfo.

operations and maintenance—reaching IDR 11.2 trillion or 57 percent of Komdigi's budget in 2024.[77] Refocusing BAKTI's role toward market facilitation, while ensuring that its wholesale arrangements support expanded connectivity in underserved regions, could improve efficiency and help attract more private investment.[78]

Investment promotion and long-term planning remain less coordinated. In the data center domain, Komdigi has a clear mandate for government data centers and cloud policy, but responsibilities for promoting private investment are distributed among the Coordinating Ministry for Economic Affairs and the Investment Coordinating Board (BKPM) and coordination mechanisms do not yet exist. Indonesia also lacks a comprehensive National Broadband Plan,[79] which could better align institutional roles, targets, and investment priorities by both the public and private sectors. Expanding a future plan to cover digital infrastructure more broadly would help Indonesia harness the synergies across the entire digital infrastructure value chain. Importantly, Indonesia's new sovereign wealth fund and state holding company, Danantara, has identified digital infrastructure, and especially sustainable data centers, as one of its priority sectors, offering a timely opportunity to mobilize more private investment.

Backbone

Indonesia's international connectivity—the first mile—has expanded rapidly but remains concentrated. The vast majority of Indonesia's international bandwidth is routed from Singapore through landing stations in Batam, creating a single-corridor dependency that raises costs and exposes the network to congestion and potential disruptions, including natural hazards. Several new submarine cable systems—including recent routes deployed by hyperscale cloud providers—now pass near Indonesia without landing, possibly reflecting concerns about permitting complexity and route restrictions. This reduces Indonesia's direct connectivity to global networks and reinforces dependence on the Singapore-Batam corridor. A recent Minister of Maritime Affairs and Fisheries Decree (KepMenKKP No. 14 of 2021) reinforces this concentration by restricting new international cables to Batam, Manado, Jayapura, and Kupang. The exclusion of Jakarta, the country's commercial and data center hub, limits investment in data-intensive industries that depend on low-latency links. A Ministry of Transportation regulation (Permenhub No. 2 of 2021), based on cabotage rules in Government Regulations No. 20 of 2010 and No. 22 of 2011, also restricts foreign-flagged vessels from undertaking submarine cable repairs, requiring special permits even though there are limited Indonesian-flagged options.[80] Allowing additional landing sites—especially Jakarta but also other commercial hubs such as Makassar, Medan, and Surabaya—would strengthen resilience, attract data center investment, and lower transit costs to rural and eastern Indonesia.[81] Streamlining approvals through a single window licensing process and allowing foreign repair vessels would further enhance competitiveness.

Indonesia's domestic backbone now reaches nearly all cities and districts, but large gaps in middle-mile connectivity, especially to villages, continue to limit rural access and service quality. Fiber ODPs exist in 98 percent of cities and districts, but only 70 percent of *kecamatan* (sub-districts) and half of villages.[82] The backbone is dominated by Telkom, complemented by PLN's fiber, the Palapa Ring, and a handful of private operators. The lack of integration into a neutral or shared backbone has led to duplication along profitable routes and weak coverage in less-commercial regions. Large parts of rural and eastern Indonesia still depend on microwave or satellite backhaul, which are more expensive, less reliable and have less capacity than fiber, leading to slower speeds, higher latency, and higher retail prices for users, as well as disruption in poor weather. Closing these middle mile gaps and improving interconnection among backbone providers will be essential to lower costs and strengthen competition.

The Palapa Ring project has helped the backbone to reach districts that were commercially unviable—but it remains underutilized. Completed in 2019, the 12,200-kilometer network connects 57 cities and remote districts in eastern and central Indonesia to

77. Ministry of Communication and Digital (Komdigi), 2024. *Laporan Kinerja Kementerian Komdigi 2024.*
78. PPP models also offer viable alternatives for bridging digital gaps with lower fiscal outlays and greater sustainability, such as a rural tower model in which private investors build and operate sites in remote areas while BAKTI provides targeted viability-gap funding.
79. The last National Broadband Plan (Presidential Regulation No. 96 of 2014) expired in 2019.
80. These rules increase restoration times and deter investment in new routes, especially by foreign carriers.
81. Indonesia can maintain appropriate permitting and security safeguards while ensuring these processes do not unnecessarily constrain foreign cable landings that are in the national interest.
82. Based on December 2024 data provided by the Telecommunications Monitoring Center, Directorate General of Digital Infrastructure, Ministry of Communications and Digital (Komdigi).

the broader commercial backbone serving the rest of the country. Although built with USO funds to deliver socio-economic benefits rather than commercial returns, tariffs are set at market levels.[83] Given that BAKTI covered the high-cost capital investment, pricing should primarily recover operational expenses and stimulate middle and last mile network deployment in underserved areas. Underutilization mainly reflects limited reach to end users and the lack of direct interconnection with wholesale internet gateways.[84] In eastern Indonesia, onward connectivity still depends on privately operated backbone segments where bilateral access agreements and the absence of transparent reference offer limit ISP participation. Moving toward open access, cost-based tariffs on commercially viable routes, while maintaining affordable pricing on BAKTI segments, would improve utilization and support wider connectivity.

Domestic interconnection through Internet Exchange Points (IXPs) remains fragmented and heavily concentrated in Jakarta, increasing costs and latency for users outside Java. Indonesia hosts nearly 20 IXPs, but most regional exchanges carry little traffic, and around three-quarters of domestic data continues to pass through Jakarta-based IXPs. This centralization means that data generated in eastern Indonesia often routes through Jakarta before reaching nearby destinations, adding delay and transit costs. Strengthening regional IXPs in Sulawesi, Kalimantan, and Papua, and co-locating them with data centers and content delivery networks (CDNs), would help keep traffic local and improve network efficiency. Promoting open peering policies and integrating IXP planning into national backbone development would reduce the need for domestic traffic to detour through Jakarta or even through international routes, and thus lower costs and increased resilience.

Mobile broadband

Indonesia's mobile broadband market is one of the most competitive in ASEAN, enabling widespread coverage and among the region's most affordable prices. There are now three mobile network operators (MNOs)—Telkomsel, Indosat, and XLSmart[85]—following a period of consolidation and market rationalization. Strong competition, reinforced by regulations and policies that mandated tower sharing and opened the market to independent tower companies (see Box B.3), has helped to push 4G coverage above 99 percent of the population and driven retail prices down to around 0.6 percent of GNI per capita—well below the International Telecommunications Union (ITU) affordability target of 2 percent. Introducing mobile number portability would further strengthen consumer choice and competitive pressure, empowering users to switch MNOs more easily without changing numbers.[86] As noted in the 'Institutional Arrangements' section, public investment through BAKTI has complemented private rollout by connecting non-commercial areas, notwithstanding BAKTI's sustainability challenges.

BOX B.3: The success and lesson of Indonesia's tower sharing reforms

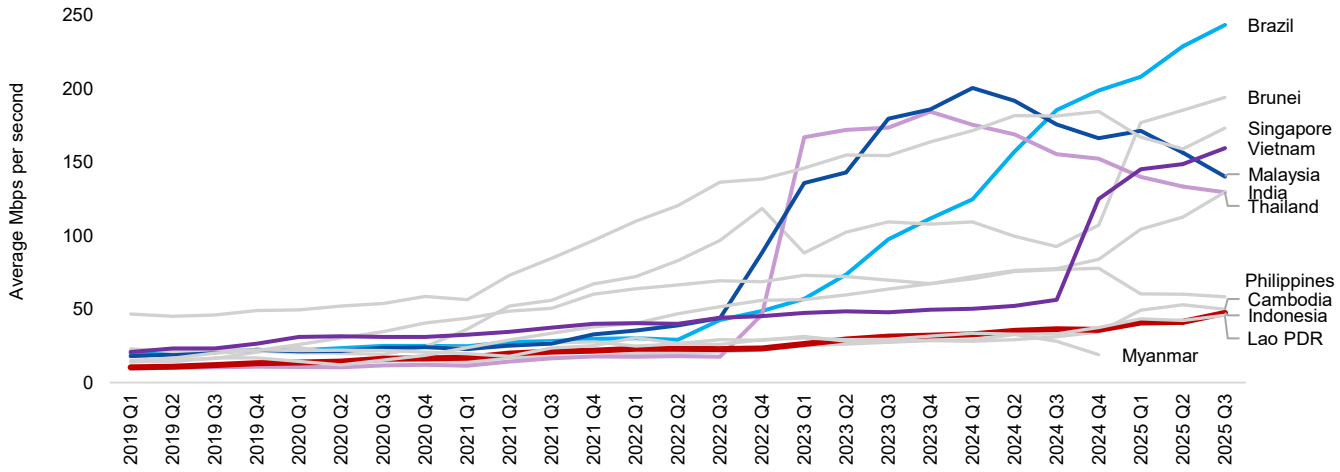
Indonesia's mobile network tower sharing framework was established through Kominfo Ministerial Regulation No. 2 of 2008, which required operators to share passive infrastructure and allowed independent tower companies to build and lease towers to multiple tenants. This regulation was later replaced by Joint Regulation of Minister of Home Affairs, Minister of Public Works, Minister of Kominfo, and Head of BKPM (No. 18 of 2009, 07/PRT/M/2009, 19 of 2009, 3 of 2009). Implementation was uneven at first, but enforcement strengthened in the late 2010s as Kominfo began conditioning new site permits on evidence of sharing. Further reforms under the Omnibus Law on Job Creation (Law No. 11 of 2020) implemented through Presidential Regulation No. 10 of 2021 on the New Investment List liberalized the telecommunications tower sector, allowing full foreign ownership and investment.

These reforms curbed duplication, lowered capital costs, and created one of the region's largest neutral-host tower markets. Today, over 90 percent of towers are owned by tower companies, providing a more efficient backbone for mobile network expansion. Based on this and international experience, applying the same open access principles to fixed broadband—through shared ducts, poles, and middle and last mile fiber networks—could boost competition, reduce costs, and improve service quality across Indonesia.

83. World Bank staff analysis comparing Palapa Ring published wholesale tariffs with other global routes, including transatlantic and U.S. domestic routes operated by private carriers with far lower per-gigabit costs, and modeling business viability for operators in low-demand districts showing that current tariffs make network expansion commercially unviable.
84. BAKTI reported low utilization of the Palapa Ring Central and East networks in 2023—37.5% and 35% of capacity—due to limited interconnection, lack of end-to-end backhaul, and high connection costs. See Bisnis.com, 2023. "*Utilisasi Palapa Ring Tengah dan Timur Rendah, BAKTI Ungkap Tantangannya*," December 17.
85. XLSmart was created out of a merger between XL and Smartfren in 2025.
86. This should be accompanied with the government offering secure, cost-efficient methods of identity verification for SIM registration.

However, Indonesia's mobile broadband speeds remain among the slowest in the region. Figure B.10 shows that while countries such as Brazil, India, Thailand, Malaysia, and Viet Nam have recorded sharp improvements following their respective 5G rollouts, Indonesia's performance has barely improved and now ranks near the bottom of ASEAN, on par with Cambodia, Lao PDR, Myanmar, and the Philippines. Median download speeds are less than half those of Malaysia and less than one-third of Singapore's, indicating network capacity lagging demand from rising data traffic. As AI adoption grows, the need and demand for low latency connectivity will become even more critical, given the need fast and stable round-trip connections; without which, applications such as real-time analytics and automated decision systems perform poorly.

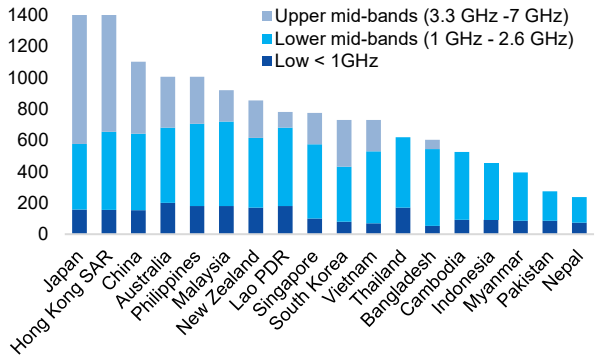
Figure B.10: Average mobile broadband speeds (2019-2025 Q3)



Source: Ookla Speedtest Global Index.

Spectrum scarcity is currently the main bottleneck holding back Indonesia's mobile broadband speeds. As shown in Figure B.11, Indonesia assigns one of the lowest amounts of low and mid band mobile spectrum[87] in the region, roughly half that of Malaysia or Singapore, and last in the G20, with key mid-band frequencies that are critical for high-capacity 5G networks largely unavailable for mobile broadband. This scarcity reflects legacy allocations for broadcasting and satellite services, slow refarming processes to clear and reallocate existing spectrum to new mobile uses, and high spectrum costs that dampen MNO demand. While refarming of 3G bands in 2022–2023 helped expand 4G capacity within existing holdings, it did not materially ease the broader spectrum deficit.

Figure B.11: Spectrum allocation for major regional economies



Source: Windsor Place Consulting (April 2025), based on regulator data (GSMA, NERA, MIC Japan, MIIT China, Komdigi, PTA, NBTC, SpectrumMonitoring etc).

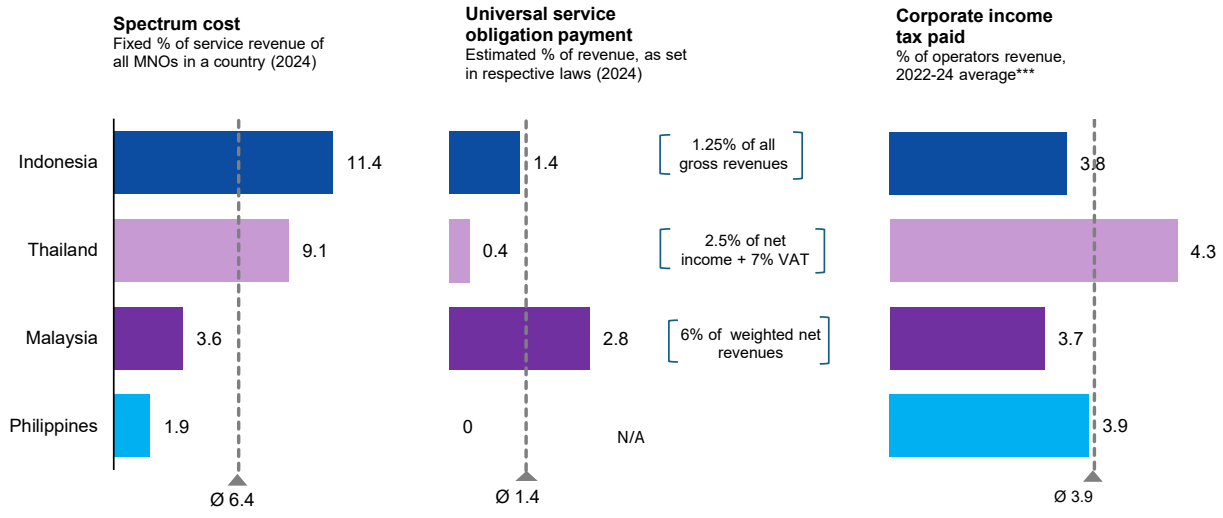
87. Spectrum is the set of radio frequencies used to carry wireless signals. If too little is allocated to mobile broadband, networks become congested and speeds decline as data demand increases. It is also a finite natural resource – a scarce input that must be allocated across multiple, often competing uses such as broadcasting, satellite, emergency communications, and mobile broadband.

Progress in clearing long-blocked spectrum bands has been slow, with only limited reallocations achieved to date. The 700 MHz band was cleared for mobile use in 2022 but still not yet commercially available to MNOs, such as through auctions, while the 2.6 GHz band was cleared in early 2025 after years of delay due to previous satellite broadcasting use. However, the 2.6 GHz allocation provides only an additional 190 MHz of mobile spectrum, which is insufficient for balanced assignment among three MNOs, who ideally require contiguous 100 MHz blocks each for efficient 5G deployment, based on international benchmarks. The 3.5 GHz band, which is the primary "capacity band"[88] for 5G worldwide, remains largely occupied by satellite connectivity, including for ATMs in urban areas that have access to better mobile and fiber networks.

Accelerating spectrum allocation for mobile broadband should now be an urgent priority. Indonesia's rapid growth in data traffic and smartphone use has outpaced the capacity of existing spectrum, leaving mobile networks congested. Delays in releasing new bands have already widened the performance gap with regional peers, where 5G is being deployed at scale, and in Viet Nam's case with a 15 percent capital expenditure subsidy. Swift allocation of additional spectrum, especially 2.6GHz and 3.5GHz bands, is essential to relieve pressure on 4G networks in urban areas and accelerate the transition to 5G. Coordinated planning across bands will be critical to ensure fairness and efficiency, including enabling each MNO to access adequate spectrum to be competitive. Establishing a credible spectrum roadmap—with predictable timelines for release, refarming, and distribution—would provide the certainty MNOs and other investors need to expand network capacity and improve service quality nationwide. It will also help position Indonesia for 6G as that technology becomes available around 2030.

High spectrum and regulatory costs continue to constrain MNOs' ability to invest and modernize. MNOs in Indonesia face among the highest cumulative costs in the region, paying 11.4 percent of revenues in spectrum usage fees—nearly twice the regional average—on top of a 1.25 percent USO levy, 0.5 percent telco operations right fee, and corporate income tax (Figure B.12). In addition, overlapping local levies and permits on towers, rights-of-way, and land use raise compliance costs and delay deployments. While reducing these fiscal burdens would free resources for capital investment, fee reductions without clear coverage obligations risk lowering public revenues with limited improvements in service. A balanced approach would shift from large upfront charges toward simpler annual fees, combined with performance-based discounts tied to explicit coverage obligations, including for closing network "blank spots" and public facility connectivity, and measurable quality gains in underserved areas.

Figure B.12: Regulatory costs for MNOs



Source: World Bank staff calculations based on MNO's annual report and Komdigi auction data.[89]

88. Coverage bands" (typically below 1 GHz, such as 700 MHz) travel long distances and penetrate buildings, making them essential for wide-area and rural coverage. "Capacity bands" (mid-band spectrum such as 3.5 GHz) provide much higher throughput and are critical for carrying large volumes of data in urban and high-traffic areas. Effective 5G networks require a mix of both.
89. MNOs include Telkomsel, IOH, XLSmart, AIS, NT, True Corp, CelcomDigi, Maxis Mobile, Telekom Malaysia, DITO, Globe Telecom, and PLDT.

Once additional mobile spectrum is released, the next challenge for improving mobile broadband speeds will be fiberization of towers. Fiber backhaul is essential for unlocking the full capacity of new spectrum because of the much higher data loads and low-latency traffic that advanced mobile networks require. Many rural and remote towers are still connected through microwave links or satellite, which are more vulnerable to congestion, weather, and distance, resulting in slower and less stable speeds for users. Public disclosures by MNOs suggest that around half or two-thirds of towers are now connected to fiber.[90] However, a 2022 industry estimate was that there were at least 25,000 sites requiring fiberization, mostly in rural areas with a likelihood for this number to increase in the subsequent years.[91]

Accelerating 5G deployment is now critical for Indonesia's competitiveness and productivity. Beyond faster speeds, 5G offers lower costs (per GB) and greater energy efficiency. Yet Indonesia has been among the slowest in ASEAN to transition to 5G, reflecting delays in spectrum allocation and investment decisions. Once additional mobile spectrum is released, MNOs will need substantial capital to upgrade radio equipment, deploy new sites, and fiberize towers. At a fundamental level, recognizing digital infrastructure as a strategic sector—rather than primarily a revenue source—can help mobilize this private investment, accelerate network modernization and expansion, and ultimately drive faster productivity gains.

Fixed broadband

Expanding fixed broadband access is essential for Indonesia's shift toward digital productivity. Fixed broadband—delivered through fiber, fixed wireless access (FWA), or satellite links—provides the stable, high-capacity connectivity required for households, businesses, and public facilities to move from basic internet access to data-driven production and service delivery. Fiber networks remain the "gold standard," capable of delivering multi-gigabit speeds with low latency and almost unlimited scalability. FWA and satellite play a complementary role by extending coverage to areas where fiber deployment is not yet viable, though their performance can fluctuate depending on spectrum availability, signal conditions, and type and number of concurrent users. Unlike mobile broadband, which primarily serves users on the move and shares limited spectrum among many users, fixed broadband is designed to support high-bandwidth, always-on applications with more reliable throughput. AI and other data-intensive tools amplify the costs of slow and unreliable broadband, making fixed connections increasingly essential for firms, schools, and health facilities. Fiber also underpins the wider connectivity ecosystem: fiberized towers and backhaul sustain 5G expansion, while also relieving pressure on mobile networks.

Coverage has expanded recently, but uptake remains shallow and uneven. 39 million homes—nearly half of all households—are within reach of a fixed broadband network, yet only 17.4 million (22 percent) are connected.[93] This leaves a "coverage gap" of 45 million homes and a "usage gap" of 21.6 million.[94] Nearly all households are electrified, suggesting that the challenge lies not in general infrastructure availability but in affordability and demand. Uptake is highest in Java-Bali (28 percent) and falls to just 4 and 3 percent in Papua and Maluku, respectively.[95] Average fixed broadband speeds (41 Mbps) are third last in ASEAN, ahead only of Myanmar and Timor-Leste. This is not due to infrastructure or technical constraints, but, instead, a commercial decision by ISPs: most offer entry-level packages at 30-50 Mbps[96] to segment customers and maintain affordability, whereas in neighboring markets, and under Komdigi's own target, the baseline is now 100 Mbps. For Telkomsel's Indihome, the overwhelming market leader, average household revenue per user (ARPU) declined to about IDR 217,000 (US$13-14) per month in 2025, down from around IDR 250,000 two years earlier, reflecting growing competition that continues to drive prices downward, but not yet at a level that makes fixed broadband widely affordable.[97] As noted earlier, the cost of fixed broadband internet in Malaysia, Thailand, and Viet Nam is half or less of Indonesia's in GNI per capita terms.

90. For example, in 2024, XL Axiata (prior to its merger with SmartFren) reported to investors that 63 percent of its sites were fiberized 2024. The most recent public information for Telkomsel is that 66 percent of its sites were fiberized in 2020.
91. Analysys Mason, 2022. *Independent telecoms market study – Report for PT Mora Telematika Indonesia.*
92. Estimated homes passed are based on Telkom's 2024 reported data, used as a proxy given its dominant market share and nationwide network.
93. The number of connected homes is based on 2024/25 subscription data from 12 major ISPs – Telkom, PLN Icon Plus, MyRepublic, XLSMART, Biznet, Oxygen, CBNfiber, SURGE, Asianet (including MNC and IOH), and Bali Fiber – compiled from Telegeography's GlobalComms database, annual reports, and press releases. However, it should be noted that industry statistics also understate total users, as many households access internet services through smaller ISPs and informal neighborhood Wi-Fi resellers that are not counted in the formal subscriber base.
94. "Coverage gap refers to households and firms that are not within reach of any fixed broadband network. "Usage gap" refers to households and firms that are covered by a fixed broadband network but have not subscribed to a service.
95. World Bank. 2023. *World Bank Economic Household Survey 2023.*
96. APJII 2025 Internet Penetration and User Behavior Survey shows 85 percent fixed broadband subscriber buy below 50 Mbps and under IDR 300,000 package.
97. Figures are based on Telkom Annual Report 2023 and Telkom Info Memo 9M/2025

Cost and mobile substitution remain the main reasons households do not subscribe to fixed broadband. 32 percent of households cite cost as the primary reason for not signing up, while another 31 percent rely mainly on mobile internet instead. In Papua, West Papua, and Maluku, more than a third of households say no fixed broadband provider is available.[98] The substitution of mobile for fixed broadband also reflects perceived value: for many users, mobile data already meets basic needs for communication, entertainment, and social media, while the additional cost of a fixed broadband is harder to justify without more content, digital services, and devices that require faster speeds. This creates a potential "chicken-and-egg" problem: low demand discourages network investment, while limited networks and low-data content keep demand from growing.

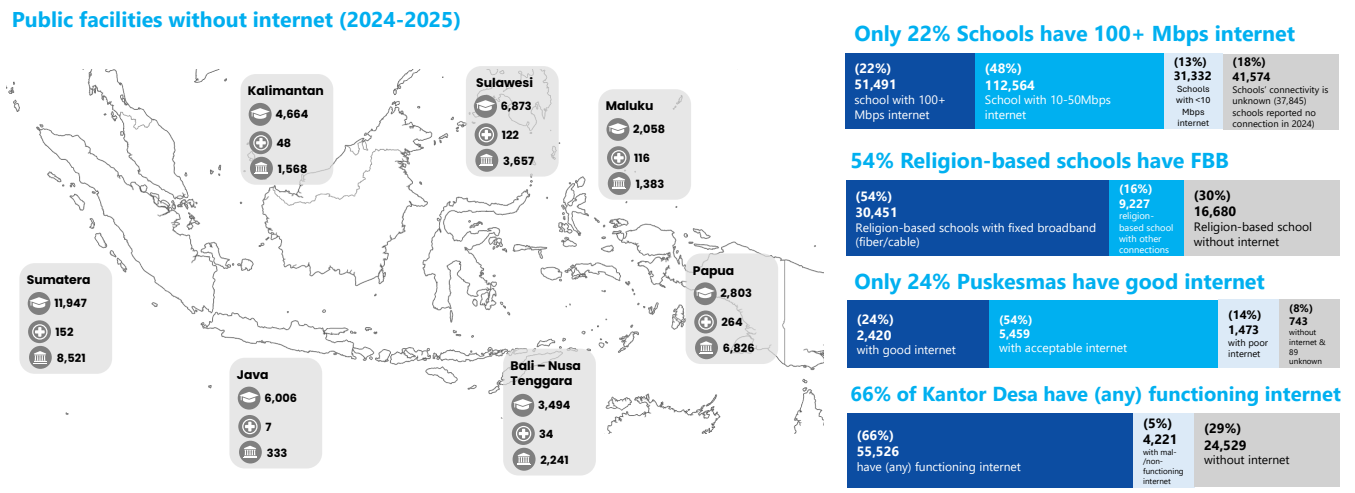
Competition has expanded in recent years but remains insufficient to counter strong market concentration. Telkom Group has roughly two-thirds of all retail fixed broadband subscriptions, followed by PLN's Iconnet at 9 percent. The remaining quarter of the market is fragmented among more than 300 active ISPs—out of about 1,200 licensed nationwide—including XL, MyRepublic, Biznet, and numerous local cooperatives. Telkom Group's extensive backbone and access network, and to a lesser extent PLN's fiber network, continue to determine the market structure. Competition remains strongest in urban areas, while smaller ISPs in secondary cities and rural districts face high entry barriers due to limited availability of passive infrastructure and continued dependence on incumbent wholesale networks, which drives up backbone access costs. A more competitive wholesale environment would expand the overall market, as lower prices and better service quality can unlock substantial new demand—benefiting all ISPs.

FWA is emerging as an important complement to fiber in expanding broadband coverage. The 2025 auction of 1.4 GHz spectrum to two ISPs marks an important step, enabling delivery of last mile connectivity and more affordable plans in rural and peri-urban areas. Surge, one of the ISPs, has announced plans for 100 Mbps at IDR 100,000 per month.[99] However, the 1.4 GHz band is not commonly used for FWA globally, and compatible customer premises equipment (CPE) remains limited and thus costly.[100] FWA can quickly bring homes and small businesses online where fiber is not yet viable, but cannot match the speed and reliability of FTTP, as performance depends on spectrum availability, signal quality, and line-of-sight. As additional and more standard spectrum bands are refarmed and MNOs deploy 5G-based FWA, performance and cost efficiency are expected to improve substantially. Even so, FWA should be seen as a complementary solution: it can extend connectivity quickly where fiber is not yet viable, while fiber will remain essential for higher speeds, capacity, and long-term network scalability. In the most remote and commercially challenging areas, however, FWA may still offer a practical enduring option.

High-speed internet at schools, health centers, and local governments is critical for Indonesia's digital transformation and improvements in service delivery, yet current levels are alarmingly low. As of 2025, only 22 percent of schools have 100 Mbps or faster connections,[101] while 48 percent have 10-50 Mbps and 17 percent (roughly 38,000 schools) remain unconnected.[102] In 2023, only 22 percent of schools had a fixed broadband connection and 36 percent were unconnected, demonstrating an improvement in access but not really an improvement in *quality*.[103] Among religion-based schools, 54 percent are connected, and 30 percent (17,000 facilities) remain unconnected, with speeds and connection type unknown.[104] Connectivity gaps are equally severe in health and village administration facilities: just 24 percent of health clinics (*puskesmas*)[105] have good internet, and only

98. World Bank, 2023. *World Bank Economic Household Survey 2023.*
99. Surge, 2025. *PT Solusi Sinergi Digital Tbk (Surge) to Roll Out "People's Internet" 5G FWA Network - IDR 100,000 for 100 Mbps Unlimited Data in Region 1 (Java, Papua & Maluku) - Covering Over 60% of Indonesia's Market Potential.* October 22.
100. Global mobile Suppliers Association's 5G *Market Snapshot (Nov 2025)* and *LTE Device Ecosystem (Sept 2025)* report no commercial networks or major device base in the 1.4 GHz TDD band, unlike the widely supported 2.3, 2.6, and 3.5 GHz bands. However, in November 2025, Surge has announced a major with Nokia for the requisite FWA equipment.
101. There is no international consensus on the minimum internet speeds for schools. The ITU-UNICEF Giga school connectivity initiative tiers schools according to 10-20 Mbps (Minimum), 50-100 Mbps (Good), and 100+ Mbps (Excellent). In OECD countries, the target for school connectivity is generally 250 Mbps to 1 Gbps.
102. World Bank staff calculations based on H1 2025 *Data Pokok Pendidikan* (DAPODIK) by the Ministry of Primary and Secondary Education (MoPSE), categorized by speed. Data on technology was not available for this period.
103. Proportions are based on 2023 DAPODIK by MoPSE, categorized by technology. Data on speeds was not available for this period.
104. Proportions are based on 2024 religion-based schools connectivity data by Ministry of Religious Affairs.

Figure B.13: Public service connectivity



Source: Ministry of Primary and Secondary Education (2025), Ministry of Religious Affairs (2024), Ministry of Health (2024), and *Potensi Desa Survey 2024.*

two-thirds of village offices (*kantor desa)*[106] have any functioning connection (see Figure B.13). In contrast, schools have universal fixed broadband connectivity in Viet Nam[107] and China[108] and 70 percent in Thailand.[109] If Indonesia is to achieve high-income status and strengthen its human capital and service delivery, it will need to invest far more in school and other public facility connectivity. Importantly, public facilities are also a source of sustainable demand that can make rural fixed broadband expansion commercially viable.

These public facilities are also the foundation for community-level digital infrastructure. Rural Indonesia has few community Wi-Fi points, digital access centers, or local training hubs, which limits the ability of citizens and MSMEs in these areas to meaningfully use digital services. Countries that successfully narrowed digital divides, such as Viet Nam, Thailand, and India, paired last-mile broadband with community access facilities and digital literacy programs. Komdigi's Village Internet (*Kampung Internet*) program[110] has shown that subsidizing internet for public facilities and such community initiatives can help not only connect those facilities but help to expand internet access in the communities around them. Currently limited to a very small scale, it could be scaled up into a transformative national program.

High deployment costs, limited infrastructure sharing, and weak wholesale competition continue to constrain middle and last mile network expansion and affordability. Civil works are expensive and time-consuming due to inconsistent and often high local right-of-way fees, which can differ among local governments. The 2020 Omnibus Law on Job Creation formally opened utility and transport infrastructure for fair shared use, lowering barriers for new entrants and improving coordination between the telecommunications, energy, and transport sectors—but it has not yet been fully enforced. The recent spin-off by Telkom of at least half of its fiber assets to its subsidiary, InfraNexia—a dedicated wholesale fiber company—can be a game-changer, if it provides transparent, non-discriminatory access. Similar approaches in other countries where wholesale fiber assets are legally or structurally separated to ensure neutral access have led to strong results for competition and retail affordability.[111] Likewise, if PLN were to open access to its poles, it could both monetize a valuable national asset and reduce rural broadband deployment costs by as much as two-thirds. For example, Nepal has achieved roughly twice Indonesia's fixed broadband penetration while maintaining much more affordable prices, in part by leveraging electricity poles and rights-of-way to lower deployment costs.

105. Ministry of Health, 2024. *SATUSEHAT data – Puskesmas connectivity.*
106. Central Bureau of Statistics (BPS), 2025. *Potensi Desa Survey 2024.*
107. VietnamNet, 2022. *Vietnam Changes Rapidly After 25 Years of Internet Connection.* Published November 21, 2022.
108. Ministry of Education of the People's Republic of China, 2023. *World Digital Education Conference 2023 Opens in Beijing.* Published February 13, 2023.
109. UNESCO, 2023. Global Education Monitoring Report 2023: *Technology in Education – A Tool on Whose Terms?*
110. *Kampung Internet* is a rebranding of the long-running *Bantuan Pemerintah Layanan Akses Internet (*BPLAI), which provides internet access subsidies for schools, health facilities, government offices, and MSMEs, in underserved areas.
111. Examples include New Zealand (Chorus), Singapore (NetLink Trust), and the United Kingdom (Openreach).

Improving in-building connectivity is also important for realizing the benefits of network investment. Many households cannot fully benefit from high-speed access because of outdated modems or legacy Wi-Fi equipment that support only older standards (typically Wi-Fi 4). As a result, effective speeds often fall short of network capacity. Komdigi's 2025 support of Wi-Fi 6E/7 making the lower 6 GHz band available for Wi-Fi provides an opportunity to raise performance standards and encourage device upgrades.[112] Promoting adoption of newer Wi-Fi 5 and 6 standards through updated technical guidelines and consumer awareness campaigns could enhance end-user speeds. Requiring fiber-ready in-building wiring in new housing and public facilities would further ensure that high-capacity networks deliver their intended performance.

Satellites provide a complementary pathway for remote and underserved regions. The government's SATRIA-1 geostationary earth orbit (GEO) satellite,[113] launched through a PPP in 2023 following a 2019 tender, has a total capacity of 150 Gbps and originally aimed to connect 150,000 public service points in unserved and underserved areas across Indonesia.[114] After launch, the target was reduced to 50,000 and then 37,000—and installation has advanced well against this adjusted scope, with 27,805 locations connected by mid-2025.[115] However, around three-fifths of activated sites are in Sumatra, Java, and Sulawesi—areas where terrestrial fiber and FWA deployment are feasible and more optimal—while less than 12 percent are in Papua and Maluku, where sparse populations and challenging terrain make satellite the most effective solution. With SATRIA-1's capacity finite,[116] refocusing its use toward public facilities in the most remote areas, particularly in eastern Indonesia, that are unlikely to be feasibly served by terrestrial technologies in the near-term would help ensure the satellite delivers the greatest impact and most efficiently. This need is significant: in Papua and Maluku alone, an estimated 4,861 schools, 380 health clinics, and 8,209 village offices still lack any internet connection (see Figure B.13), with many more relying on limited mobile coverage.

The 2024 launch of Starlink's low earth orbit (LEO) satellites[117] has been positively disruptive for the market in Indonesia. Demand has surged—not only in rural areas lacking reliable terrestrial networks, but also in cities like Jakarta, where Starlink was temporarily oversubscribed until additional capacity was added in mid-2025.[118] This strong uptake, despite prices of IDR 400,000-700,000 per month, highlights the appetite among households and firms for faster and more reliable internet. Starlink and similar LEO systems are also increasingly being used to provide satellite backhaul for mobile towers in remote regions, helping extend terrestrial coverage and provide a reliable backup option. Over the longer term, LEO systems are best positioned to serve rural communities and maritime regions, while GEO and medium earth orbit (MEO) satellites will continue to play important roles in providing backhaul, broadcast services, and resilient connectivity for remote public facilities. Satellite capacity is also essential for emergency communications and redundancy in a country highly exposed to natural disasters. Emerging direct-to-device (D2D) technologies[119] could further extend connectivity by allowing standard smartphones to connect directly to satellites, improving coverage in remote areas, at sea, and during emergencies.

Indonesia's fixed broadband market is entering a critical phase in which improved wholesale access and stronger competition will be central to expanding nationwide connectivity. Strengthening both urban and rural markets will require a modern wholesale policy and regulatory framework that opens opportunities for fair, transparent access to essential infrastructure, such as Telkom's fiber network and PLN's poles, and ensures neutral interconnection among operators to unlock greater private investment and competition. As noted earlier, Telkom's restructuring of its fiber assets into InfraNexia is a key opportunity to achieve these outcomes. This would mirror the shared infrastructure principles that underpinned strong competition and affordability in mobile broadband, extending them to the fixed sector. It will also be important for Komdigi to address anti-

112. Komdigi Press Release, 2025. *Menkomdigi resmikan Wi-Fi 6E dan Wi-Fi 7, percepat transformasi digital Indonesia.* Published February 8, 2025.
113. GEO satellites orbit at 36,000 km above the earth and cover wide areas, making them well suited for providing backhaul and public-service connectivity in remote regions, though with higher latency (typically 500–700 milliseconds), making them most suitable where terrestrial networks are not yet feasible.
114. Asian Infrastructure Investment Bank (AIIB), 2020. *Indonesia Multifunctional Satellite Public–Private Partnership (PPP) Project, Project Summary.*
115. Bisnes Tekno, 2025. *"27.805 Titik Terkoneksi Internet Satelit Satria-1 Bakti Mei 2025"* June 8, 2025.
116. For example, if 30,000 connections are working simultaneously, that would translate to a maximum average busy hour speed of 5 Mbps, well below the effective 100 Mbps, 1 Gbps or 10 Gbps access speeds that would be recommended for public facilities (depending on size).
117. LEO satellites operate at 500-2,000 km above the earth and offer much lower latency (20-40 milliseconds) and faster setup times but have limited total capacity per beam and require dense constellations.
118. OpenSignal, 2025. *Starlink in Indonesia: bridging the digital gap but at what cost?*
119. D2D technologies enable satellites to connect directly to standard smartphones or IoT devices without the need for separate satellite terminals or ground antennas.

competitive behavior where it occurs and enforcing open access principles in ways that preserve incentives for private operators to invest. Local governments, meanwhile, can support passive infrastructure sharing, such as shared ducts and utility corridors, through coordinated planning, streamlined permitting, and targeted incentives for joint builds. Achieving universal connectivity efficiently will require FTTP where feasible, FWA where it is not, and satellite in the most remote areas.

The government can play a catalytic role in expanding fixed broadband infrastructure by making rural deployment commercially viable for private investment. International experience shows that private investment alone rarely delivers nationwide fiber networks—particularly beyond dense and commercially attractive areas—underscoring the need for a proactive public role. A key bottleneck for rural expansion is the absence of predictable revenue. In Indonesia, annual public procurement cycles and the lack of multi-year subscription contracts mean ISPs cannot rely on stable demand from public facilities, making it difficult to justify new network investment even if the demand is there. Under its mandate to coordinate national digital infrastructure and advance universal connectivity and working with other national and local government stakeholders, Komdigi could orchestrate the aggregation of public facility demand across sectors nationwide, use multi-year contracts to incentivize network expansion into areas without fiber, facilitate access to PLN's poles, and require open access for new fiber networks. By pooling demand and providing predictable, multi-year revenue streams, these measures would strengthen the commercial case for private operators to invest in underserved regions. Such an approach of leveraging public resources has been deployed successfully across the world, such as in India, New Zealand, and Thailand (*Net Pracharat* – see Box B.4).

BOX B.4: Thailand's *Net Pracharat* – a playbook for Indonesia to expand fixed broadband in rural areas?



Thailand's *Net Pracharat* (People's Internet) program demonstrates how fixed broadband can expand in rural areas efficiently when the state orchestrates, shares infrastructure, and anchors demand.[120] Approved by Cabinet in December 2016, it began with a national mapping of unserved villages, followed by a commitment to connect all by fiber or satellite. Led by the Ministry of Digital Economy and Society (MDES), the SOE telecommunications company (TOT) deployed open access fiber using electricity poles and rights-of-way from the Metropolitan and Provincial Electricity Authorities, with permits fast-tracked and fees waived. The National Broadcasting and Telecommunications Commission (NBTC) filled residual gaps by satellite. The central budget financed rollout and USO funds guaranteed five years of maintenance to allow private participation. By December 2017, fiber reached 24,700 villages and extended to schools, clinics, and community offices as anchor sites. The network was open to all licensed ISPs, stimulating local service provision and competition. Digital skills training reached over one million people within a year.

This offers a useful model for Indonesia, showing how coordinated planning across government, electricity, and telecommunications sectors, under Komdigi's policy leadership, could accelerate broadband rollout.

120. Thailand Ministry of Digital Economy and Society (MDES), 2018. *Thailand Village Broadband Internet Project: Net Pracharat.*

Data centers and cloud

Data centers are critical enablers of productivity in a digital economy, allowing firms and governments to store, process, and analyze data efficiently, supporting automation, logistics optimization, digital finance, and e-commerce—all of which can raise productivity and create higher-skilled jobs. Data centers also facilitate technology transfer from foreign investment: global operators bring expertise in energy management, cybersecurity, and operations that can strengthen Indonesia's domestic capabilities. As demand for computing power grows—driven by cloud services, data-intensive applications, and AI—Indonesia will need to expand domestic data center capacity to meet growing local needs. At the same time, Indonesia has a strategic opportunity to attract regional workloads, bringing investment, better jobs, and deeper integration into regional digital value chains. Achieving this requires strong integration with regional and global networks because cloud and platform companies place regional capacity where latency is lowest and international routes are most diverse. High-quality interconnection is therefore essential for Indonesia to compete as a regional hub and to support local innovation.

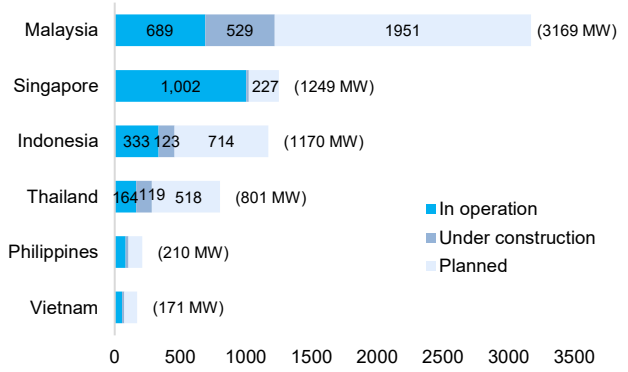
Indonesia's data center market is growing but capacity remains modest and low in per capita terms compared to ASEAN peers. Total operational capacity in Indonesia reached around 333 MW in mid-2025 (1.2 watts per capita) and is projected to rise to 456 MW by end-2025, well behind Malaysia, which is expected to rise to 1,216 MW in the same period (22 watts per capita) (Figure B.14).[121] In 2025, Thailand's pipeline capacity grew faster than Indonesia. Industry forecasts have Indonesia reaching the RPJMN 2029 target of 2,000 MW (6.9 watts per capita) a year late, in 2030. Over half of existing capacity (57 percent) is in Jabodetabek,[122] followed by Batam (7 percent), Surabaya (6 percent), and other cities across Java, Sumatra, and Bali (30 percent) (Figure B.15). This concentration reflects proximity to customers and backbone connectivity, but it also creates dependence, puts pressure on local electricity networks, and limits resilience.

Reliable, scalable, and increasingly renewable electricity has become a decisive factor in global data center location decisions, especially for AI-ready data centers. AI workloads are far more energy-intensive than traditional applications, sharply increasing demand for reliable, high-capacity and renewable power in data center hubs. In Indonesia, long lead times for electricity connections—often up to three years in Jabodetabek—combined with limited access to renewable energy constrain the expansion of existing hubs and weaken competitiveness relative to Malaysia and Thailand, which offer faster grid connections and clearer renewable energy pathways. Batam has strong potential to evolve into a renewable energy–powered data center hub, supported by emerging regional collaboration initiatives with Singapore and Malaysia under the framework of the ASEAN Power Grid. Over time, and as demand grows and grid readiness improves, locations with substantial geothermal, hydro, or solar resources—such as North Sumatra and parts of Sulawesi—could emerge as future secondary hubs. Leveraging these resources, alongside grid upgrades and improved planning, will be essential to attract hyperscale investment and position Indonesia as a competitive regional hub.

Growth is further constrained by uncertainty regarding data regulations, complex permitting, and fragmented investment facilitation. Overlapping general and sectoral regulations on data governance and cross-border flows, compounded by limited implementing guidelines and uncertainty about how rules will be enforced or evolved, raise compliance costs and discourage investment. Some data center operators have reported needing more than 100 permissions.[123] Although some permits have been streamlined through the Online Single Submission (OSS) Risk-Based Approach, others that are medium- and high-risk have processes outside of the OSS, leading to cumulative timeline of 12-24 months, compared with 3-9 months in Malaysia and Thailand. Combined with the current exclusion of Jakarta as a designated international submarine cable landing site and restrictions on foreign-flagged cable-repair vessels, these challenges constrain investor confidence and delay project delivery. As discussed in the 'Backbone' section, international connectivity and domestic IXPs are essential for data centers to deliver low-latency performance and for Indonesia's firms to participate more in the global digital economy.

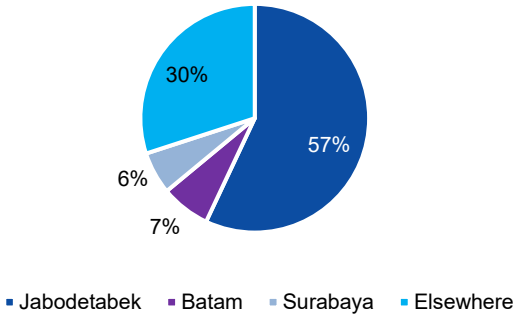
121. Cushman & Wakefield, 2025. *Asia Pacific Data Center Update, 1H 2025.*
122. Jabodetabek refers to the greater Jakarta area, including Jakarta, Bogor, Depok, Tangerang, and Bekasi.
123. World Bank Group, forthcoming. *Country Private Sector Diagnostic – Indonesia (Data Centers Deep Dive).*

Figure B.14: Data center capacity in major ASEAN economies (H1 2025)



Source: Cushman & Wakefield, 2025.

Figure B.15: Data centers by location



Source: Telegeography, 2025.

Regional competitors are advancing rapidly with more cohesive investment attraction. When Singapore had a moratorium on new data centers during 2019-2022 due to concerns over land availability, high electricity use, and environmental sustainability, Malaysia positioned itself as the region's primary alternative, offering a one-stop Digital Investment Office, renewable energy access, and an infrastructure upgrade program for Johor, the data center hub. Thailand is also scaling up through the Eastern Economic Corridor of Digital, combining tax incentives, full foreign ownership, and renewable energy access. Singapore has recently announced a 700 MW Green Data Center Park in Jurong Island. These moves illustrate how strategic coordination between energy, investment, and digital authorities can accelerate private investment. For Indonesia to compete in attracting investment, it will need to better align regulation, planning, and the provisioning of energy and water.

As the government considers its strategy for public sector data hosting and PDN, there is an opportunity to better leverage private investment while safeguarding data sovereignty. The original PDN vision aimed to consolidate thousands of government systems into three government-managed PDN facilities, aiming to reduce inefficient public ICT spending. One PDN facility has been completed in Jabodetabek but the project for the second, in Batam, was cancelled in 2025. There is an opportunity for a clearer and more practical model for government data hosting—one that defines the respective roles of PDN, agency data centers, and certified commercial cloud services, and allows them to operate together in a coordinated in a secure way.[124] A clearer framework would also create space for PPP arrangements where private providers can support secure and cost-effective data center and cloud capacity. While Minister of Komdigi Regulation No. 5 of 2025 establishes a risk-based data classification framework across government, there is still no comprehensive regulatory framework governing the use of commercial cloud services,[125] creating ambiguity and uncertainty for both government agencies and the private sector.

Accelerating data center growth will require investment incentives, regulatory clarity, and infrastructure readiness. Providing fiscal incentives and fully streamlining approvals through the OSS system would shorten timelines and attract more global operators. Clear, risk-based rules on cross-border data flows would strengthen legal certainty for regional customers. Batam could also be considered as a "trusted data zone,"[126] enabling secure cross-border data exchange with neighboring economies and enhancing its role as a regional hub. Expanding renewable energy access, including through power-purchase agreements (PPAs), and ensuring timely electricity and water connections would improve reliability and sustainability. Reforming the approach to PDN offers an opportunity to promote secure cloud adoption across government in ways that will increase private investment and contribute to better digital

124. This approach is consistent with international good practice, where commercial cloud is routinely used for non-sensitive workloads because it offers greater resilience, scalability, and cost efficiency.
125. Minister of Kominfo Decree No. 519 of 2024 on the Operation of the PDN provides partial guidance on third-party cloud services, but its scope and enforceability are limited.
126. "A "trusted data zone" is an area that enables secure, seamless cross-border data flows under shared safeguards. In a place like Batam, this would allow data to move more easily to and from regional partners under mutually recognized privacy and security rules, reducing the uncertainty that often restricts cross-border data transfers."

public services. With its interest in sustainable data centers, Danantara could also play a role in crowding in private capital.

Device affordability

High smartphone adoption reflects improving device affordability, with opportunities to make it more inclusive. 74 percent of adults own a smartphone, below Viet Nam and Thailand, which are above 85 percent.[127] 67 percent of non-mobile phone owners cite price as the main barrier.[128] Low-end smartphones cost about 42 percent of the country's average monthly salary,[129] higher than in India (32 percent), the Philippines (26 percent), and Thailand (12 percent).[130] At this level, devices are increasingly affordable in urban areas with higher incomes but remain costly in poorer, rural regions. The growing secondhand market is helping narrow this gap, with global shipments projected to rise 5 percent annually and outpacing new phones (1 percent) through 2029.[131] Indonesia could improve device affordability by lowering tariffs, especially for entry-level devices, and expanding local manufacturing, as most devices remain imported.

4. Conclusion and Recommendations

Digital transformation can make an important contribution to Indonesia's future growth but unlocking its economic potential will require a new focus toward higher-quality digital infrastructure. Indonesia has expanded access rapidly, but the next phase will require closing persistent coverage gaps while simultaneously improving service quality, affordability and data center capacity so that digital tools can be used productively nationwide. Predictable regulation, stronger competition, wider infrastructure sharing, and a more proactive government role—including leveraging public sector resources and demand—will all be essential to mobilize the much greater private investment needed to deliver. This also means treating digital infrastructure as a strategic enabling sector, whose growth is critical for the wider economy—and not primarily as a revenue source. Without timely reforms, Indonesia risks losing ground to regional competitors racing to capture digital infrastructure investments, especially to prepare for the AI-driven economy.

The following four objectives and seven recommendations set out the highest-impact near-term reforms to simultaneously strengthen Indonesia's digital infrastructure and close the remaining gaps in access. Together, they address the core bottlenecks that limit the ability of digital infrastructure to drive productivity, growth, and job creation in Indonesia. While not exhaustive, these priorities focus on the reforms with the greatest potential to deliver near-term impact and crowd in greater private investment. Looking ahead, it will also be important to develop a more comprehensive national digital infrastructure plan to guide long-term investment and coordination.

At the same time, the impact of better digital infrastructure will be greatest when accompanied by complementary reforms and investments across the digital ecosystem. Prioritization should be given to advancing digital skills, improving governance and regulatory predictability, accelerating the digital transformation of government, safeguarding digital trust through stronger personal data protection and cybersecurity frameworks, and scaling DPI. Together, these measures will help ensure that improved connectivity and expanded data center capacity translate into wider access to economic opportunities, stronger job creation, higher productivity, and faster growth of Indonesia's digital economy.

127. World Bank, 2025. *Global Findex Survey – 2024.*
128. ibid.
129. World Bank Staff calculation based on the Global Digital Inclusion Partnership (GDIP) device affordability measurement methodology.
130. Global Digital Inclusion Partnership (GDIP), 2024. *The price of the world in your pocket: a price only some can afford.*
131. International Data Corporation (IDC), 2025. *The Smartphone Shifts: Why Second-Hand Devices Are Outpacing New Ones*. IDC Blog, August 27, 2025.

Table B.1: Proposed objectives and recommendations for strengthening Indonesia's digital infrastructure

Objective	Recommendation	Additional information
Multiply mobile internet speeds	1. Assign available 700 MHz and 2.6 GHz band spectrum to mobile broadband, and release and assign the 3.5 GHz band capacity in urban areas to accelerate 5G deployment.	• Ensure fair distribution of spectrum capacity among MNOs to maximize competition. • Lower spectrum fees and and link them to clear coverage and performance obligations to incentivize efficient network investment.
Make fixed broadband universal and affordable	2. Mobilize private investment in rural broadband by aggregating predictable, multi-year demand from schools, health clinics, and village offices to increase commercial viability.	• Refocus SATRIA-1 capacity away from areas already or feasibly served by fiber and FWA to the most remote regions, while preserving capacity for emergency communications and network redundancy. • Reorienting BAKTI from network operations to market facilitation can help strengthen its financial sustainability and mobilize greater private investment.
	3. Promote fair open access to wholesale fiber networks and passive infrastructure—including PLN's poles—and standardized rights-of-way procedures to improve competition and reduce deployment costs.	
Expand data center and cloud capacity to support the digital economy and AI era	4. Streamline data center permitting and improve coordination across special economic zone (SEZ) policy, electricity provision, and incentive frameworks to reduce delays and improve the investment environment.	• Reliable and timely access to electricity and water, including renewable energy, is essential for supporting large-scale data center growth.
	5. Set clear, risk-based rules for data storage and cross-border data flows—including through issuance of the Government Regulation on Personal Data Protection and alignment of other regulations—while promoting secure government adoption of cloud services to anchor demand.	• Developing Batam as a trusted data zone could support secure cross-border data flows and help attract regional data center investment. • A hybrid PDN model can enable secure government cloud use and attract private investment, while protecting data.
Strengthen and diversify Indonesia's digital backbone	6. Amend Minister of Maritime Affairs and Fisheries Decree No. 14/2021 to allow additional international submarine cable landing sites, including Jakarta and other major commercial hubs.	• Permitting and security safeguards can be applied while avoiding unnecessary barriers to cable landings in the national interest. • Allowing foreign repair vessels to more easily operate would shorten restoration times and improve network resilience.
	7. Review Palapa Ring wholesale tariffs to ensure open and sustainable access, particularly to improve affordability and connectivity in eastern Indonesia.	• Tariffs should recover operating costs and stimulate network deployment, backed by transparent reference offers and direct interconnection to boost competition.

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